MIL-STD-1651

27 June 1975

MILITARY STANDARD

INSERT ARRANGEMENTS FOR MIL-C-5015, MIL-C-22992 (CLASSES C, J, AND R), AND MIL-C-83723 (SERIES II) ELECTRICAL CONNECTORS

FSC 5935

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Insert Arrangements for MIL-C-5015, MIL-C-22992 (Classes C, J, and R), and MIL-C-83723 (Series II) Electrical Connectors

MIL-STD-1651

1. This military standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions, or deletions should be addressed to the Commander, Naval Air Systems Command, Code Air 533D, Washington, DC 20360.

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1. SCOPE

1.1 This standard covers insert arrangements for use with MIL-C-5015, MIL-C-22992 (classes C, J, and R), and MIL-C-83723 (series II) electrical connectors.

2. REFERENCED DOCUMENTS

2.1 The issues of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

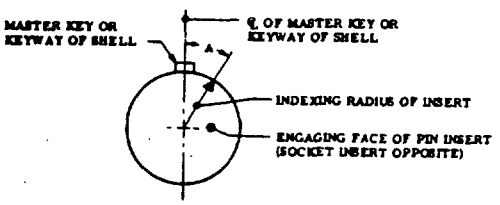
MILITARY

MIL-C-5015 - Connectors, Electric, "AN" Type.
MIL-C-22992 - Connectors, Plugs and Receptacles, Electrical, Waterproof, Quick Disconnect, Heavy Duty Type, General Specification for.
MIL-C-83723 - Connectors, Electrical, (Circular, Environment Resisting), Receptacles and Plugs, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 The terms used in this standard are generally accepted by the electrical and electronics industries and commonly used in electrical connector engineering practice.



4. GENERAL REQUIREMENTS

Not applicable.

5. DETAIL REQUIREMENTS

5.1 Dimensions. Dimensions shall be in accordance with the applicable section of this standard and the following dimensional data:

(a) Dimensioning and tolerancing in accordance with ANSI Y14.5-1973. Dimensions are true position and are in inches.
(b) Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
(c) Dimensions and markings shown are for engaging face of pin insert. Socket insert is opposite.
(d) The following tolerances apply to insert installed in shell:
1. The center of each hole in insert for crimp contact connectors shall be located at true position within .014 dia. ⊕ .014 dia.
2. The center of engaging end of each contact shall be located in true position within .024 dia. ⊕ .024 dia.
(e) Unless otherwise indicated, dimensions are symmetrical about centerline. ▲ indicates ℄ of insert arrangement.
(f) Each insert arrangement is shown in the "normal position" in the shell.

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5.2 Marking. Marking shall be in accordance with MIL-C-5015, MIL-C-22992, or MIL-C-83723, as applicable, and as shown in the applicable section of this military standard.

5.3 Contacts. Solder contacts shall be in accordance with MIL-C-5015, MIL-C-22992, or MIL-C-83723, as applicable. Crimp contacts shall be in accordance with table I. Contact sizes are designated by the symbols shown in table I.

TABLE I. Contact designation.

Crimp or solder contacts		Crimp contact part numbers		
		MIL-C-5015		MIL-C-83723 (series II)
Size	Symbol	MS3400's connectors	MS3450's connectors	
		MIL-C-23216 contacts	MIL-C-23216 contacts	
0		MS90453-0-0 MS90454-0-0	MS3162-0-0 MS3163-0-0	M83723-29T0 M83723-30T0
4		MS90453-4-4 MS90454-4-4	MS3162-4-4 MS3163-4-4	M83723-29T4 M83723-30T4
8		MS90453-8-8 MS90454-8-8	MS3162-8-8 MS3163-8-8	M83723-29T8 M83723-30T8
12		MS90453-12-12 MS90454-12-12	MS3162-12-12 MS3163-12-12	M83723-29T12 M83723-30T12
16		MS90453-16-16 MS90454-16-16	MS3162-16-16 MS3163-16-16	M83723-29T16 M83723-30T16
8 (shielded)		MS14056		N/A
Video coax		MS33695		

5.4 International standardization agreements. Certain insert arrangements of this standard, identified by *, #, and †, are the subject of the related international standardization agreements listed under "International Interest". When revision or cancellation of this standard is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including Departmental Standardization Offices, if required.

INTL INTEREST

* ASCC 12/44
NEPR 28
† NEPR 44

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Custodians:
Army - EL
Navy - AS
Air Force - 85

Review activities:
Army - MU
Navy - EC
Air Force - 11, 15, 17, 80
DSA - ES

Preparing activity:
Navy - AS

Agent:
DSA - ES

(Project 5935-2026)

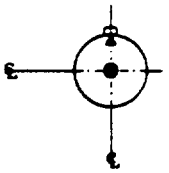
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SECTION 10

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 8, 8S

NOTES:
1. Shell size 8 is identical to shell size 8S and applies only to MIL-C-83723, series II.
2. Shell size 8S applies only to MIL-C-5015.
3. Section 10 is not applicable to MIL-C-22992.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
8, 8S	-1	A	1	16		---	MS33680-1

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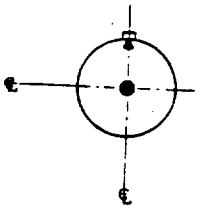
SECTION 20

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 10, 10S, 10SL, AND 11

NOTES:

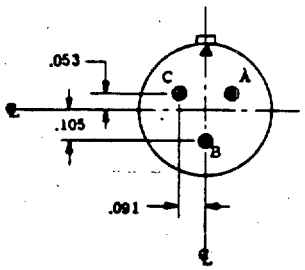
1. Shell size 10 is identical to 10S, and applies only to MIL-C-83723, series II.
2. Shell size 10S applies only to MIL-C-5015.
3. Shell size 10SL applies only to MIL-C-5015.
4. Shell size 11 is identical to 10SL, and applies only to MIL-C-83723, series II.
5. Section 20 is not applicable to MIL-C-22992.



ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
10, 10S	-2	A	1	16		---	MS33680-2

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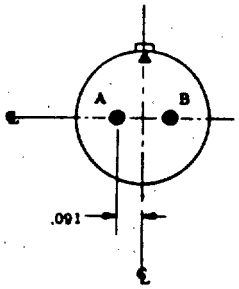


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

* #

Shell size	Arrange-ment number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Connector class	Supersedes document
10SL, 11	-3	INST	3	16	●	All	K	MS33680-3
		A					All others	

INACTIVE FOR DESIGN
AFTER 17 DEC 1958



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
10SL, 11	-4	A	2	16	●	All	MS33680-4

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THIS ARRANGEMENT NUMBER NOT TO BE USED

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
10, 10S	-5	---	---	---	---	---	MS33680-5

CANCELLED AFTER 17 DEC 1958

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
10, 10S	-6	---	---	---	---	---	MS33680-6

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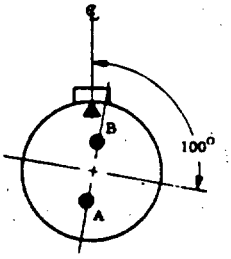
SECTION 30

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 12, 12S, AND 13

NOTES:
1. Shell size 12 applies to MIL-C-5015, MIL-C-22992, and MIL-C-83723, series II.
2. Shell size 12S applies only to MIL-C-5015.
3. Shell size 13 is identical to 12S, and applies only to MIL-C-83723, series II.

INACTIVE FOR DESIGN
AFTER 8 MAR 1945



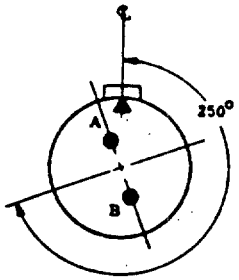
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 12S-3 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12S, 13	-1	A	2	16		A11	MS33681-1

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Check the source to verify that this is the current version before use.

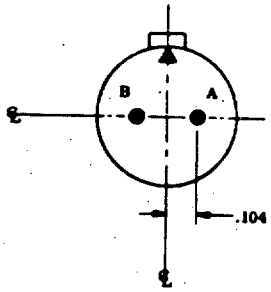
INACTIVE FOR DESIGN AFTER 8 MAR 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 12S-3 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12S, 13	-2	A	2	16	●	All	MS33681-2



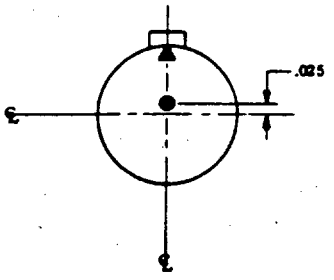
FOR OTHER INSERT POSITIONS SEE 12S-1 AND 12S-2

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
70	145	215	290

* #

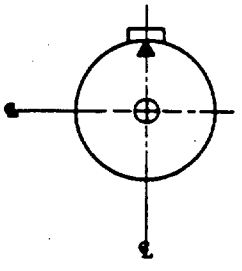
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12S, 13	-3	A	2	16	●	All	MS33681-3

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12S, 13	-4	D	1	16	●	---	MS33681-4



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12	-5	D	1	12	⊕	---	MS33681-5

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CANCELLED
AFTER
8 MAR 1945

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12S, 13	-6	---	---	---	---	---	MS33681-6

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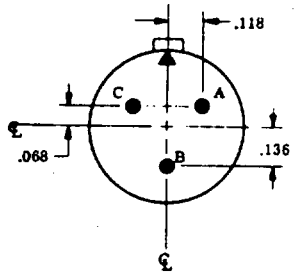
SECTION 40

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 14, 14S, AND 15

NOTES:
1. Shell size 14 applies to MIL-C-5015, MIL-C-22992, and MIL-C-83723, series II.
2. Shell size 14S applies only to MIL-C-5015.
3. Shell size 15 is identical to 14S, and applies only to MIL-C-83723, series II.

INACTIVE FOR DESIGN
AFTER 17 APR 1945

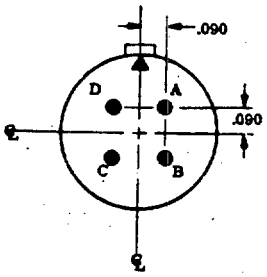


FOR OTHER INSERT POSITIONS SEE 14S-12 AND 14S-13

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-1	A	3	16		All	MS33682-1

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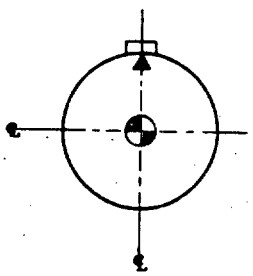


FOR OTHER INSERT POSITIONS SEE 14S-10 AND 14S-11

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	120	240	—

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-2	INST	4	16		All	MS33682-2



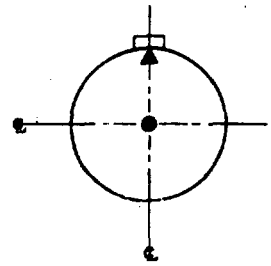
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

†

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14	-3	A	1	8		---	MS33682-3

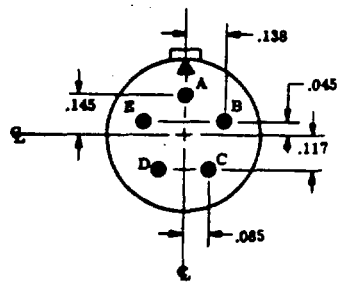
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INACTIVE FOR DESIGN AFTER 13 MAY 1954



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-4	D	1	16		---	MS33682-4

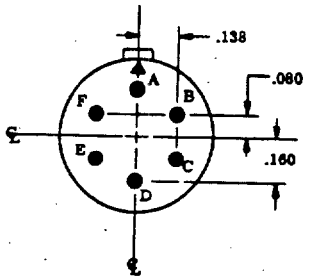


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	110	—	—

†

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-5	INST	5	16		A11	MS33682-5

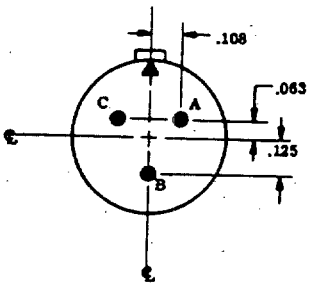
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

* #

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-6	INST	6	16	●	All	MS33682-6



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	—

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-7	A	3	16	●	All	MS33682-7

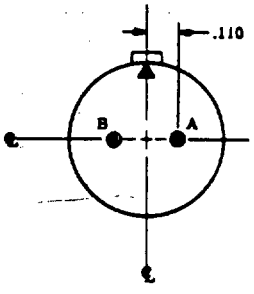
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THIS ARRANGEMENT NUMBER NOT TO BE USED

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-8	---	---	---	---	---	MS33682-8

INACTIVE FOR DESIGN AFTER 13 MAY 1954

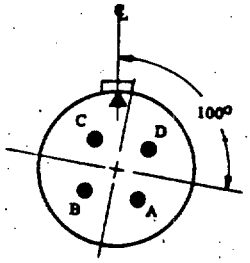


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-9	A	2	16		A11	MS33682-9

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN AFTER 17 APR 1945

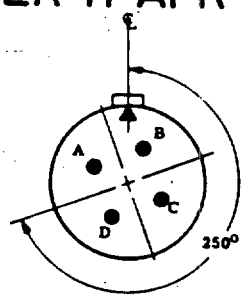


CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 14S-2 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-10	INST	4	16	●	All	MS33682-10

INACTIVE FOR DESIGN AFTER 17 APR 1945



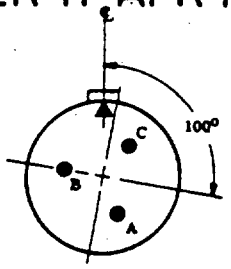
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 14S-2 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-11	INST	4	16	●	All	MS33682-11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN AFTER 17 APR 1945

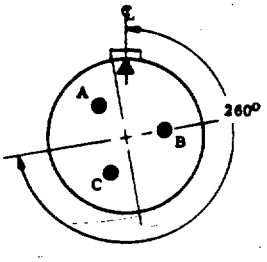


CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 14S-1 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-12	A	3	16	●	All	MS33682-12

INACTIVE FOR DESIGN AFTER 17 APR 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 14S-1 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-13	A	3	16	●	All	MS33682-13

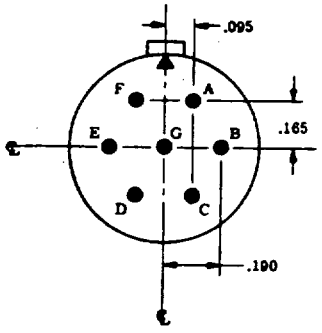
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Check the source to verify that this is the current version before use.

SECTION 50

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 16, 16S, AND 17

NOTES:
1. Shell size 16 applies to MIL-C-5015, MIL-C-22992, and MIL-C-83723, series II.
2. Shell size 16S applies only to MIL-C-5015.
3. Shell size 17 is identical to 16S, and applies only to MIL-C-83723, series II.

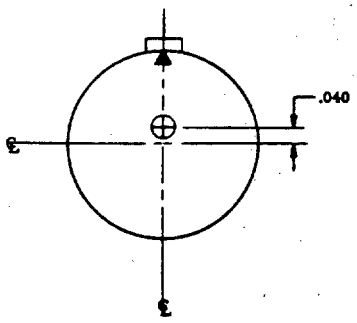


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	—	—	280

* #

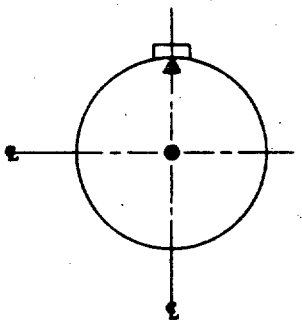
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-1	A	7	16	●	All	MS33683-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

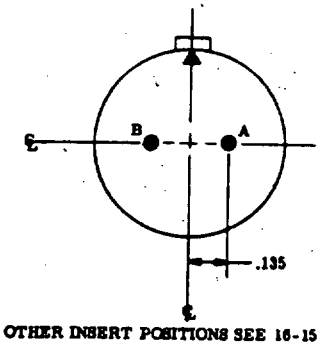
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-2	E	1	12	⊕	---	MS33683-2



ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-3	B	1	16	●	---	MS33683-3

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Check the source to verify that this is the current version before use.

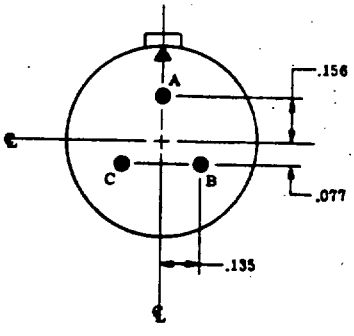


FOR OTHER INSERT POSITIONS SEE 16-15 AND 16-16

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-4	D	2	16		All	MS33683-4

INACTIVE FOR DESIGN
AFTER 13 MAY 1954

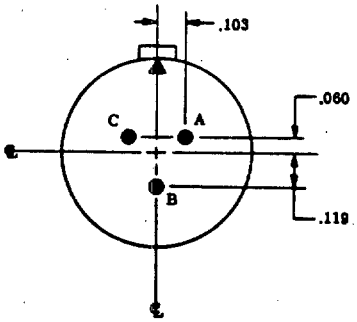


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-5	A	3	16		All	MS33683-5

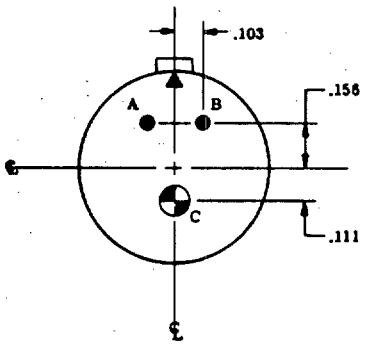
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 30 JUL 1951



ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
90	180	270	—

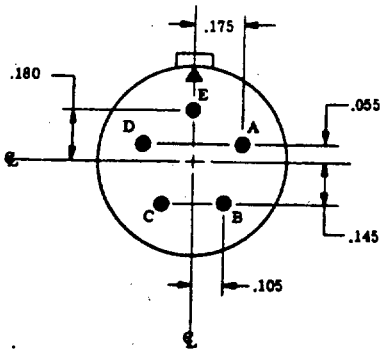
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-6	A	3	16		All	MS33683-6



ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
80	110	250	280

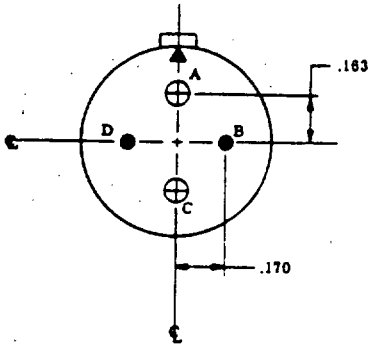
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-7	A	2	16		All	MS33683-7
			1	8			

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Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	170	265	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-8	A	5	16	●	All	MS33683-8

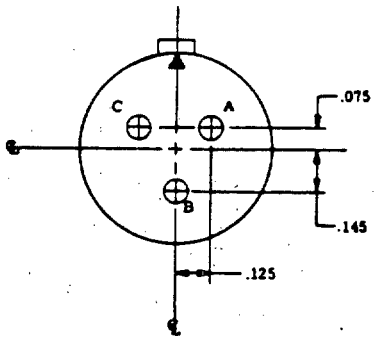


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-9	A	2	16	●	All	MS33683-9
			2	12	⊕		

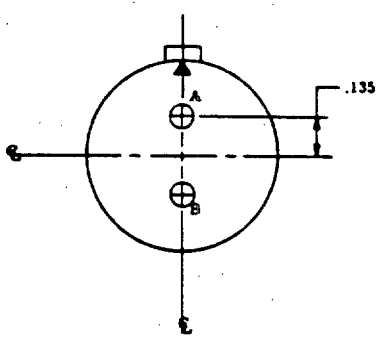
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
90	180	270	—

* # †

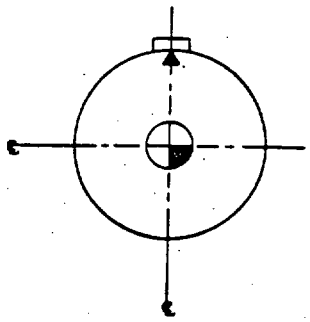
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-10	A	3	12	⊕	A11	MS33683-10



ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
35	110	250	325

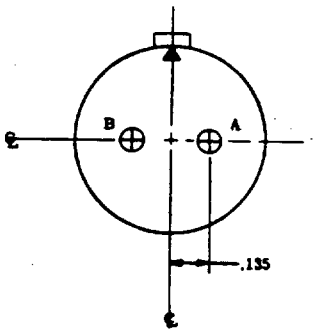
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-11	A	2	12	⊕	A11	MS33683-11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-12	A	1	4	⊕	---	MS33683-12



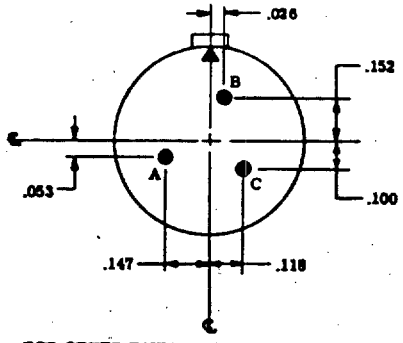
CONTACT A - IRON
CONTACT B - CONSTANTAN

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35.	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-13	A	2	12	⊕	A11	MS33683-13

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 27APR 1945

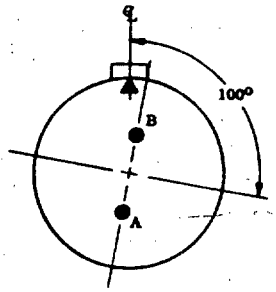


FOR OTHER INSERT POSITION SEE 16S-17

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-14	D	3	16	●	All	MS33683-14

INACTIVE FOR DESIGN AFTER 27 APR 1945



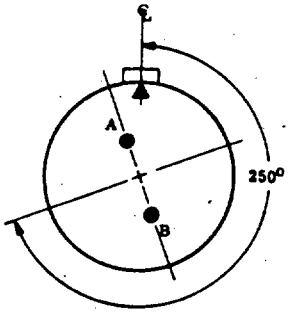
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 16S-4 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-15	D	2	16	●	All	MS33683-15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 27 APR 1945

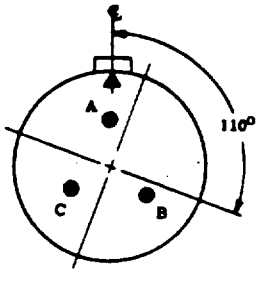


CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 16S-4 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-16	D	2	16	●	All	MS33683-16

INACTIVE FOR DESIGN AFTER 27 APR 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 16S-14 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-17	D	3	16	●	All	MS33683-17

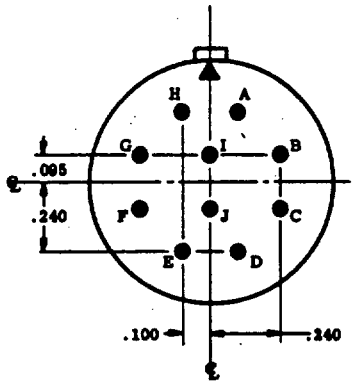
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

SECTION 60

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 18

INACTIVE FOR DESIGN OF AIR FORCE AIRBORNE EQUIPMENTS AFTER 28 MAY 1971.



FOR OTHER INSERT POSITIONS SEE 18-23 AND 18-24

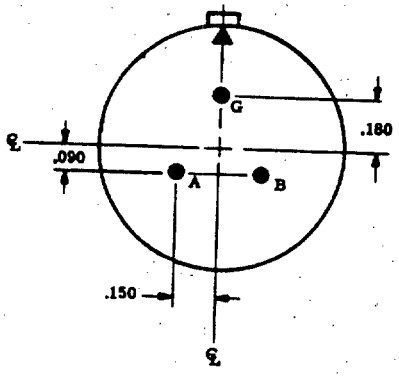
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
70	145	215	290

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-1	A	4	16		B,C,F,G	MS33684-1
		INST	6			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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CANCELED AFTER
19 MAY 1945



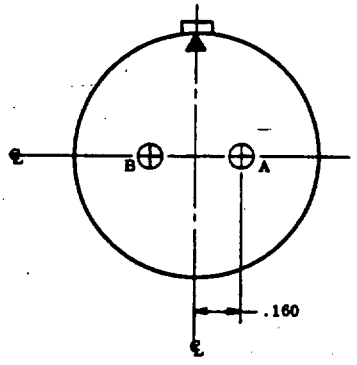
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-2	24V	3	16		All	MS33684-2

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INACTIVE FOR DESIGN
AFTER 13 MAY 1954



FOR OTHER INSERT POSITIONS SEE 18-25 AND 18-26

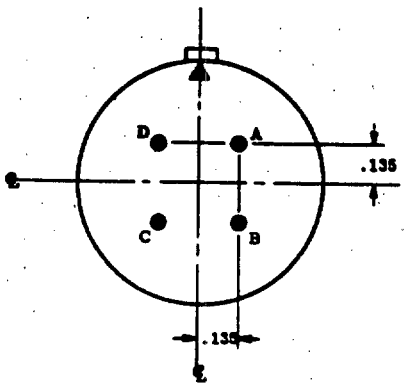
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-3	D	2	12	⊕	A11	MS33684-3

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INACTIVE FOR DESIGN OF AIR FORCE AIRBORNE EQUIPMENTS AFTER 28 MAY 1971.

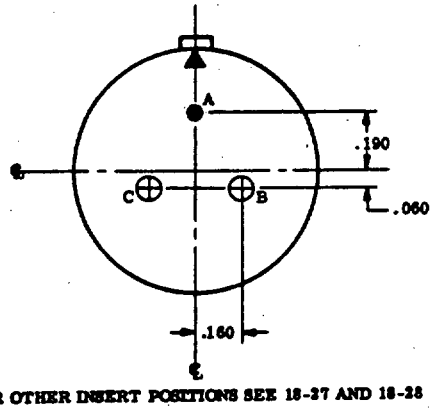


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-4	D	4	16	●	All	MS33684-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN OF AIR FORCE AIRBORNE EQUIPMENTS AFTER 28 MAY 1971.

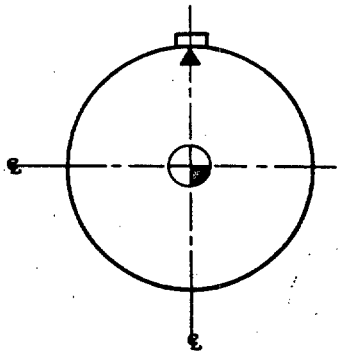


FOR OTHER INSERT POSITIONS SEE 18-27 AND 18-28

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

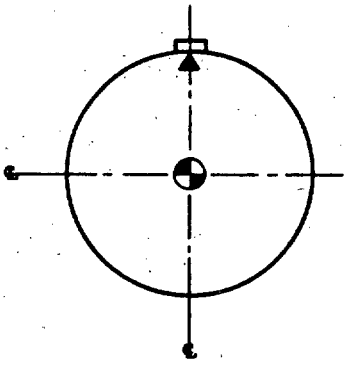
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-5	D	1	16	●	All	MS33684-5
			2	12	⊕		

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Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-6	D	1	4		---	MS33684-6

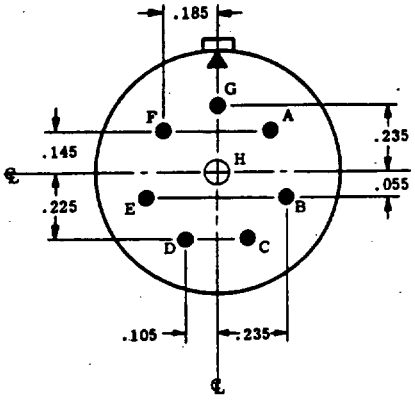


ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-7	B	1	8		---	MS33684-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

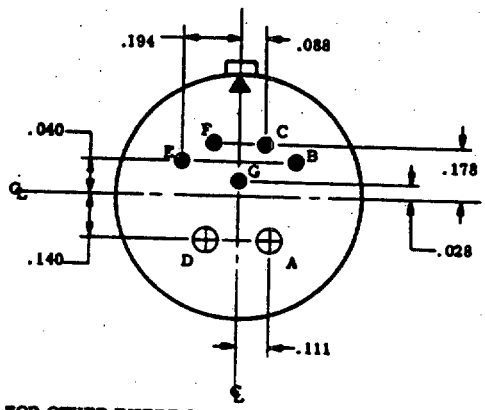
INACTIVE FOR DESIGN OF AIR FORCE AIRBORNE EQUIPMENTS AFTER 28 MAY 1971.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	—	—	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-8	A	7	16	●	A11	MS33684-8
			1	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



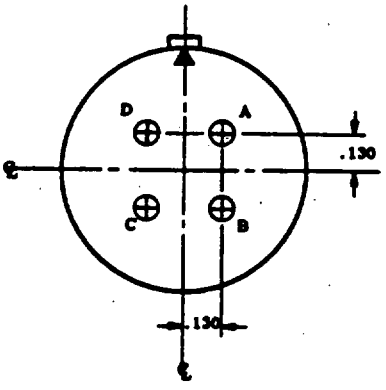
FOR OTHER INSERT POSITIONS SEE 18-17 AND 18-18

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-9	INST	5	16	●	All	MS33684-9
			2	12	⊕		

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INACTIVE FOR DESIGN AFTER 13 MAY 1954



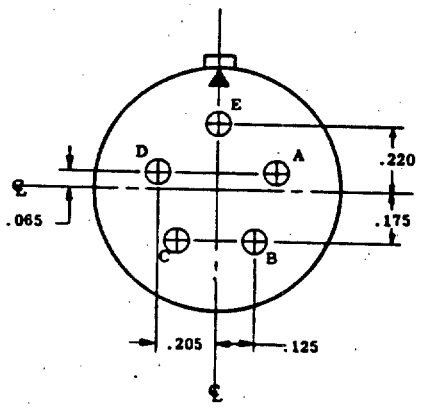
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	120	240	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-10	A	4	12	⊕	All	MS33684-10

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INACTIVE FOR DESIGN OF AIR FORCE AIRBORNE EQUIPMENTS AFTER 28 MAY 1971.



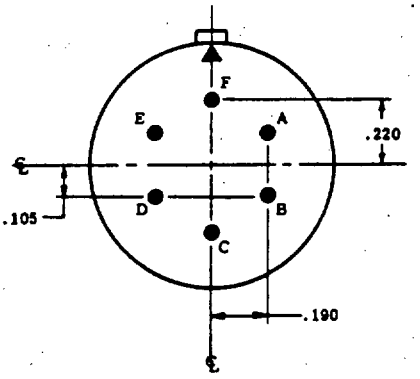
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	170	265	—

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-11	A	5	12	⊕	All	MS33684-11

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INACTIVE FOR DESIGN OF AIR FORCE AIRBORNE EQUIPMENTS AFTER 13 MAY 1954.



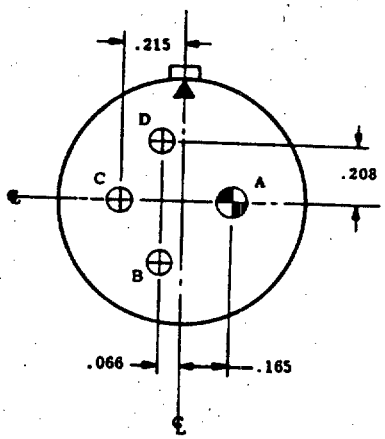
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	—	—	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-12	A	6	16	●	A11	MS33684-12

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INACTIVE FOR DESIGN OF AIR FORCE EQUIPMENTS AFTER 28 MAY 1971.

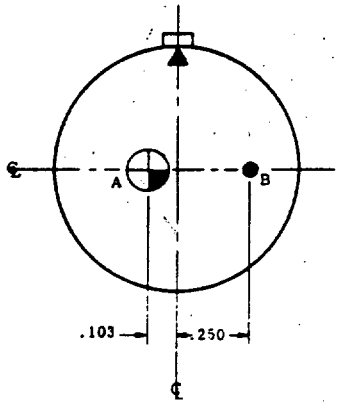


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-13	A	3	12	⊕	All	MS33684-13
			1	8	◔		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN OF AIR [illegible]
EQUIPMENTS AFTER 28 MAY 1971.

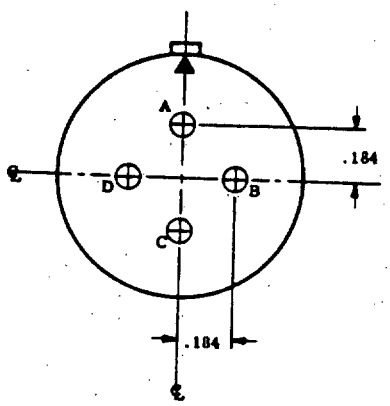


ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-14	A	1	16		All	MS33684-14
			1	4			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN OF AIR FORCE
AIRBORNE EQUIPMENTS AFTER 28 MAY 1971.



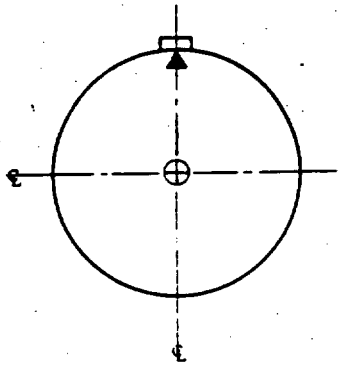
CONTACTS A AND C-IRON
CONTACTS B AND D-CONSTANTAN

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	120	240	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-15	A	4	12	⊕	All	MS33684-15

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Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN OF AIR FORCE
AIRBORNE EQUIPMENTS AFTER 28 MAY 1971.



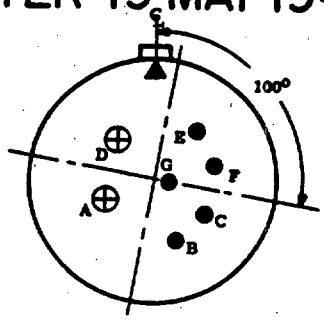
FOR INSERT DESIGN
SEE MS33694

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell siz.	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-16	C	1	12	⊕	---	MS33684-16

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Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-9 EXCEPT FOR POSITION OF INSERT IN SHELL.

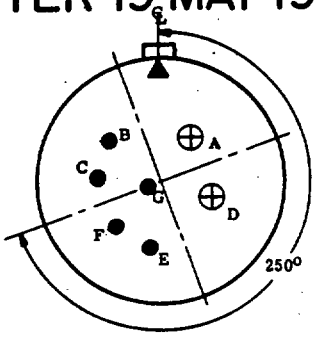
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-17	INST	5	16	●	All	MS33684-17
			2	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN
AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-9 EXCEPT FOR POSITION OF INSERT IN SHELL.

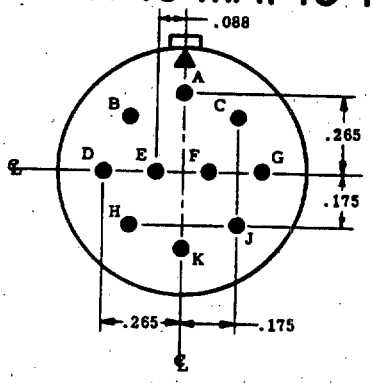
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-18	INST	5	16	●	All	MS33684-18
			2	12	⊕		

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INACTIVE FOR DESIGN
AFTER 19 MAY 1945



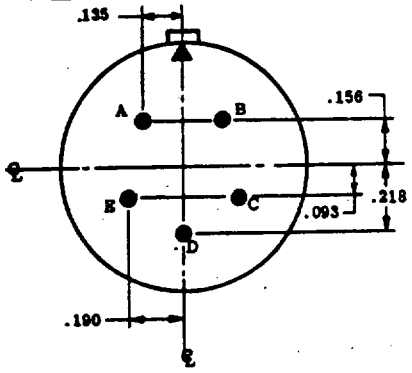
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	120	240	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-19	A	10	16	●	All	MS33684-19

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INACTIVE FOR DESIGN
AFTER 13 MAY 1954



FOR OTHER INSERT POSITIONS SEE 18-30 AND 18-31

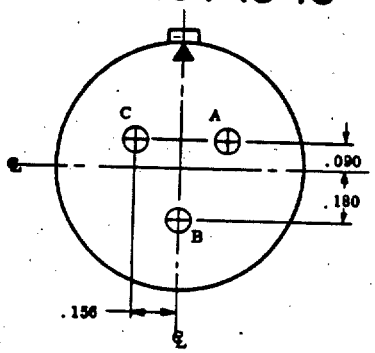
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-20	A	5	16	●	All	MS33684-20

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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CANCELED AFTER
25 NOV 1949



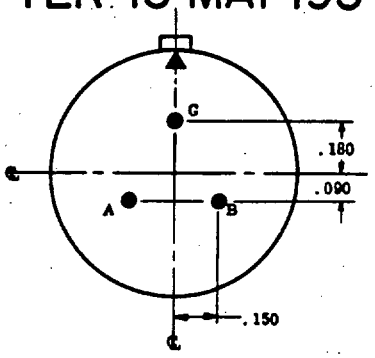
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-21	A	3	12	⊕	All	MS33684-21

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INACTIVE FOR DESIGN AFTER 13 MAY 1954



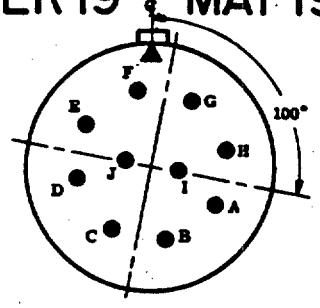
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-22	D	3	16	●	All	MS33684-22

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INACTIVE FOR DESIGN
AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-1 EXCEPT FOR POSITION OF INSERT IN SHELL.

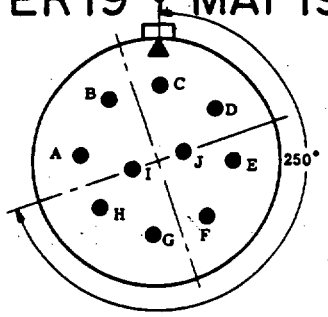
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-23	A	4	16		B,C,F,G	MS33684-23
		INST	6			All others	

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INACTIVE FOR DESIGN
AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-1 EXCEPT FOR POSITION OF INSERT IN SHELL.

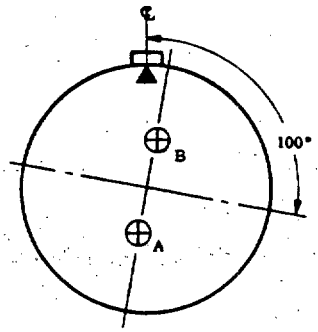
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-24	A	4	16	●	B,C,F,G	MS33684-24
		INST	6			All others	

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INACTIVE FOR DESIGN AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-3 EXCEPT FOR POSITION OF INSERT IN SHELL.

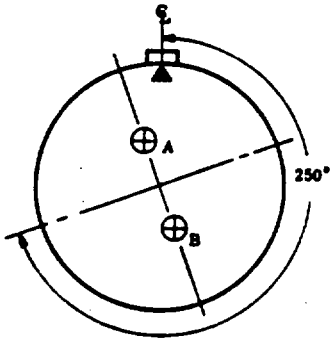
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-25	D	2	12	⊕	All	MS33684-25

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INACTIVE FOR DESIGN AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-3 EXCEPT FOR POSITION OF INSERT IN SHELL.

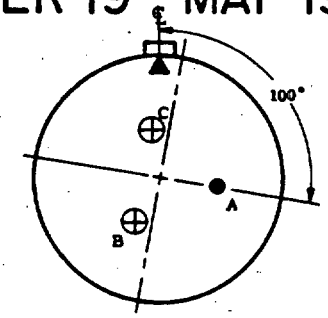
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-26	D	2	12	⊕	A11	MS33684-26

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INACTIVE FOR DESIGN AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-5 EXCEPT FOR POSITION OF INSERT IN SHELL.

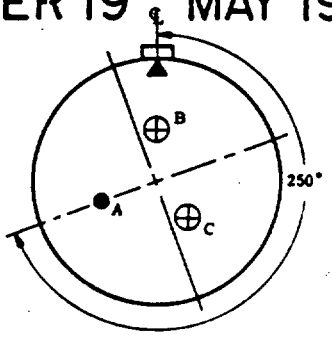
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-27	D	1	16	●	A11	MS33684-27
			2	12	⊕		

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INACTIVE FOR DESIGN AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-5 EXCEPT FOR POSITION OF INSERT IN SHELL.

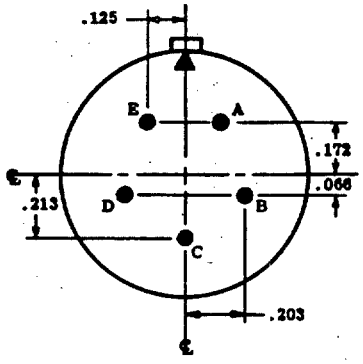
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(N c applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-28	D	1	16	●	All	MS33684-28
			2	12	⊕		

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INACTIVE FOR DESIGN AFTER 13 MAY 1954



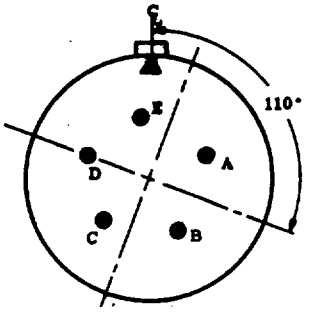
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-29	A	5	16	●	A11	MS33684-29

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INACTIVE FOR DESIGN
AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-20 EXCEPT FOR POSITION OF INSERT IN SHELL.

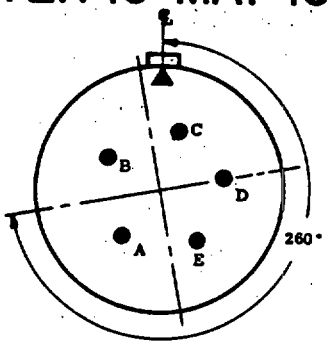
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-30	A	5	16	●	A11	MS33684-30

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INACTIVE FOR DESIGN
AFTER 19 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-20 EXCEPT FOR POSITION OF INSERT IN SHELL.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-31	A	5	16	●	All	MS33684-31

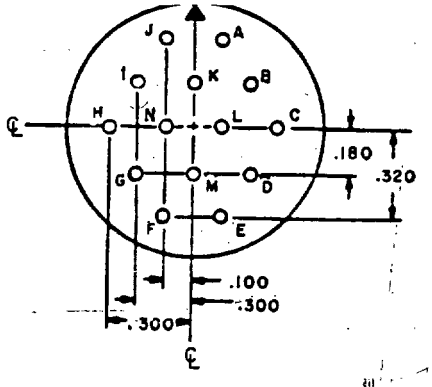
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Check the source to verify that this is the current version before use.

SECTION 70

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 20

CANCELED AFTER
1 JUN 1945

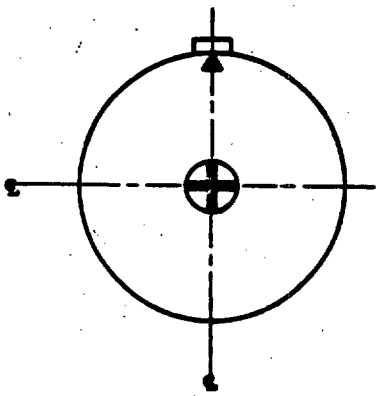


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-1	INST	14	20 (.0625D)	◯	All	MS33685-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

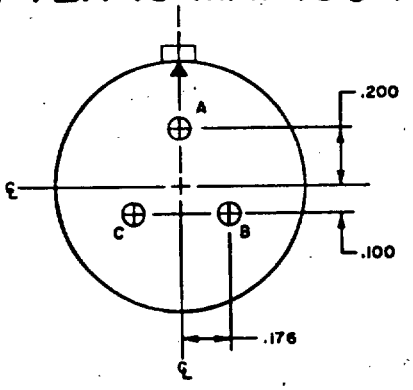


ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-2	D	1	0	⊕	---	MS33685-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN
AFTER 13 MAY 1954

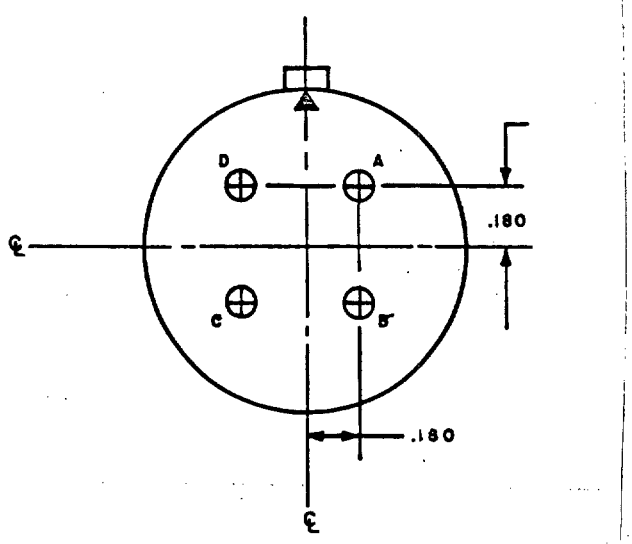


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-3	D	3	12	⊕	All	MS3368S-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

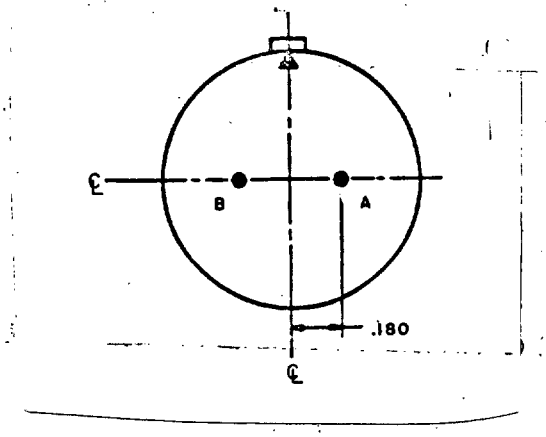


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
45	110	250	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-4	D	4	12	⊕	A11	MS33685-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN
AFTER 17 DEC 1958



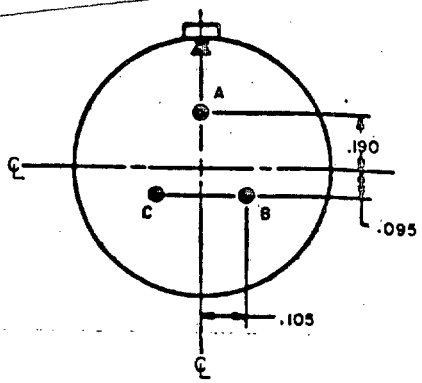
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-5	E	2	16	●	A11	MS33685-5

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INACTIVE FOR DESIGN
AFTER 13 MAY 1994

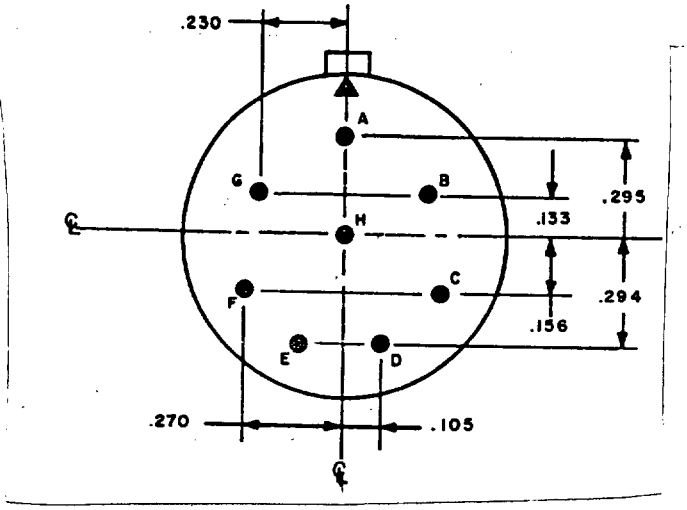


ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-6	D	3	16	●	A11	MS33685-6

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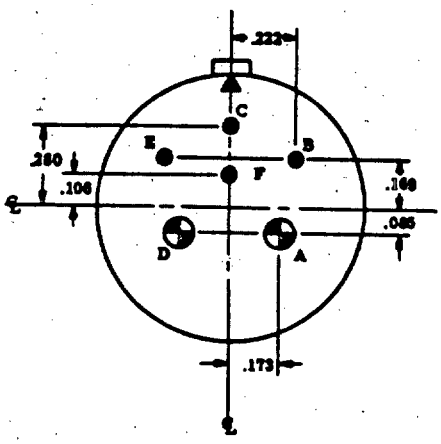


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITION SEE 20-33

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-7	D	4	16	●	A,B,G,H	MS333685-7
		A	4			All others	

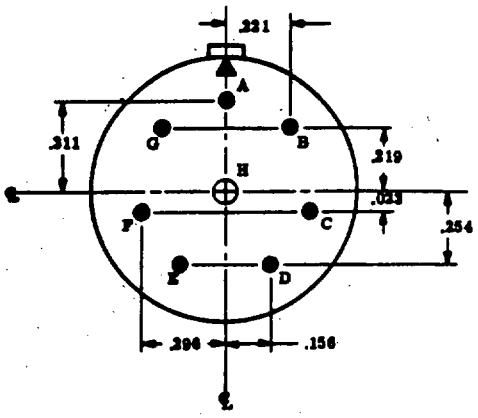
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-8	INST	4	16	●	A11	MS33685-8
			2	8	◒		

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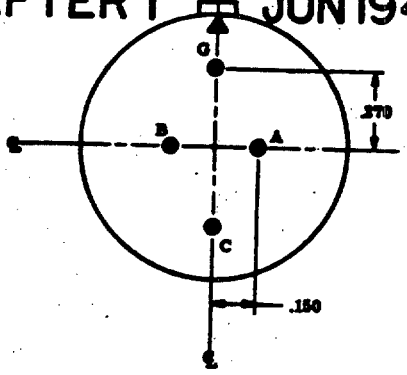


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-9	D	1	12	⊕	H	MS33685-9
		A	7	16	●	All others	

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INACTIVE FOR DESIGN
AFTER 1 JUN 1945



SOCKET CONTACTS A AND B SHALL GROUND TO CONTACT G WHEN PIN CONTACTS ARE REMOVED

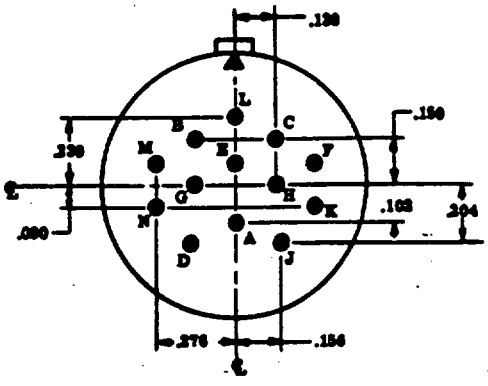
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
.0	-10	A	4	16		All	MS33685-10

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Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN
AFTER 1 JUN 1945



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

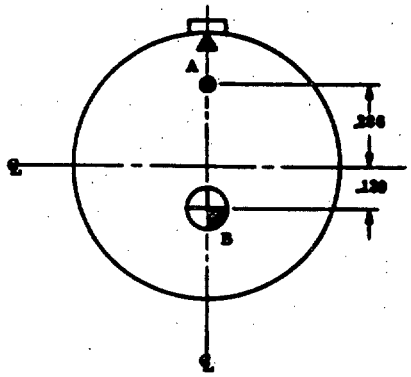
FOR OTHER INSERT POSITIONS SEE 20-25 AND 20-30

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-11	INST	13	16	●	All	MS33685-11

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INACTIVE FOR DESIGN
AFTER 13 MAY 1954



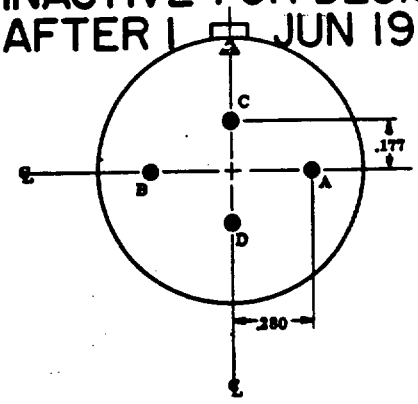
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-12	A	1	16	●	A11	MS3368S-12
			1	4	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN
AFTER 1 JUN 1945



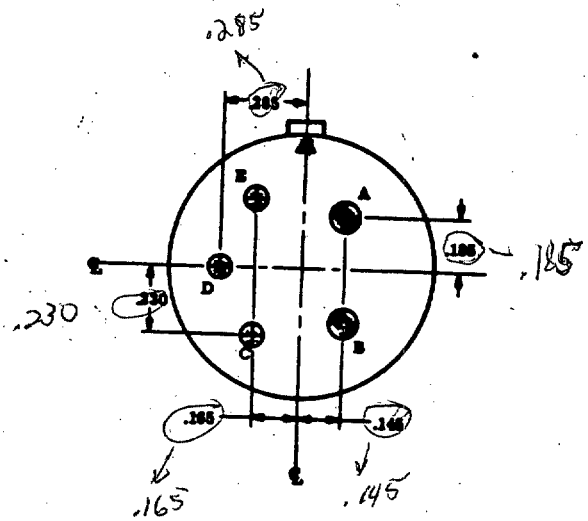
SOCKETS C AND D TO MAKE METALLIC CONTACT TO EACH OTHER WHEN PINS ARE REMOVED

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-13	A	4	16	●	A11	MS33685-13

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

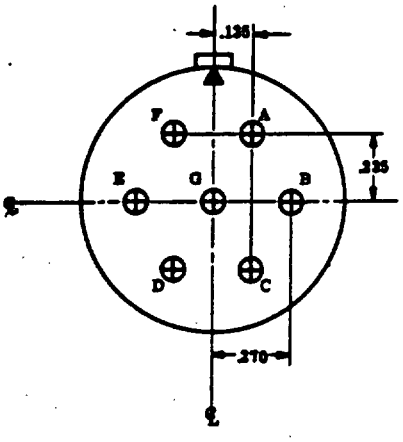


ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
80	110	250	280

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-14	A	3	12	⊕	All	MS33685-14
			2	8	◔		

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Check the source to verify that this is the current version before use.

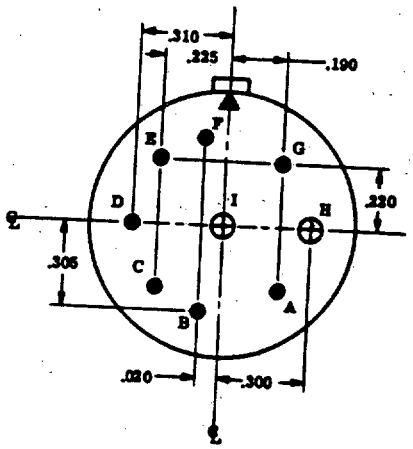


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	—	—	280

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-15	A	7	12	⊕	A11	MS3368S-15

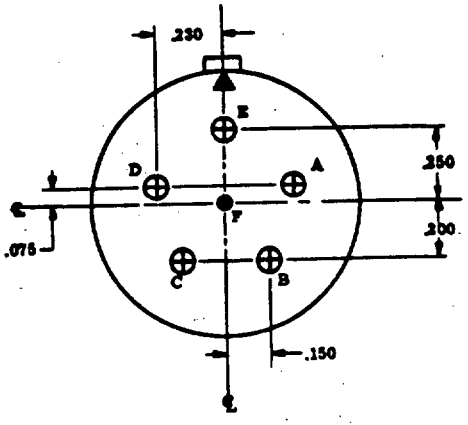
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-16	A	7	16	●	All	MS33685-16
			2	12	⊕		

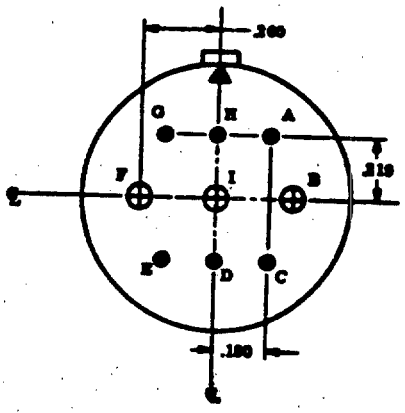
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-17	A	1	16	●	A11	MS33685-17
			5	12	⊕		

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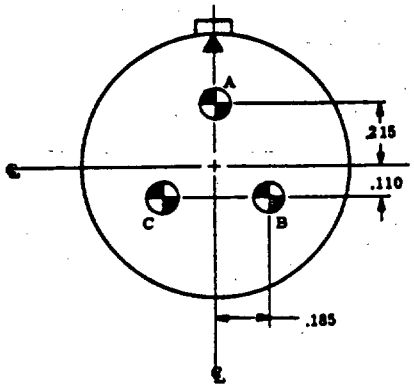
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
35	110	250	325

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-18	A	6	16	●	All	MS33685-18
			3	12	⊕		

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INACTIVE FOR DESIGN
AFTER 13 MAY 1954



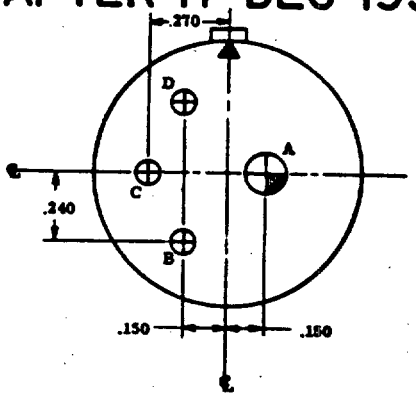
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	—

(No applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-19	A	3	8		A11	MS33685-19

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INACTIVE FOR DESIGN
AFTER 17 DEC 1958

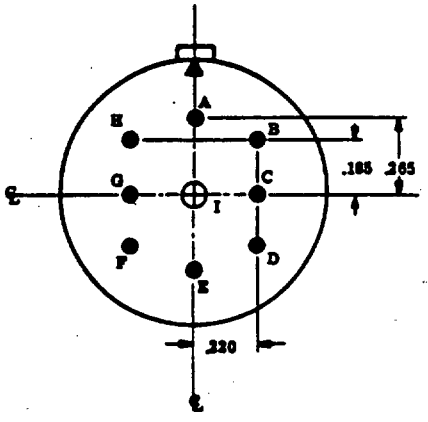


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-20	A	3	12	⊕	All	MS33685-20
			1	4	◒		

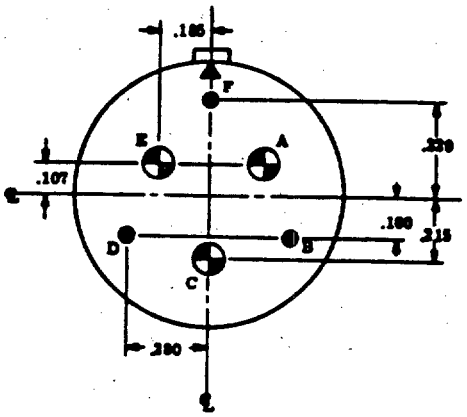
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-21	A	8	16	●	All	MS33685-21
			1	12	⊕		

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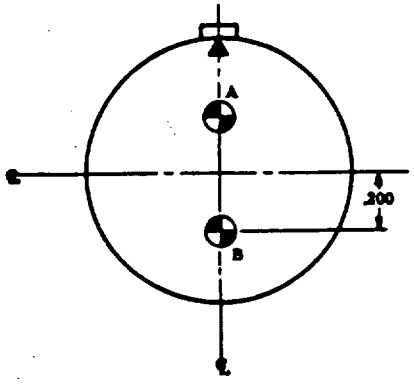
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
80	110	250	280

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-22	A	3	16		All	MS33685-22
			3	8			

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INACTIVE FOR DESIGN
AFTER 13 MAY 1954

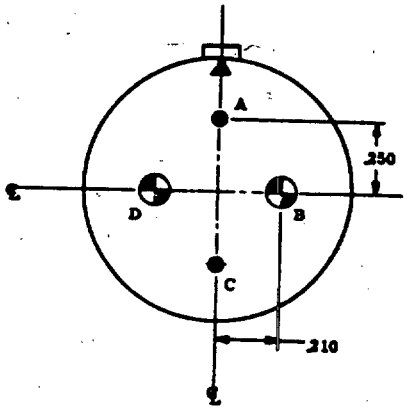


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-23	A	2	8		All	MS33685-23

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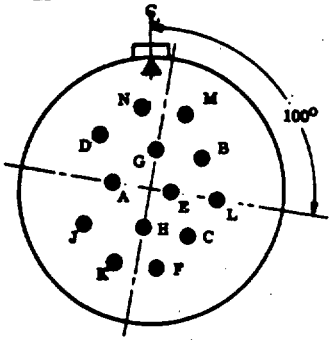


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-24	A	2	16		A11	MS3368S-24
			2	8			

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INACTIVE FOR DESIGN
AFTER 1 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 20-11 EXCEPT FOR POSITION OF INSERT IN SHELL

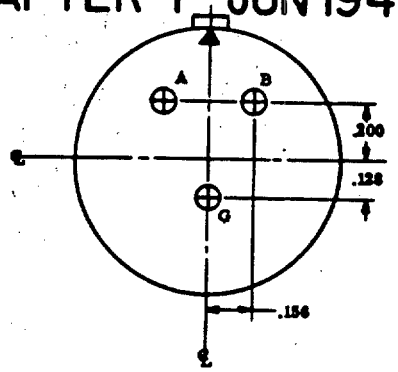
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-25	INST	13	16	●	All	MS33685-25

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INACTIVE FOR DESIGN
AFTER 1 JUN 1945



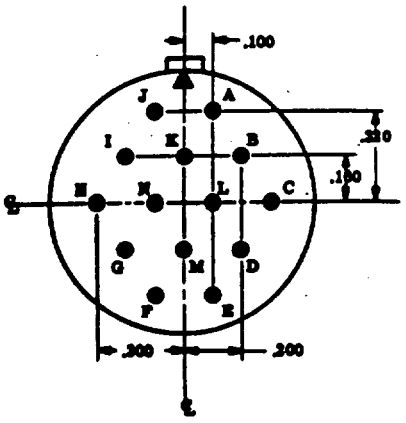
SOCKET CONTACTS A AND B GROUND
TO CONTACT G WHEN PINS ARE REMOVED

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-26	A	3	12	⊕	A11	MS33685-26

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

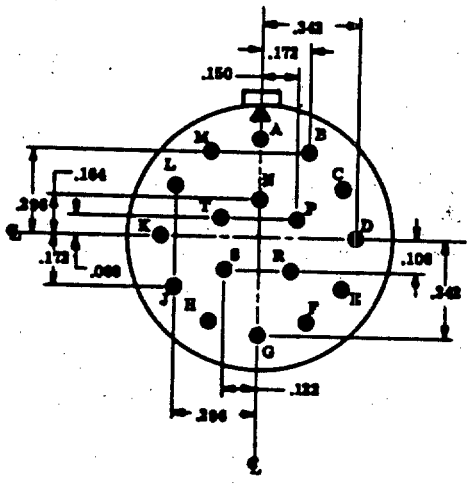
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-27	A	14	16	●	A11	MS33685-27

CANCELLED
AFTER
1 JUN 1945

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-28	---	---	---	---	---	MS33685-28

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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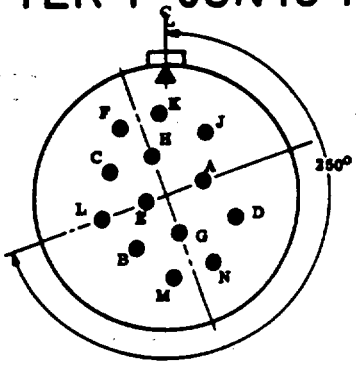
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	—	—	280

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-29	A	17	16	●	A11	MS33685-29

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INACTIVE FOR DESIGN
AFTER 1 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 20-11 EXCEPT FOR POSITION OF INSERT IN SHELL

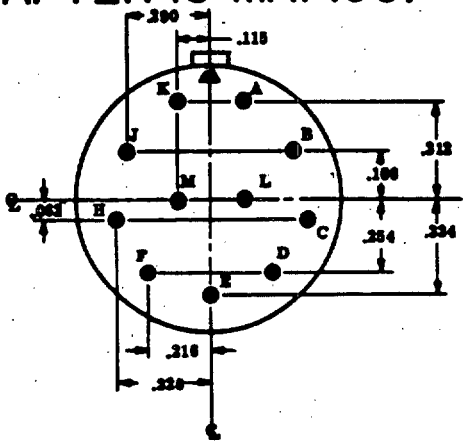
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-30	INST	13	16	●	A11	MS33685-30

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN
AFTER 18 MAY 1951



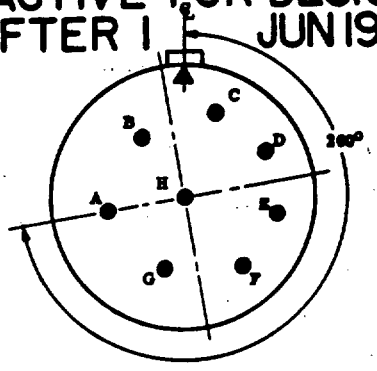
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-31	A	11	16		A11	MS33685-31

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 1 JUN 1945



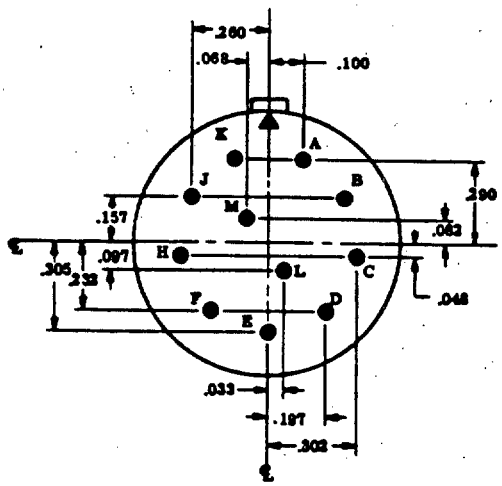
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 20-7 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-32	D	4	16	●	A,B,G,H	MS33685-32
		A	4			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-33	A	11	16		All	MS33685-33

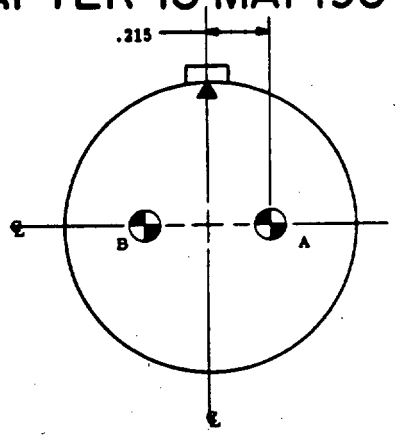
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

SECTION 80

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 22

INACTIVE FOR DESIGN
AFTER 13 MAY 1954

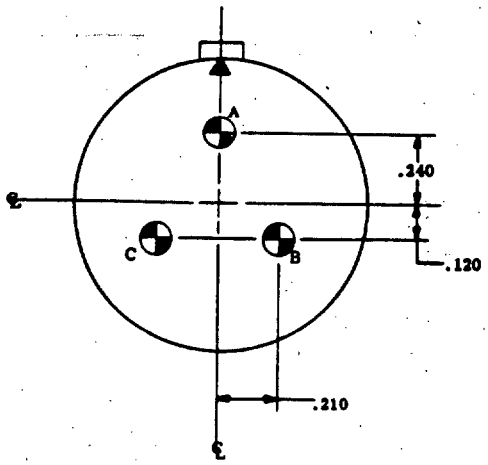


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-1	D	2	8		A11	MS33686-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

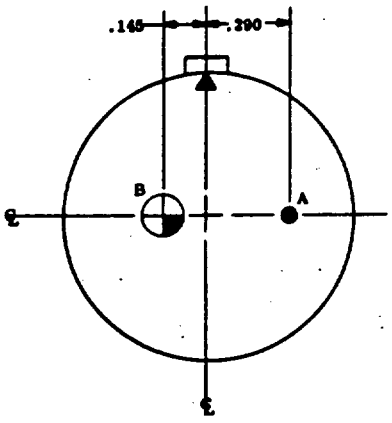


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-2	D	3	8		All	MS33686-2

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Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 17 DEC 1958



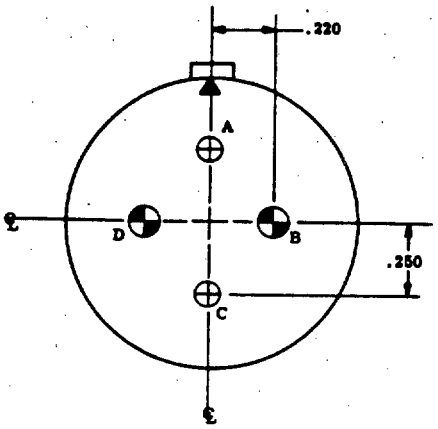
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-3	D	1	16	●	A11	MS33686-3
			1	4	⊕		

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INACTIVE FOR DESIGN AFTER 13 MAY 1954

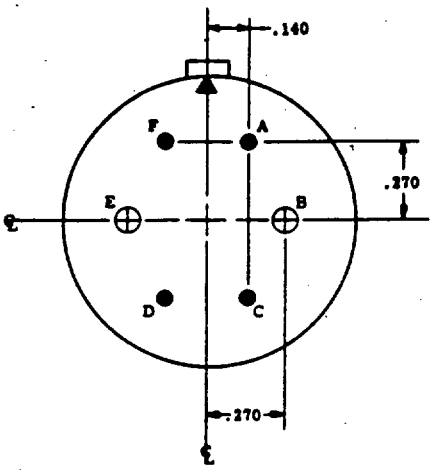


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-4	A	2	12	⊕	All	MS33686-4
			2	8	◔		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



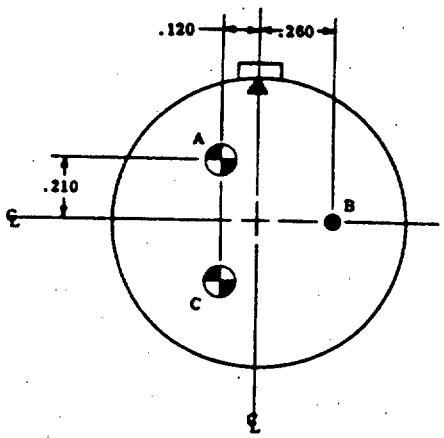
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

FOR OTHER INSERT POSITION SEE 22-32

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-5	D	4	16	●	A11	MS33686-5
			2	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN AFTER 13 MAY 1954

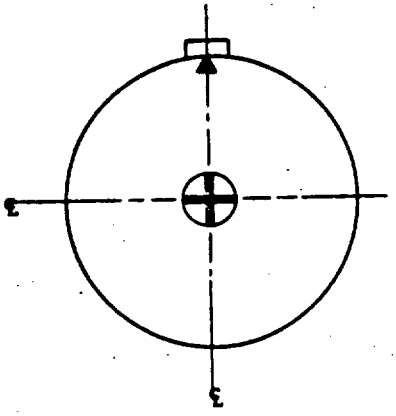


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-6	D	1	16		All	MS33686-6
			2	8			

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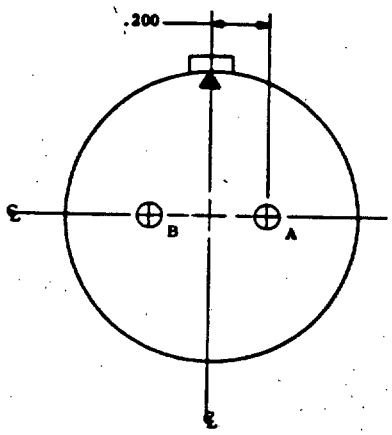


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-7	E	1	0	⊕	---	MS33686-7

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INACTIVE FOR DESIGN AFTER 13 MAY 1954

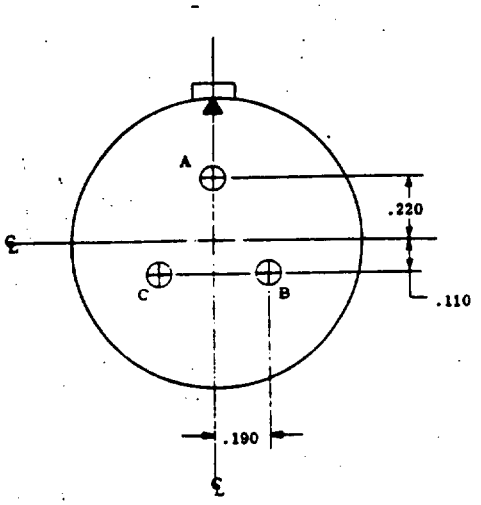


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-8	E	2	12	⊕	All	MS33686-8

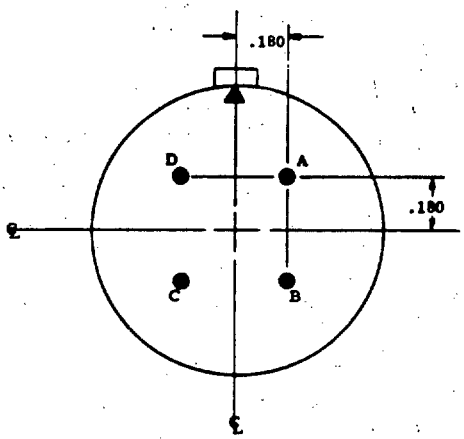
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-9	E	3	12	⊕	A11	MS33686-9

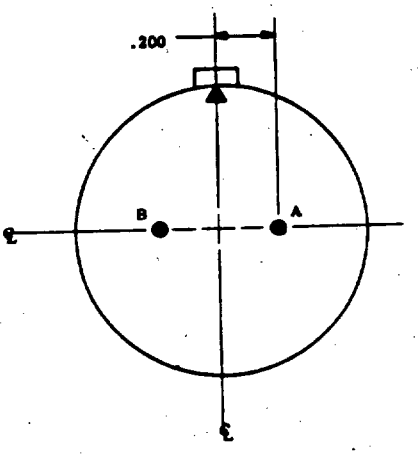
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-10	E	4	16	●	All	MS33686-10

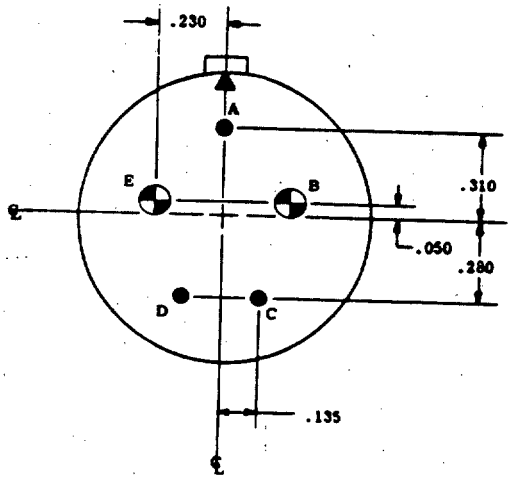
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-11	B	2	16	●	A11	MS33686-11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

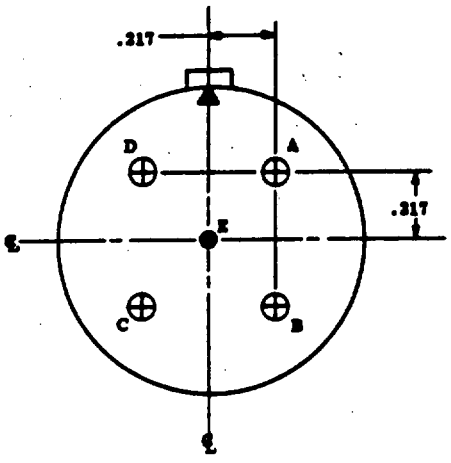


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-12	D	3	16		All	MS33686-12
			2	8			

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INACTIVE FOR DESIGN AFTER 13 MAY 1954



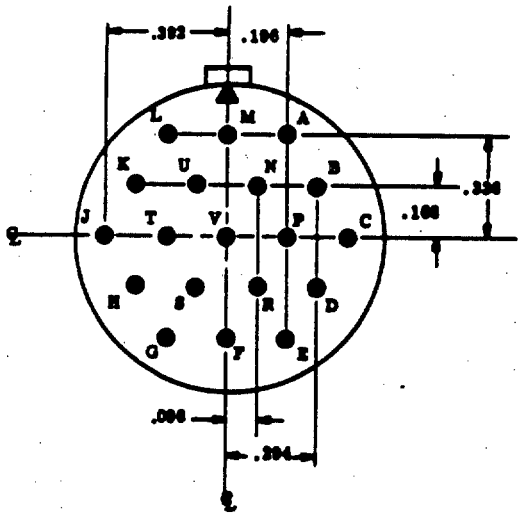
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-13	D	1	16	●	E	MS33686-13
		A	4	12	⊕	All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE POSITIONS X AND Y CANCELED AFTER 26 JUN 1968



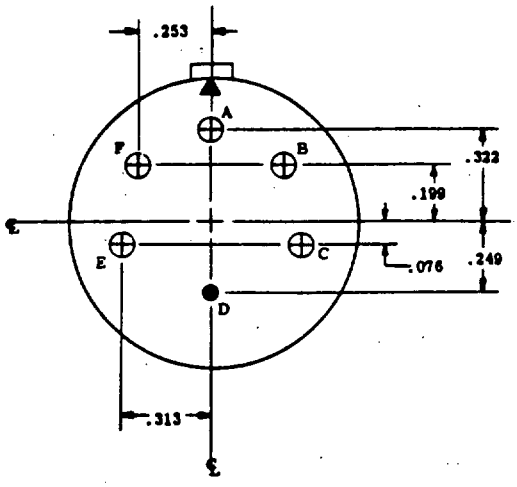
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITION SEE 22-30

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-14	A	19	16		A11	MS33686-14

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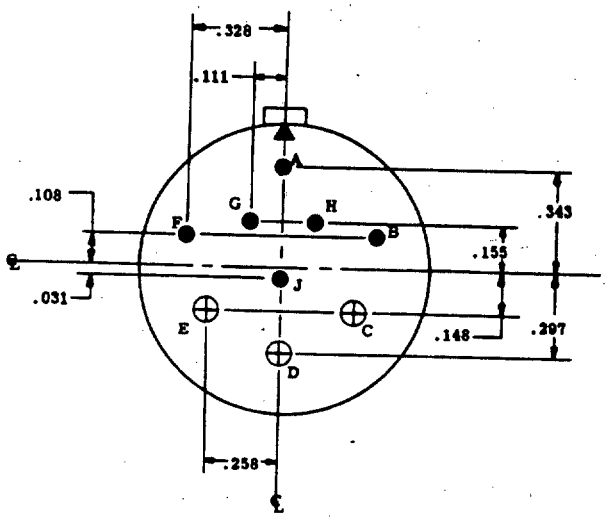


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-15	E	1	16	●	D	MS33686-15
		A	5	12	⊕	All others	

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INACTIVE FOR DESIGN AFTER 13 MAY 1954

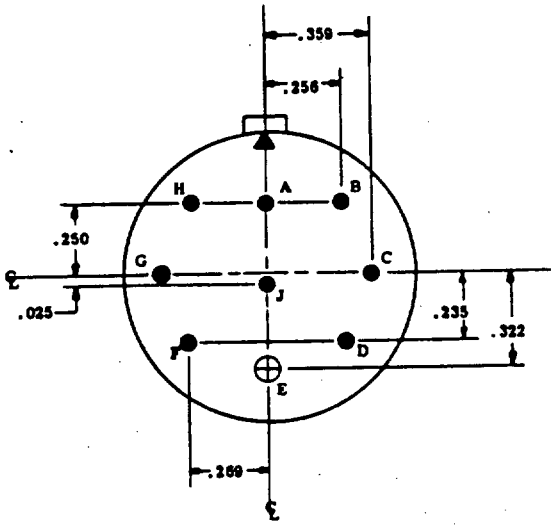


ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-16	A	6	16	●	All	MS33686-16
			3	12	⊕		

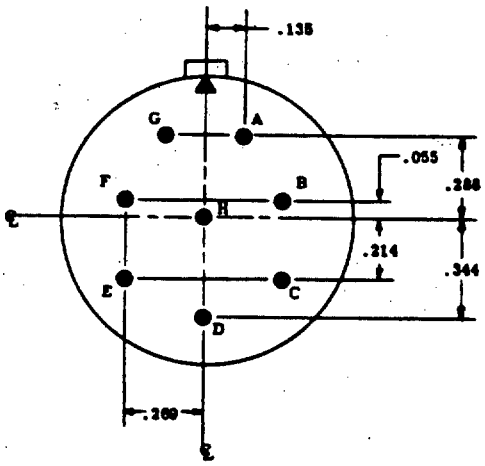
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-17	D	1	16	●	A	MS33686-17
		A	7			All others	
			1	12	⊕		

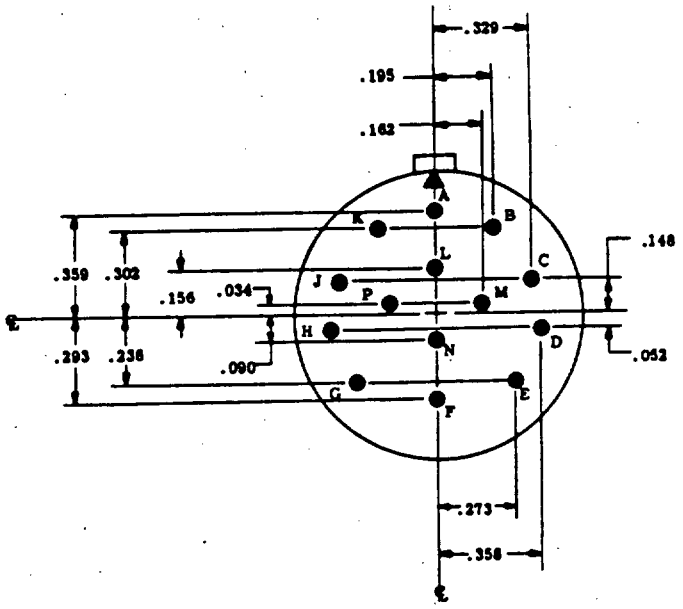
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-18	A	3	16		C,D,E	MS33686-18
		D	5			All others	

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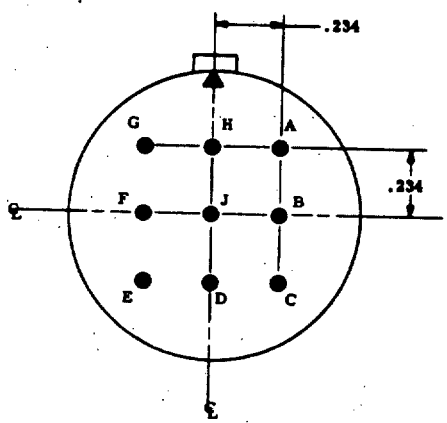


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-19	A	14	16	●	A11	MS33686-19

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INACTIVE FOR DESIGN AFTER 13 MAY 1954

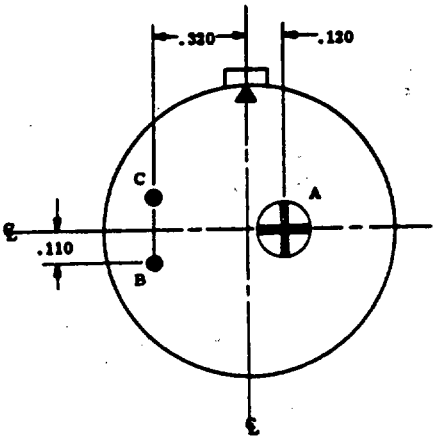


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-20	A	9	16	●	All	MS33686-20

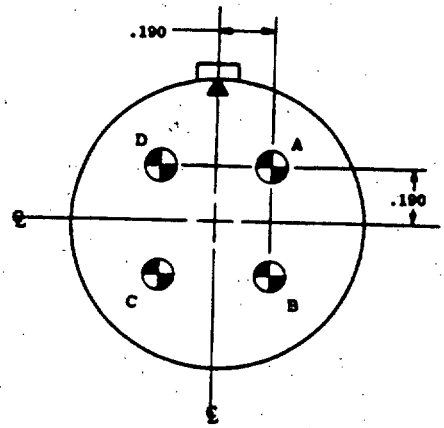
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-21	A	2	16	●	All	MS33686-21
			1	0	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

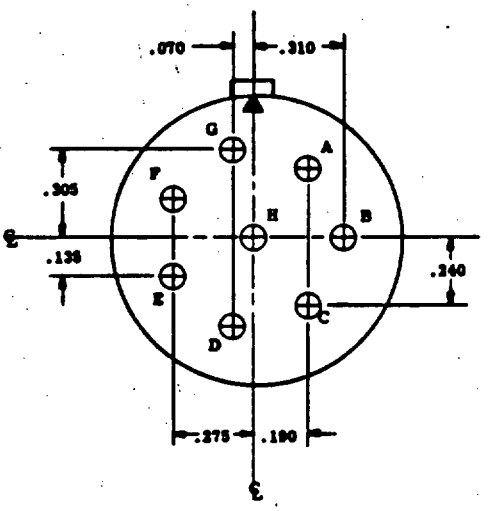


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
——	110	250	——

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-22	A	4	8		All	MS33686-22

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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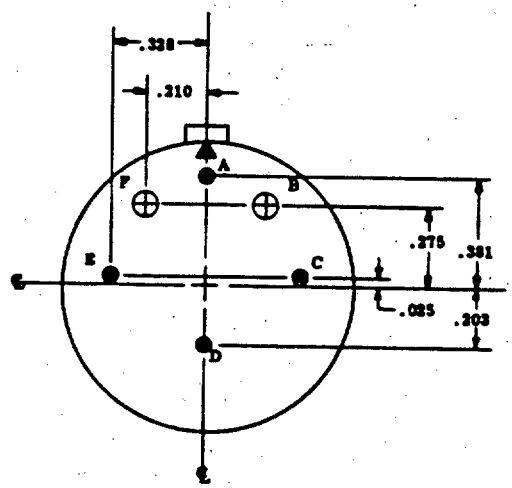
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	—	250	—

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-23	D	1	12	⊕	H	MS33686-23
		A	7			All others	

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INACTIVE FOR DESIGN AFTER 13 MAY 1954



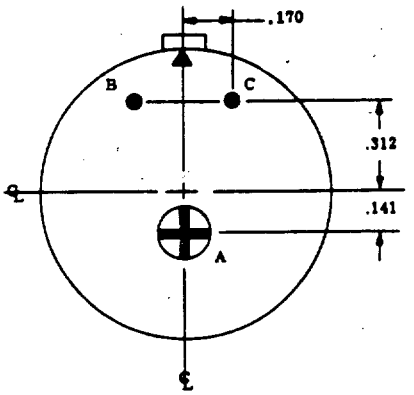
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-24	D	3	16	●	C,D,E	MS33686-24
		A	1			All others	
			2	12	⊕		

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INACTIVE FOR DESIGN
AFTER 31 JUL 1951



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

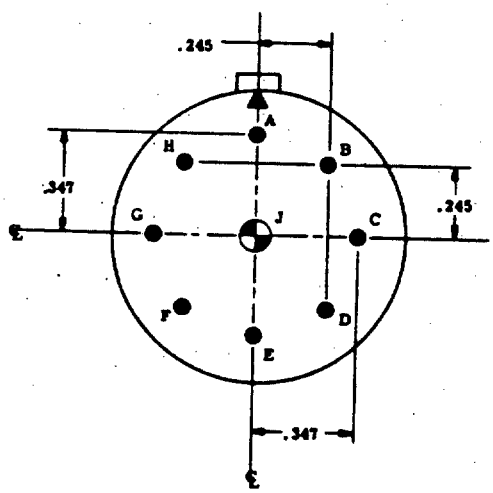
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-25	A	2	16	●	A11	MS33686-25
			1	0	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

THIS ARRANGEMENT NUMBER NOT TO BE USED

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-26	---	---	---	---	---	MS33686-26

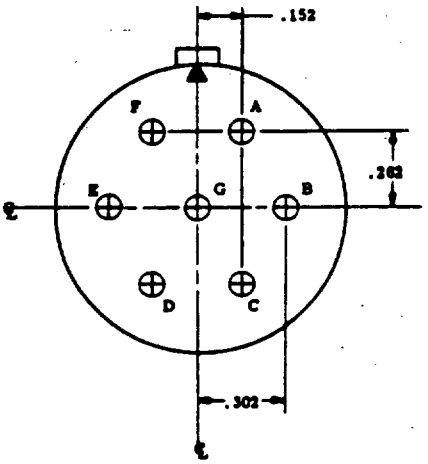


ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
80	——	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-27	D	1	8		J	MS33686-27
		A	8	16		All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 13 MAY 1954



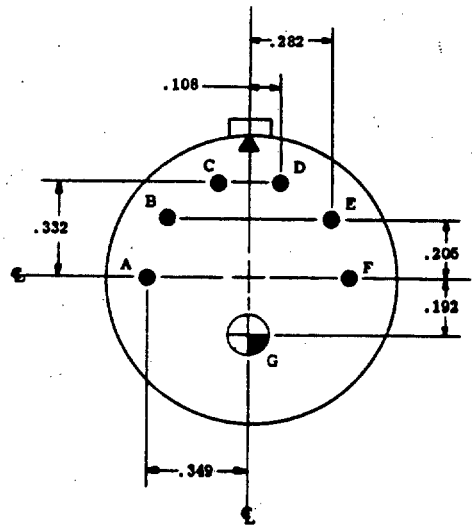
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	——	——	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-28	A	7	12	⊕	A11	MS33686-28

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INACTIVE FOR DESIGN AFTER 17 DEC 1958



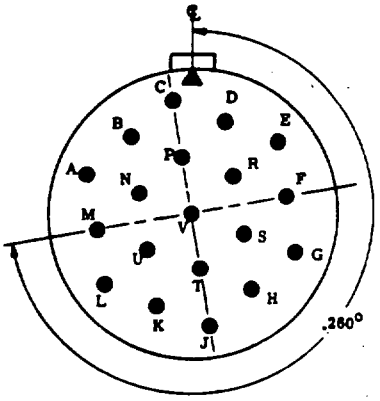
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-29	A	6	16	●	All	MS33686-29
			1	4	⊕		

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INACTIVE FOR DESIGN AFTER 26 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 22-14 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-30	A	19	16	●	All	MS33686-30

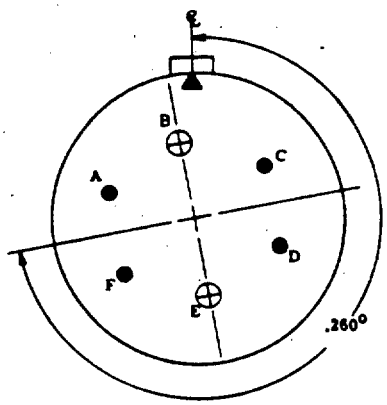
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THIS ARRANGEMENT NUMBER NOT TO BE USED

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-31	---	---	---	---	---	MS33686-31

INACTIVE FOR DESIGN AFTER 26 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS ARE IDENTICAL TO 22-5 EXCEPT FOR POSITION OF INSERT IN SHELL

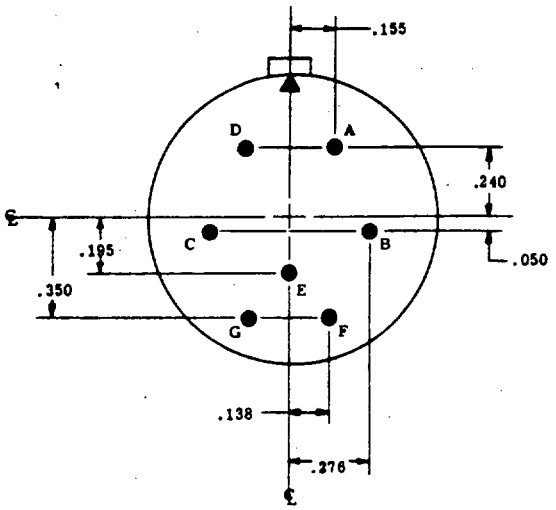
(Not applicable to MIL-C-22992)

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-32	D	4	16	●	All	MS33686-32
			2	12	⊕		

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INACTIVE FOR DESIGN AFTER 13 MAY 1954



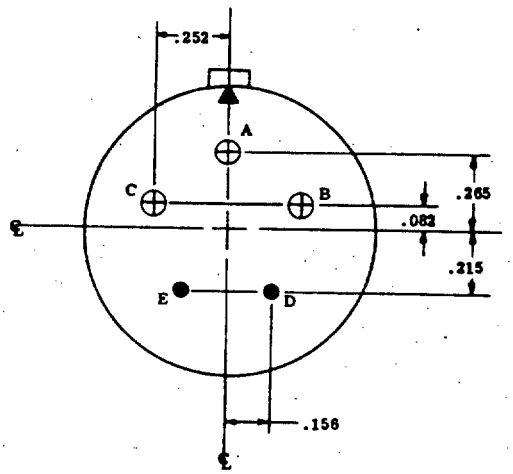
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-33	A	3	16		E,F,G	MS33686-33
		D	4			All others	

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INACTIVE FOR DESIGN AFTER 17 DEC 1958



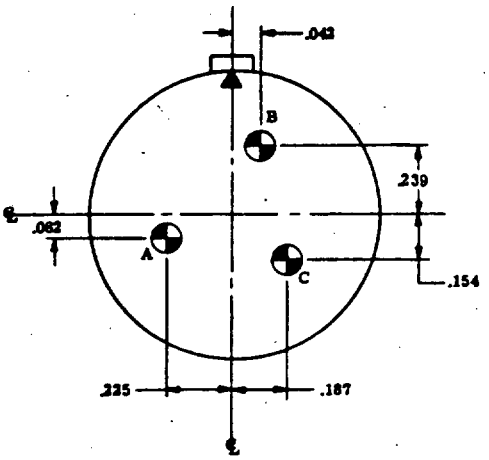
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-34	D	2	16	●	All	MS33686-34
			3	12	⊕		

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INACTIVE FOR DESIGN AFTER 26 JUN 1945

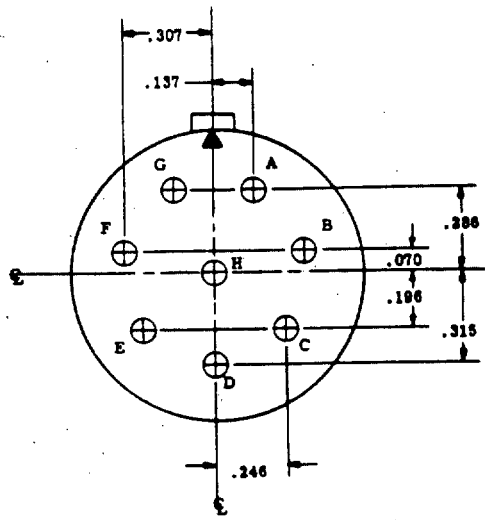


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-35	D	3	8		All	MS33686-35

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CONTACTS A, C, E AND G - IRON
CONTACTS B, D, F AND H - CONSTANTAN

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	——	270	——

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-36	D	1	12	⊕	H	MS33686-36
		A	7			All others	

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CANCELLED
AFTER
26 JUN 1945

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-37	---	---	---	---	---	MS33686-37

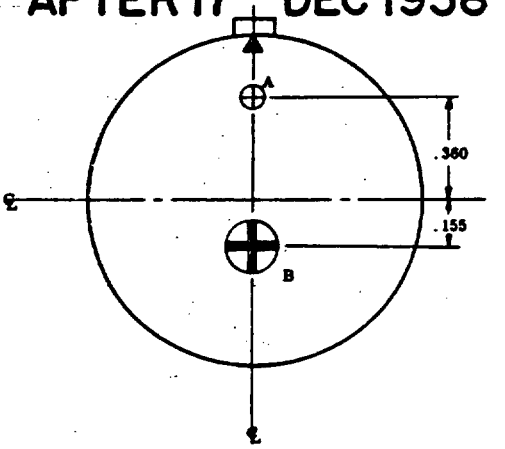
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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SECTION 90

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 24

INACTIVE FOR DESIGN
AFTER 17 DEC 1958

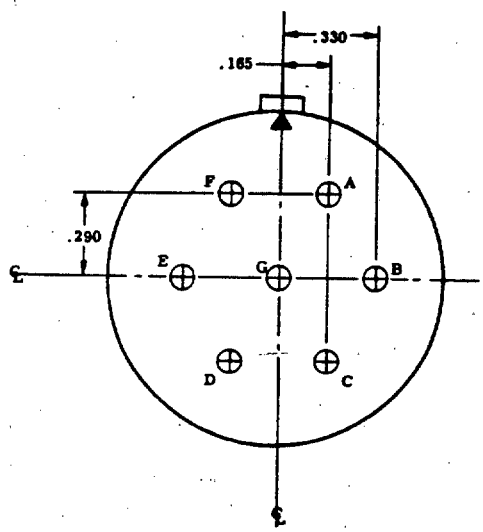


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-1	D	1	12	⊕	All	MS33687-1
			1	0	⊕		

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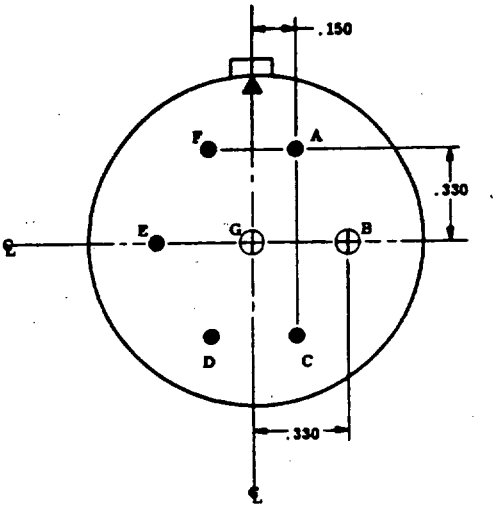


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	——	——	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-2	D	7	12	⊕	All	MS33687-2

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INACTIVE FOR DESIGN AFTER 13 MAY 1954

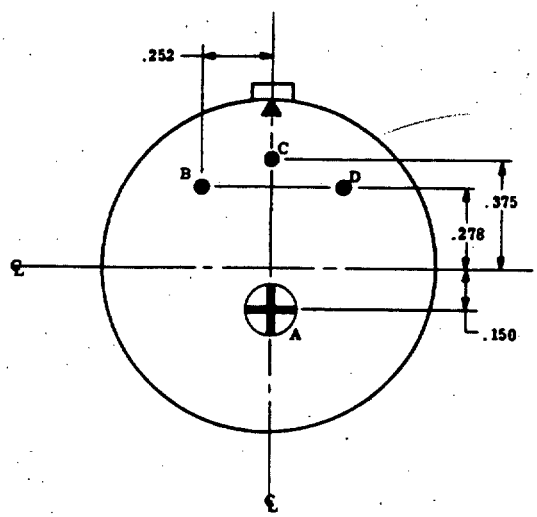


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-3	D	5	16	●	All	MS33687-3
			2	12	⊕		

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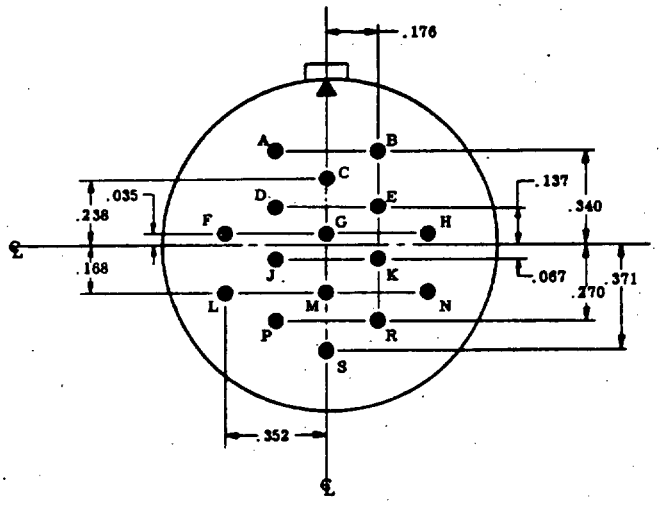
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-4	D	3	16	●	All	MS33687-4
			1	0	⊕		

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INACTIVE FOR DESIGN AFTER 13 MAY 1954



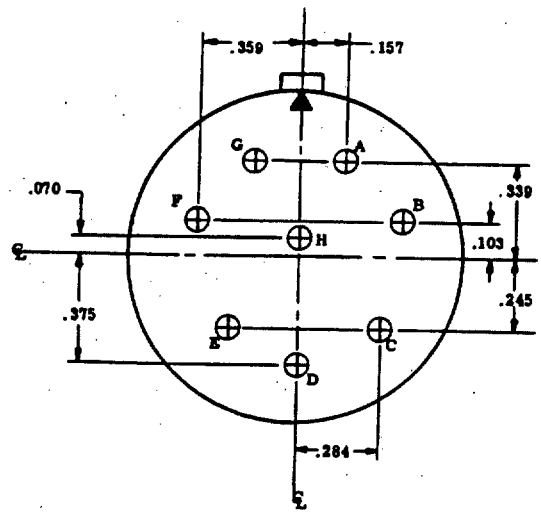
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITIONS SEE 24-15 AND 24-24

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-5	A	16	16	●	All	MS33687-5

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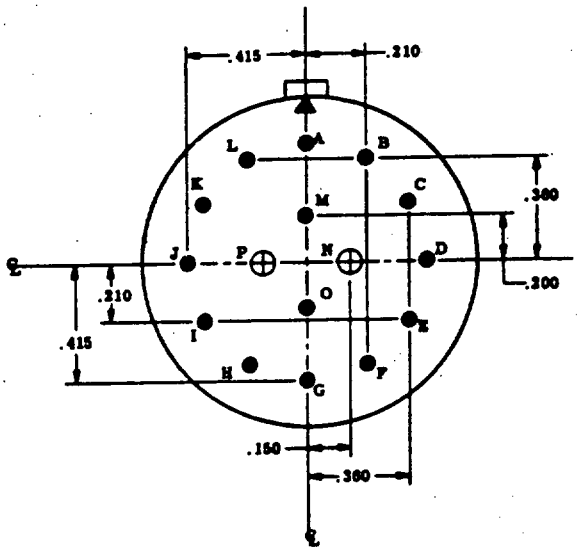


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITIONS SEE 24-25 AND 24-26

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-6	D	3	12	⊕	A,G,H	MS33687-6
		A	5			All others	

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-7	A	14	16	●	A11	MS33687-7
			2	12	⊕		

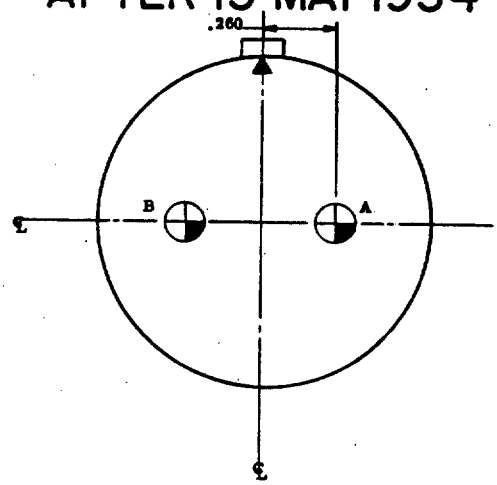
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CANCELLED
AFTER
26 JUN 1945

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-8	---	---	---	---	---	MS33687-8

INACTIVE FOR DESIGN
AFTER 13 MAY 1954

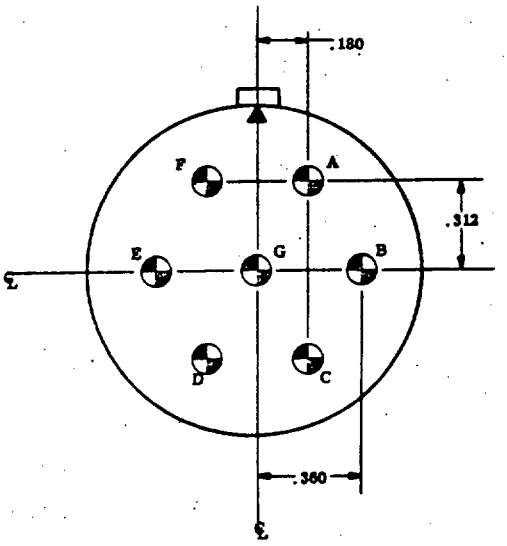


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-9	A	2	4	⊕	All	MS33687-9

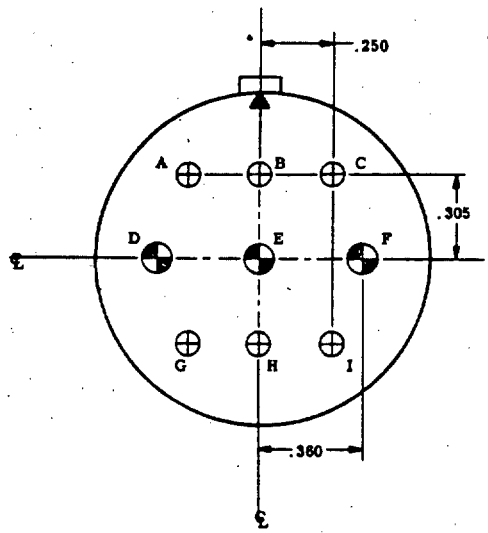
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	——	——	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-10	A	7	8		All	MS33687-10

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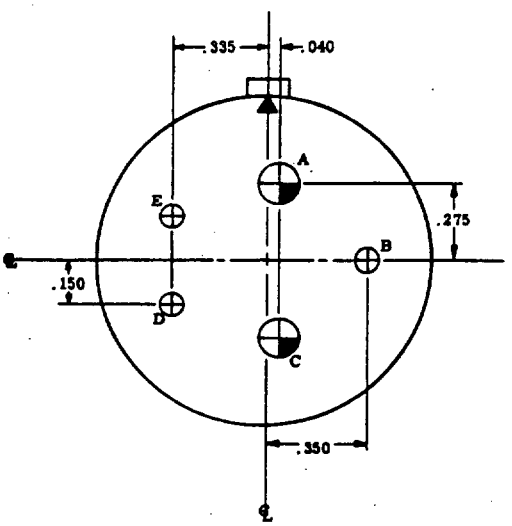


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-11	A	6	12	⊕	All	MS33687-11
			3	8	◔		

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-12	A	3	12	⊕	All	MS33687-12
			2	4	◕		

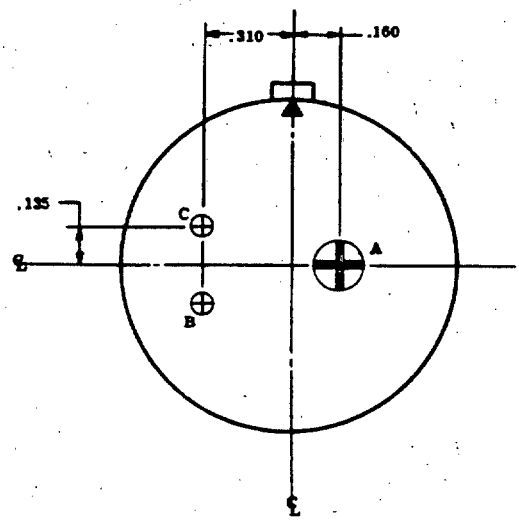
CANCELLED
AFTER
26 JUN 1945

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-13	---	---	---	---	---	MS33687-13

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INACTIVE FOR DESIGN AFTER 17 DEC 1958



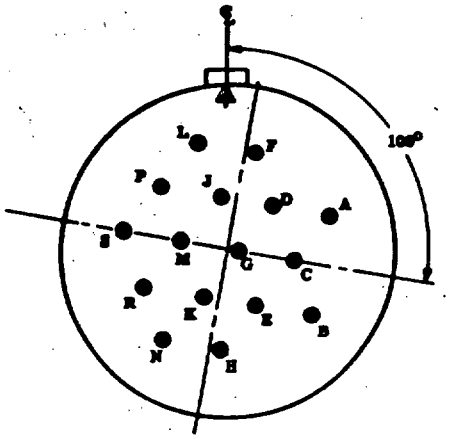
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-14	A	2	12	⊕	A11	MS33687-14
			1	0	⊕		

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INACTIVE FOR DESIGN AFTER 26 JUN 1945



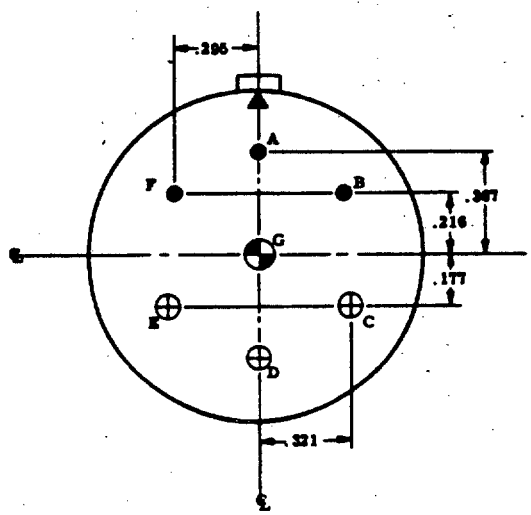
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 24-5 EXCEPT FOR POSITION OF INSERT IN SHELL

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-15	A	16	16	●	All	MS33687-15

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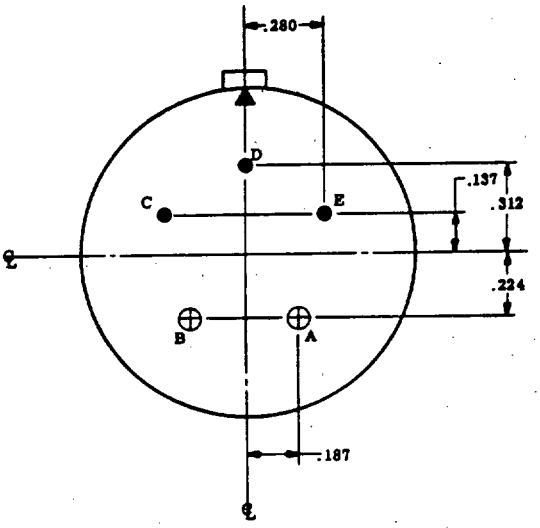


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-16	D	3	16		A,B,F,G	MS33687-16
			1	8			
		A	3	12		All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN AFTER 13 MAY 1954



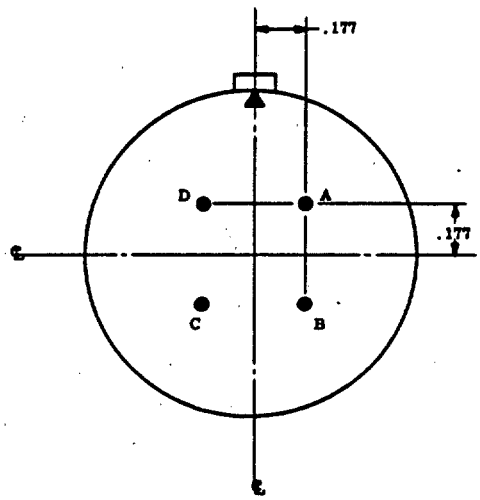
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-17	D	3	16	●	A11	MS33687-17
			2	12	⊕		

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INACTIVE FOR DESIGN AFTER 26 JUN 1945



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

SOCKET CONTACTS A & B SHORT TOGETHER AND C & D SHORT TOGETHER WHEN PIN CONTACTS ARE REMOVED

(Not applicable to MIL-C-22992)

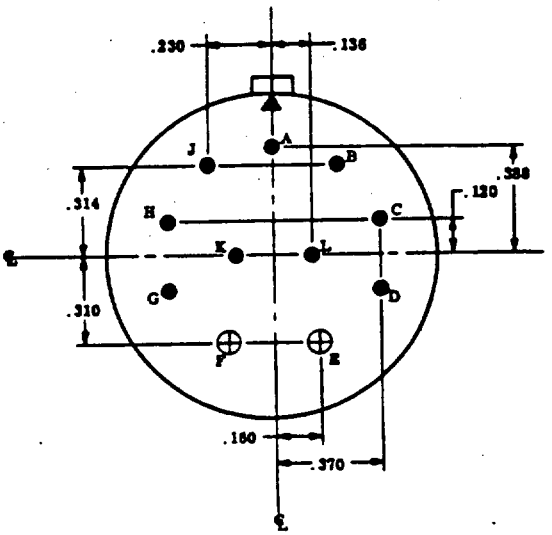
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-18	---	4	16		A11	MS33687-18

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CANCELLED
AFTER
26 JUN 1945

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

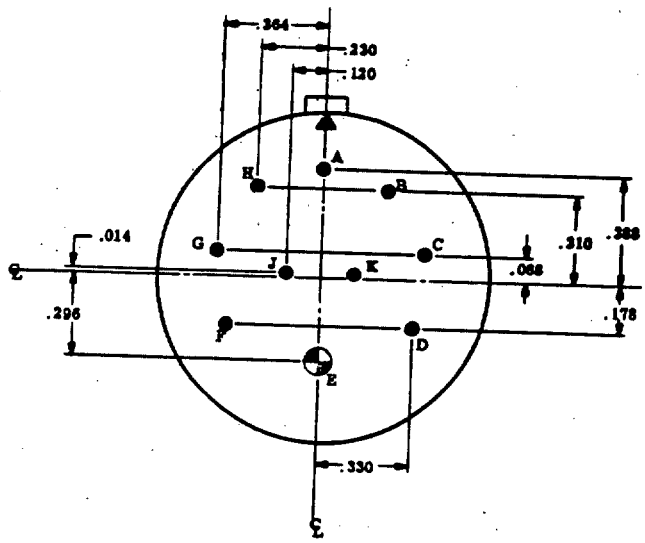
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-19	---	---	---	---	---	MS33687-19



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-20	D	9	16	●	A11	MS33687-20
			2	12	⊕		

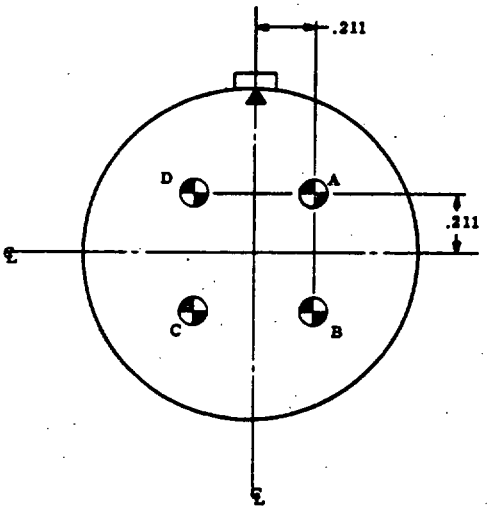
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-21	D	9	16		All	MS33687-21
			1	8			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

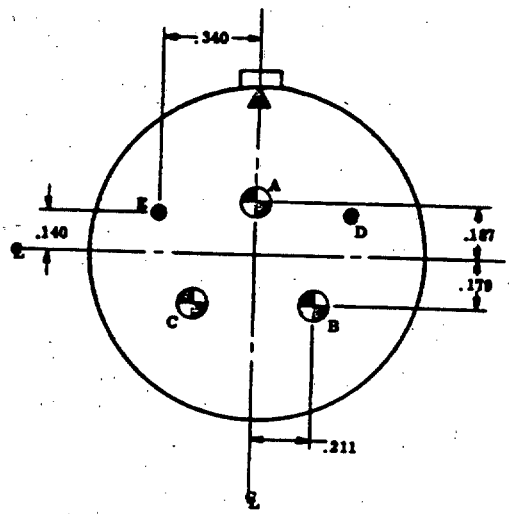


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
45	110	250	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-22	D	4	8		A11	MS33687-22

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INACTIVE FOR DESIGN AFTER 17 DEC 1958



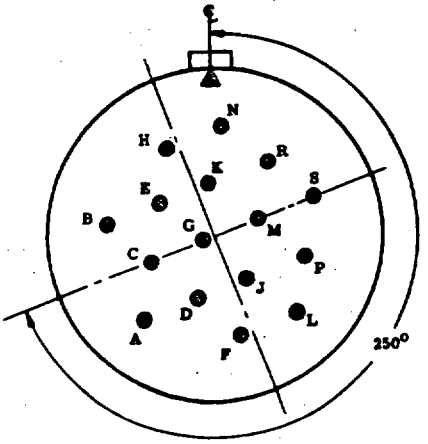
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-23	D	2	16		All	MS33687-23
			3	8			

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INACTIVE FOR DESIGN AFTER 26 JUN 1945



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

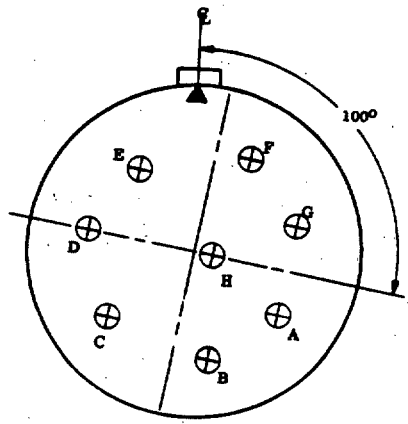
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 24-5 EXCEPT FOR POSITION OF INSERT IN SHELL.

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-24	A	16	16		A11	MS33687-24

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Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 26 JUN 1945



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

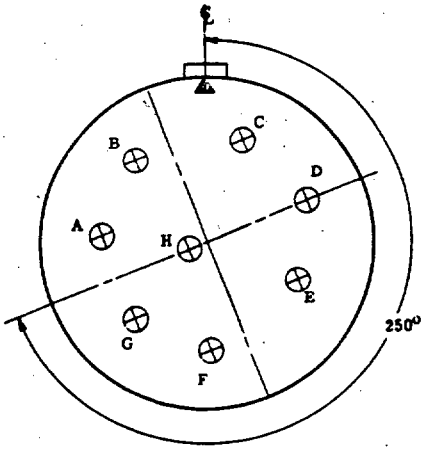
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 24-6 EXCEPT FOR POSITION OF INSERT IN SHELL.

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-25	D	3	12	⊕	A,G,H	MS33687-25
		A	5			All others	

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INACTIVE FOR DESIGN
AFTER 26 JUN 1945



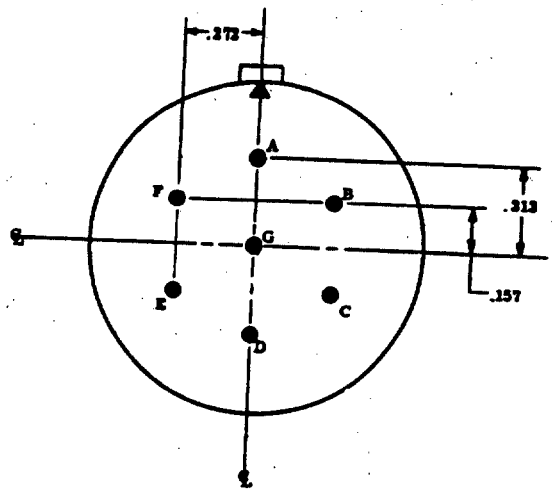
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 24-6 EXCEPT FOR POSITION OF INSERT IN SHELL.

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-26	D	3	12	⊕	A,G,H	M33687-26
		A	5			All others	

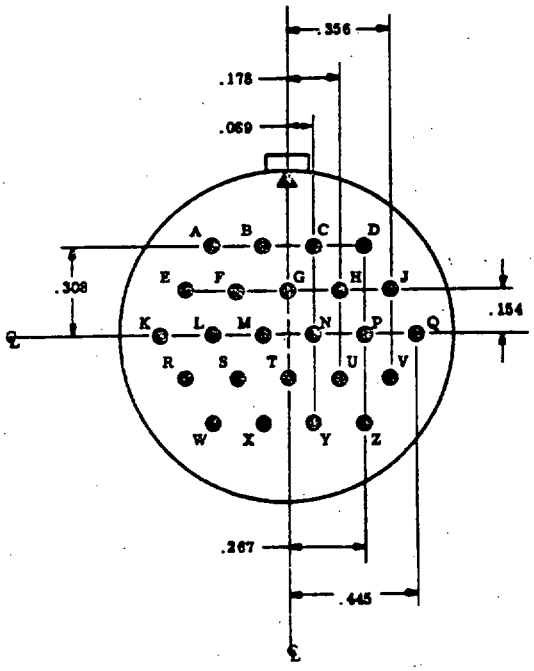
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	—	—	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-27	E	7	16		All	MS33687-27

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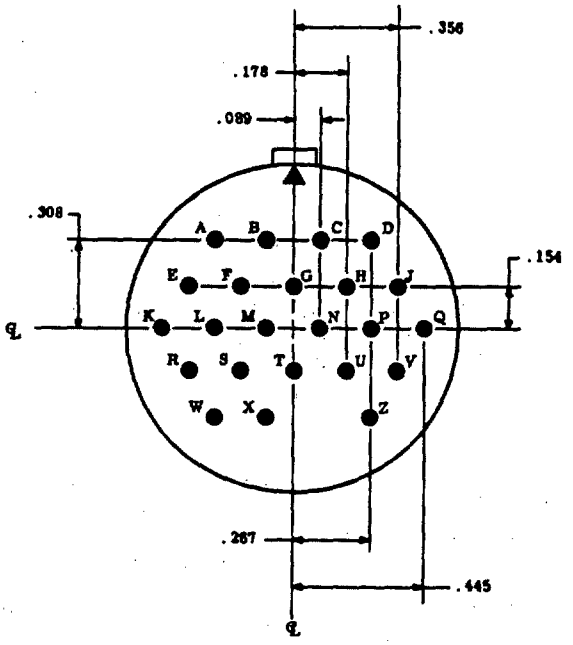


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-28	INST	24	16	●	A11	MS33687-28

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	145	240	300

NOT APPLICABLE FOR DESIGN OF AIR FORCE AIRBORNE EQUIPMENTS.

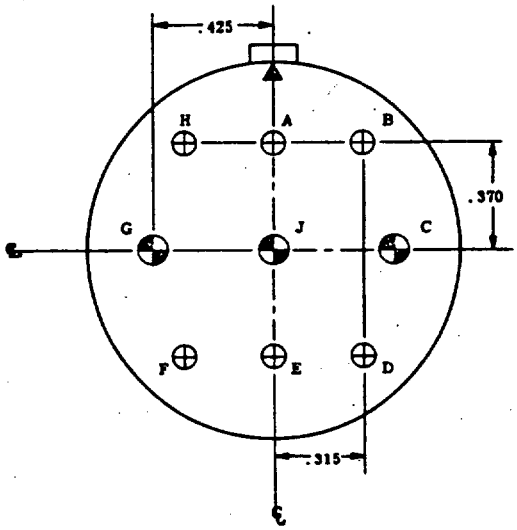
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-80	INST	23	16		A11	MS33687-80

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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SECTION 100

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

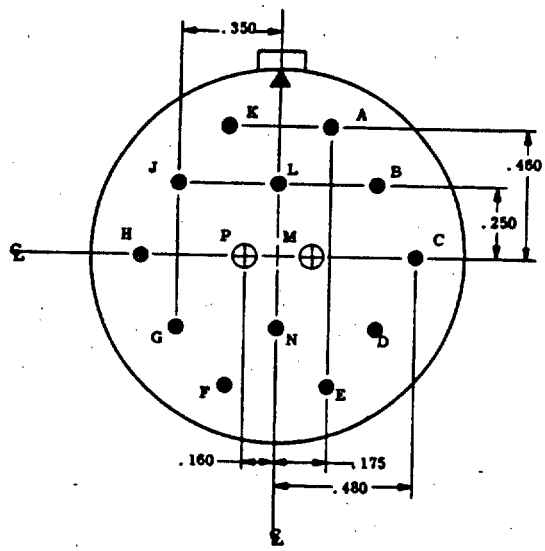
SHELL SIZE 28



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-1	D	2	12	⊕	A,E	MS33688-1
		A	4			B,D,F,H	
		D	1	8	◔	J	
		A	2			G,C	

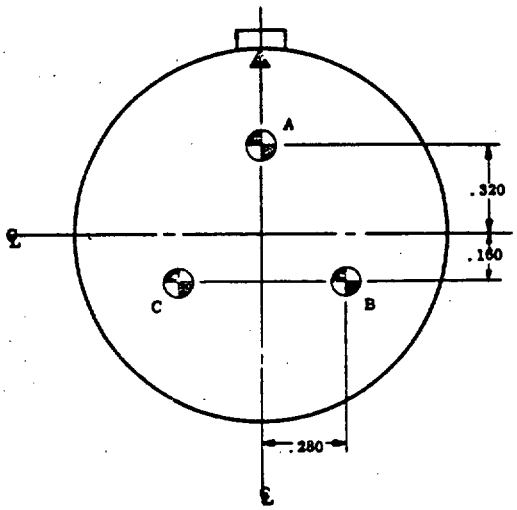
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-2	D	12	16	●	All	MS33688-2
			2	12	⊕		

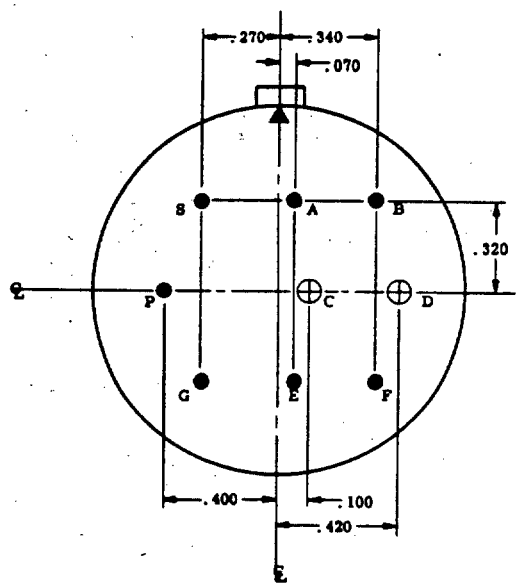
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-3	E	3	8		A11	MS33688-3

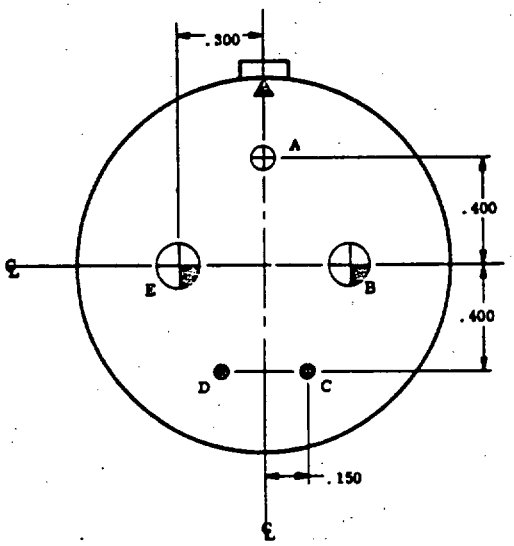
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-4	E	3	16	●	G,P,S	MS33688-4
		D	4			A,B,E,F	
			2	12	⊕	C,D	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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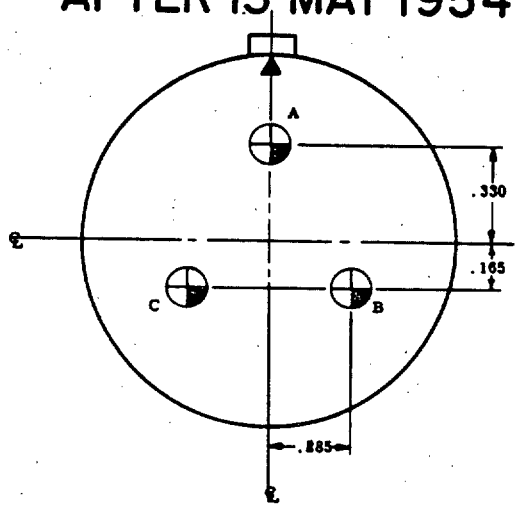


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-5	D	2	16	●	All	MS33688-5
			1	12	⊕		
			2	4	◒		

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INACTIVE FOR DESIGN
AFTER 13 MAY 1954



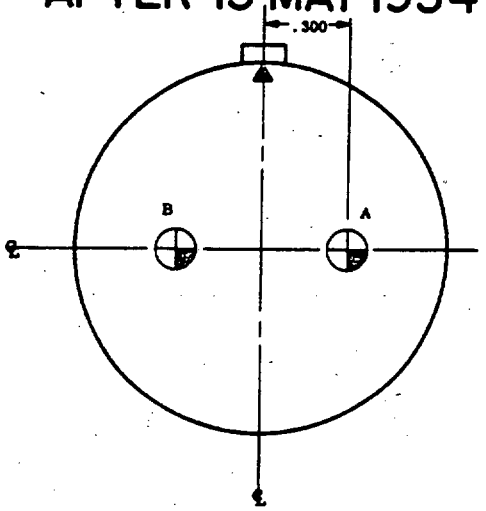
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-6	D	3	4		All	MS33688-6

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INACTIVE FOR DESIGN
AFTER 13 MAY 1954

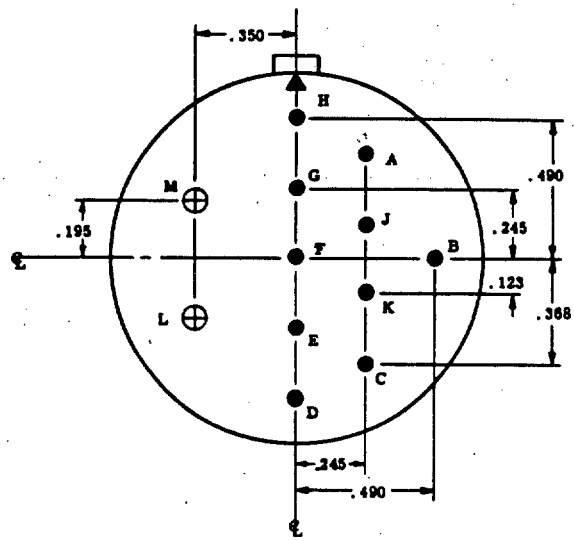


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-7	D	2	4	⊕	All	MS33688-7

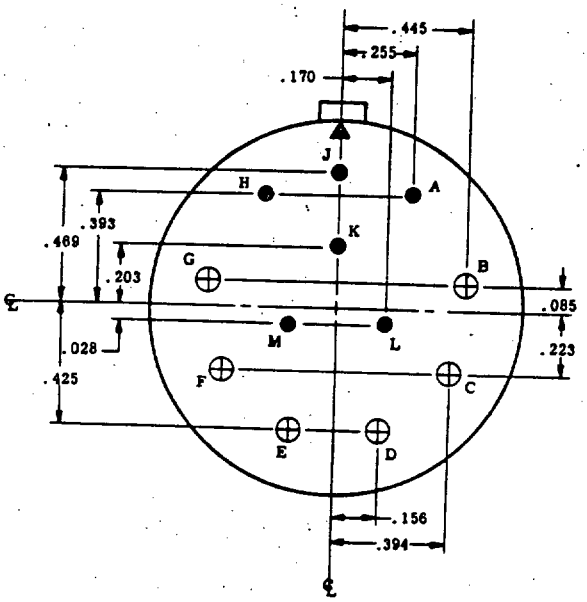
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-8	E	2	12	⊕	L,M	MS33688-8
		D	1	16	●	B	
		A	9			All others	

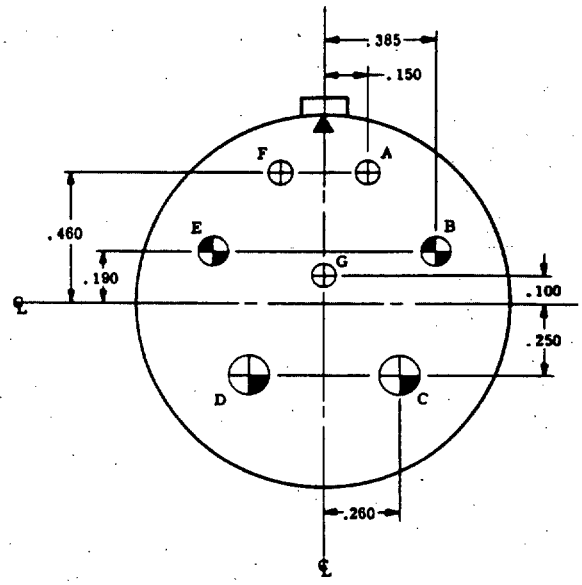
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-9	D	6	16	●	A11	MS33688-9
			6	12	⊕		

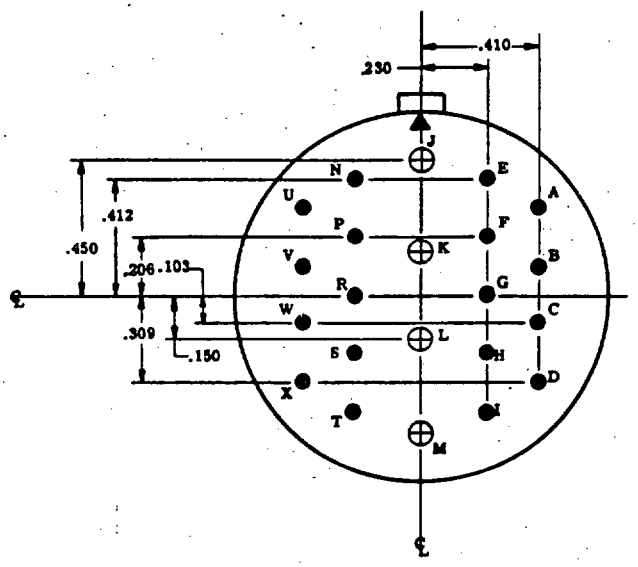
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS.			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-10	D	1	12	⊕	G	MS33688-10
		A	2			A,F	
			2	8	◕	B,E	
			2	4	◔	C,D	

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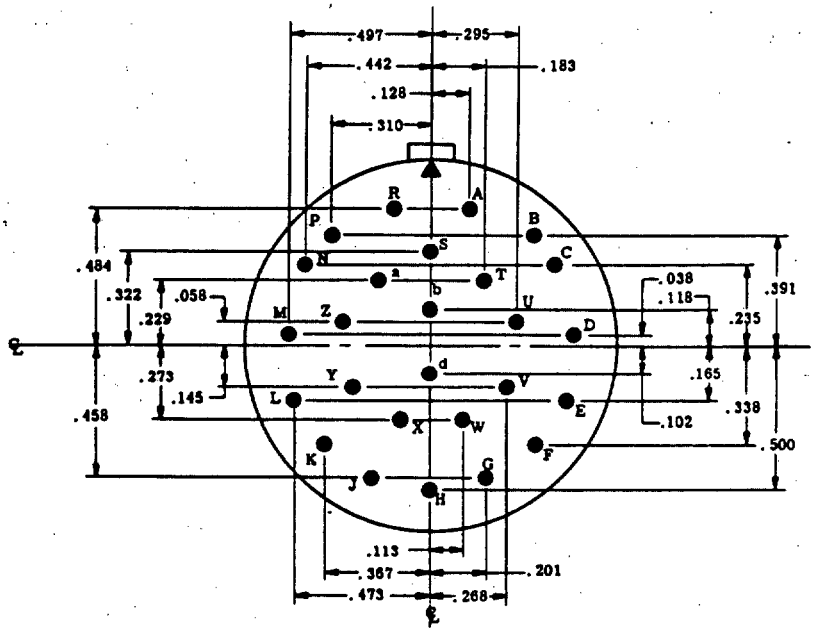


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-11	A	18	16	●	A11	MS33688-11
			4	12	⊕		

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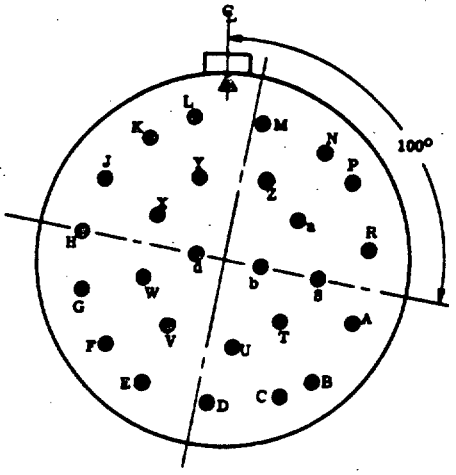
FOR OTHER INSERT POSITION SEE 28-13

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-12	A	26	16		A11	MS33688-12

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INACTIVE FOR DESIGN AFTER 26 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 28-12 EXCEPT FOR POSITION OF INSERT IN SHELL

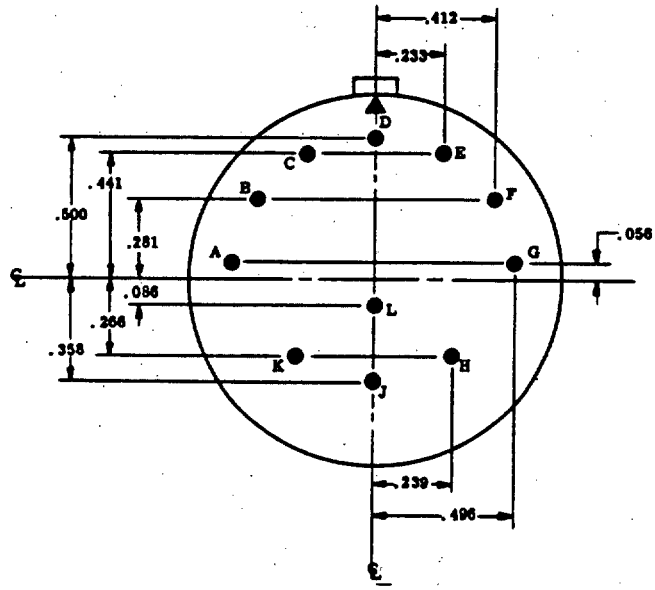
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-13	A	26	16	●	All	MS33688-13

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INACTIVE FOR DESIGN AFTER 13 MAY 1954



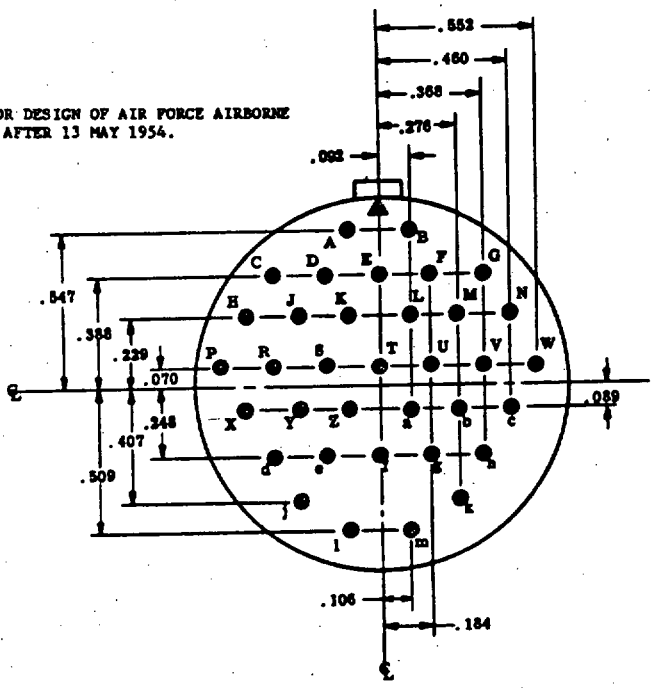
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-14	D	11	16		A11	MS33688-14

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Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN OF AIR FORCE AIRBORNE EQUIPMENTS AFTER 13 MAY 1954.

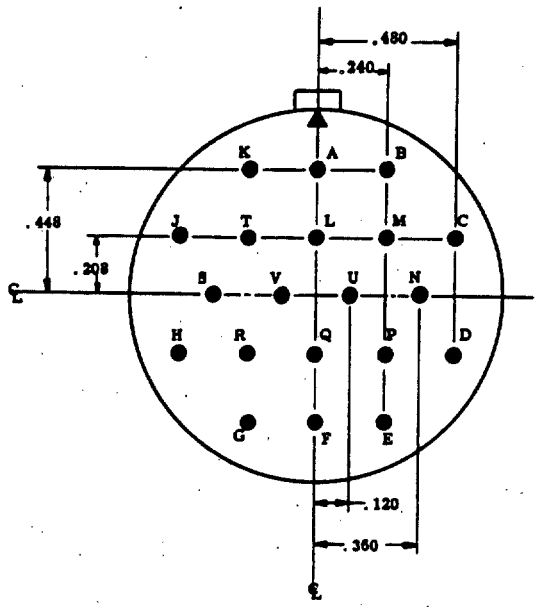


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-15	A	35	16		A11	MS33688-15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN AFTER 13 MAY 1954

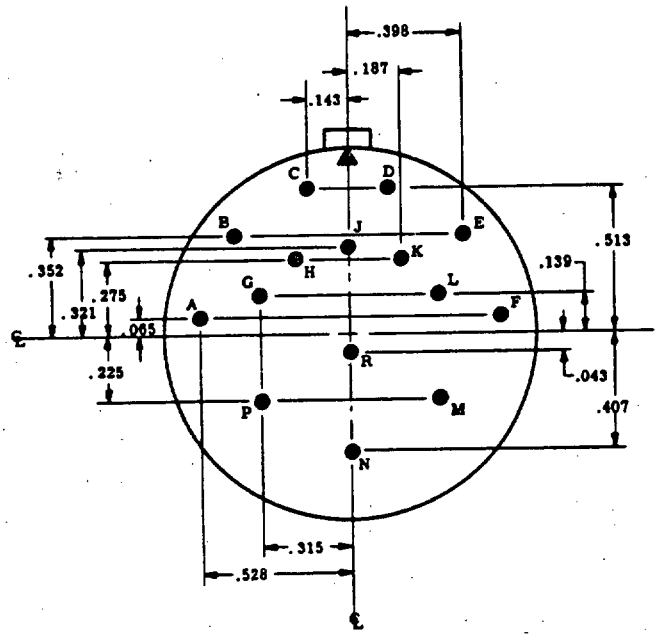


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-16	A	20	16	●	All	MS33688-16

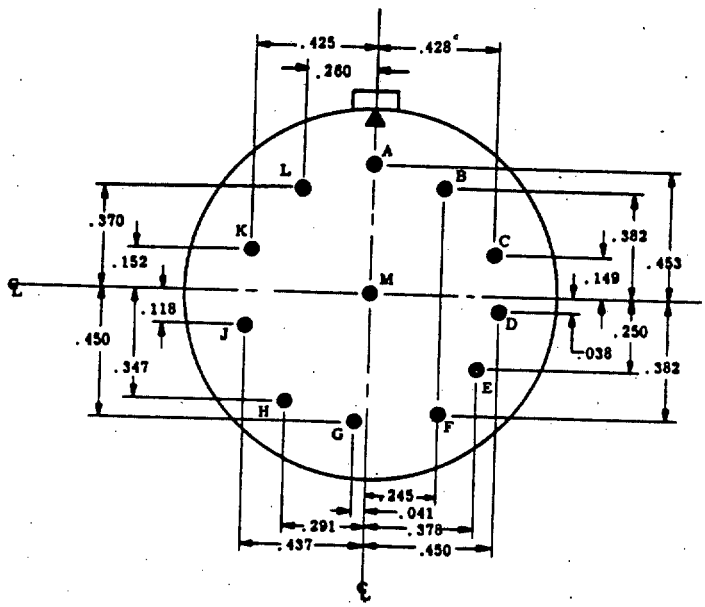
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-17	B	1	16	●	R	MS33688-17
		D	3			M,N,P	
		A	11			All others	

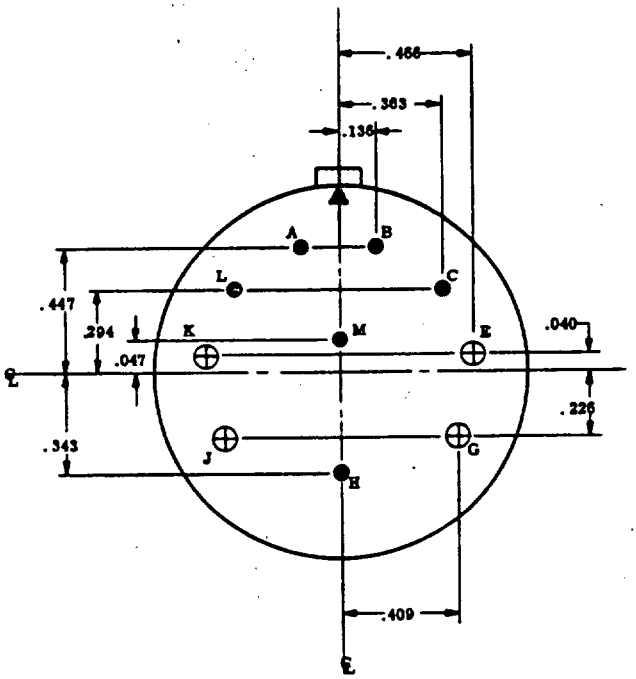
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-18	C	1	16		M	MS33688-18
		A	2			A,B	
		INST	4			C,D,E,F	
		D	5			All others	

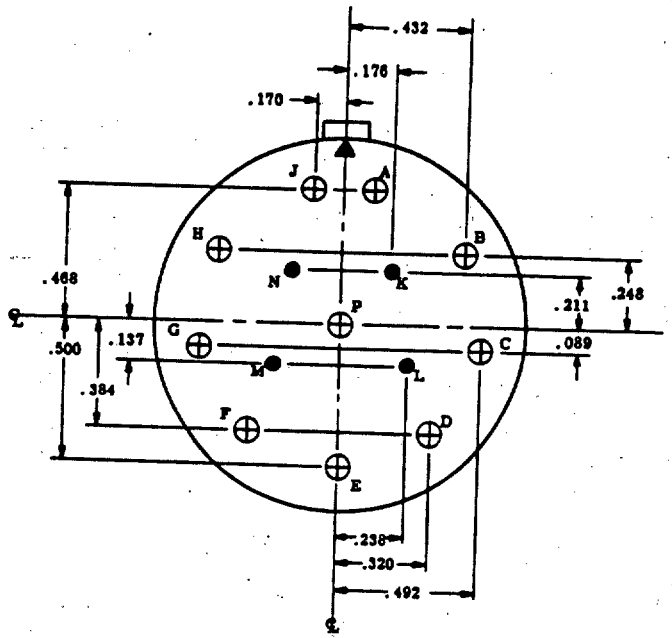
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-19	B	2	16	●	H,M	MS33688-19
		D	2			A,B	
		A	2			All others	
			4	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

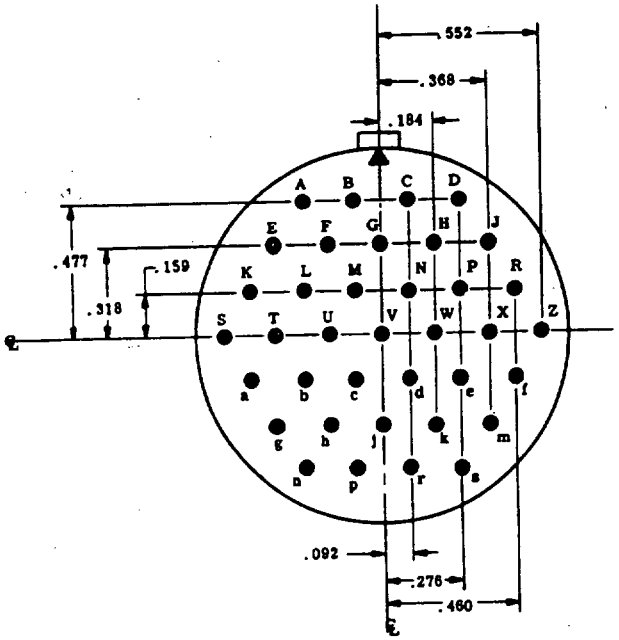


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-20	A	4	16	●	All	MS33688-20
			10	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

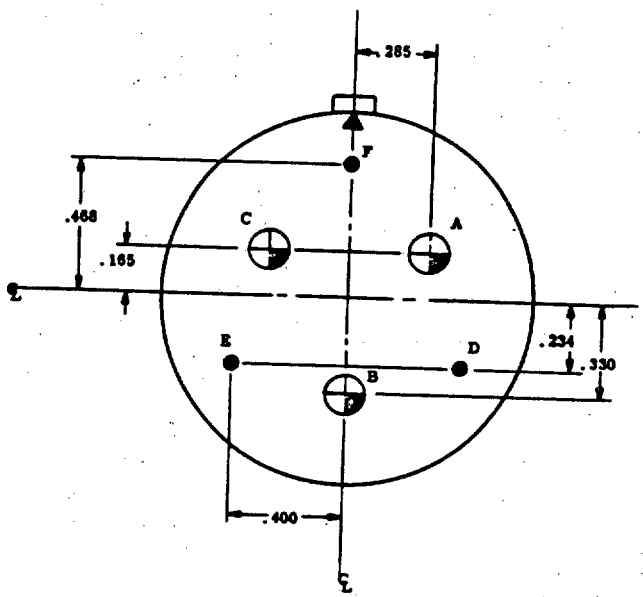


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-21	A	37	16	●	All	MS33688-21

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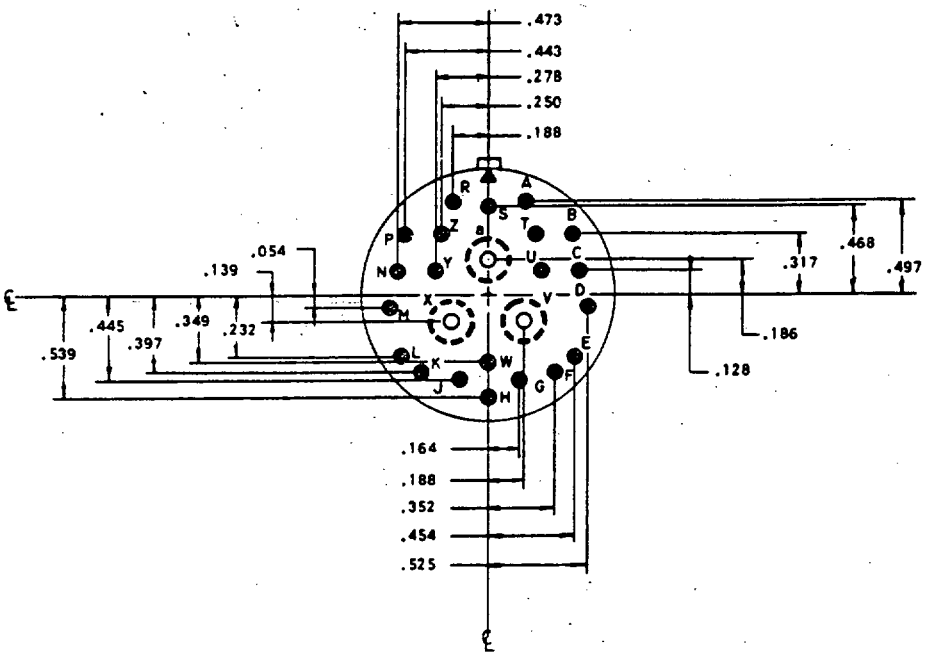


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-22	D	3	16	●	A11	MS33688-22
			3	4	⊕		

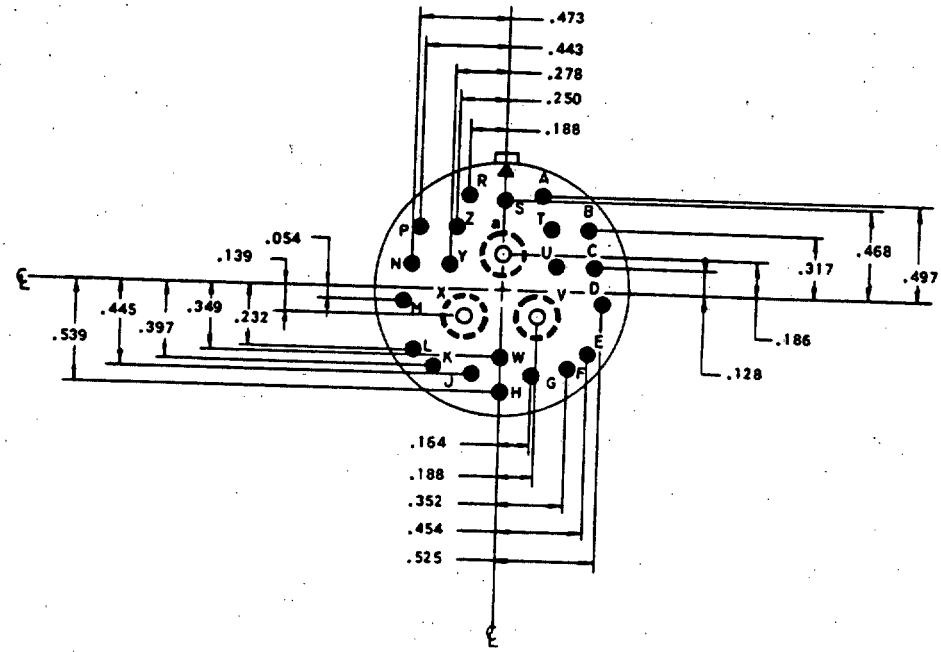
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-23		3	8 (shielded) MS14056-2		V,X,a	MS33688-23
		INST	21	16		All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

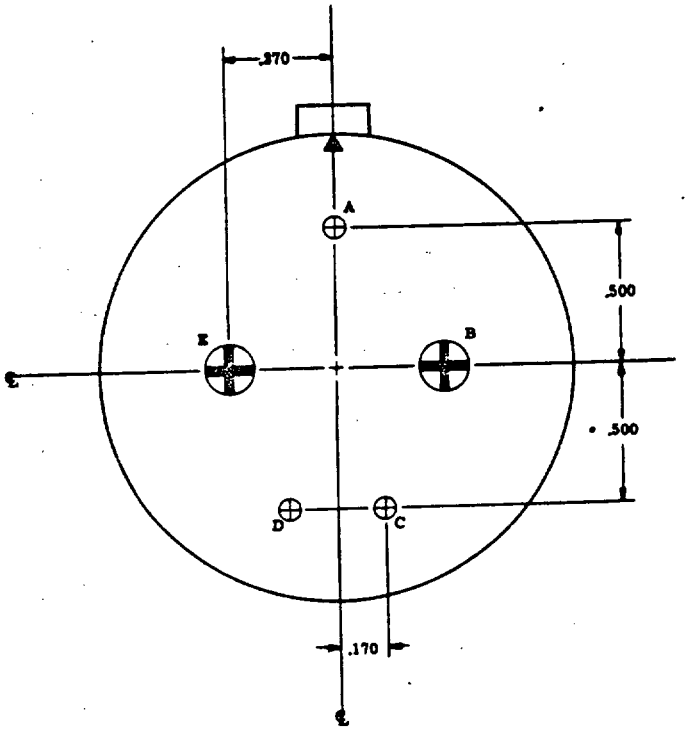
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-24		3	8 (shielded) MS14056-1	○	V,X,a	MS33688-24
		INST	21	16	●	All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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SECTION 110

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 32

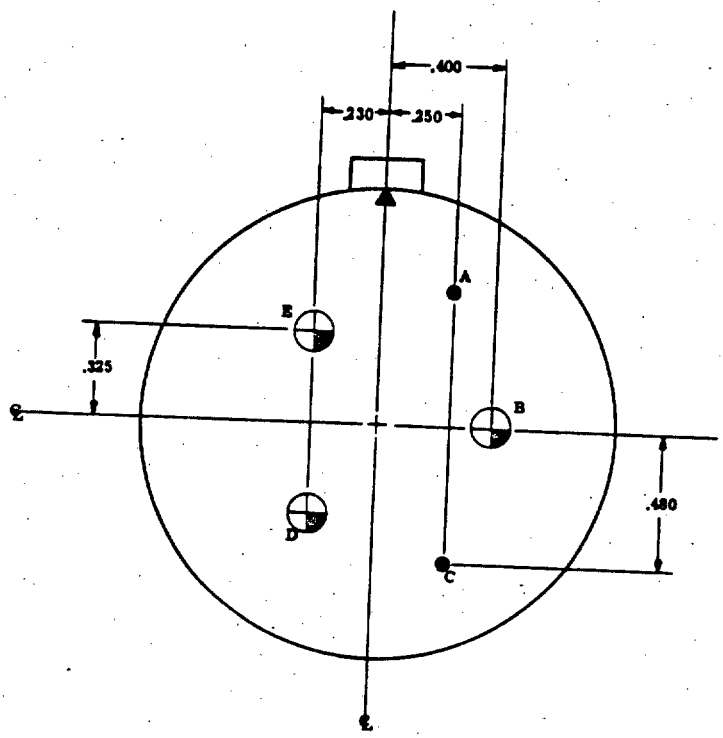


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITION SEE 32-19

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-1	E	1	12	⊕	A	MS33689-1
		D	2			D,C	
			2	0	⊕	B,E	

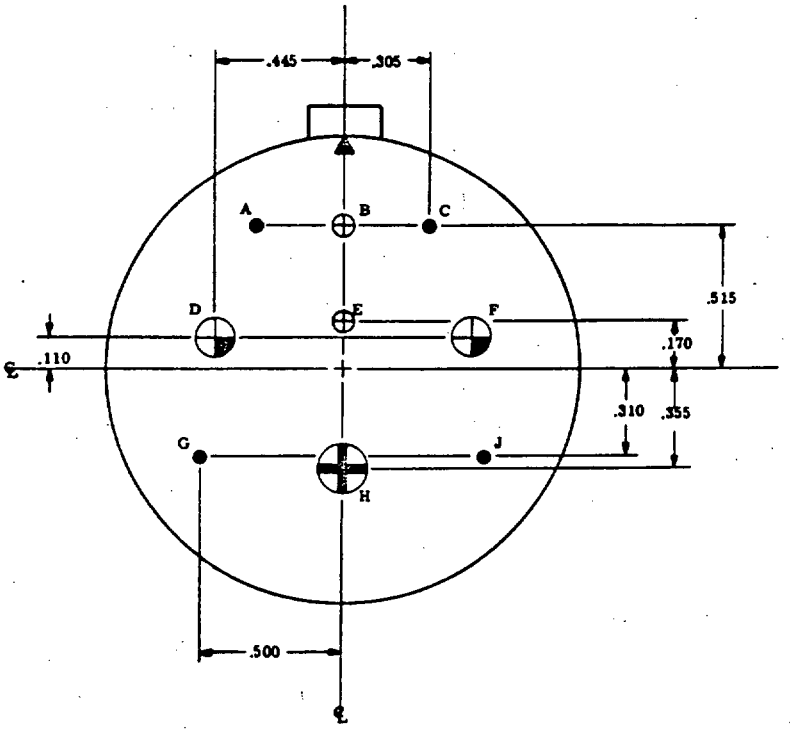
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-2	E	2	16		A11	MS33689-2
			3	4			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

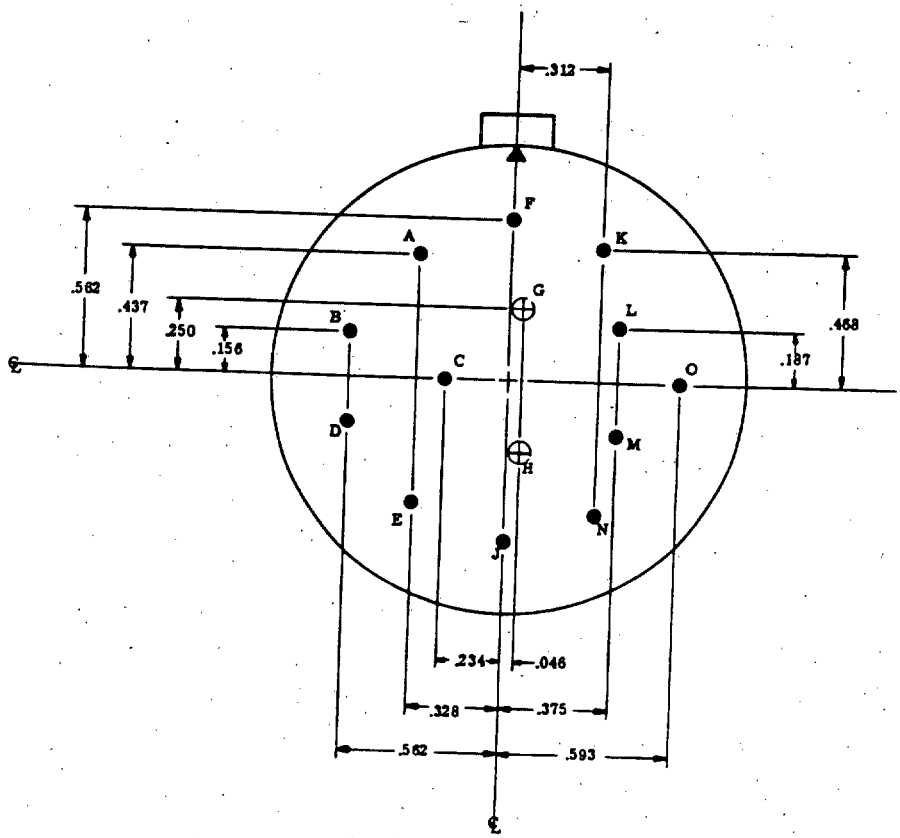


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-3	D	4	16		All	MS33689-3
			2	12			
			2	4			
			1	0			

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Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 13 MAY 1954



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

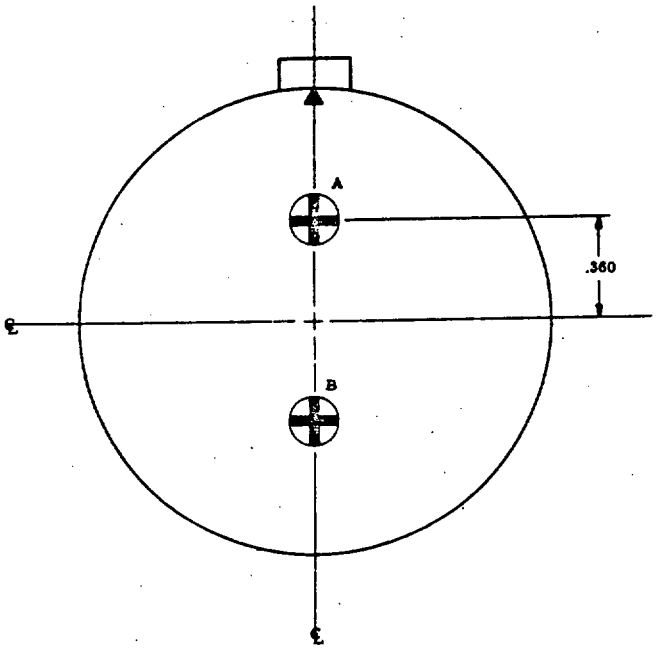
FOR OTHER INSERT POSITION SEE 32-18

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-4	D	2	12	⊕	G,H	MS33689-4
		A	4	16	●	F,J,K,N	
		D	8			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 13 MAY 1954

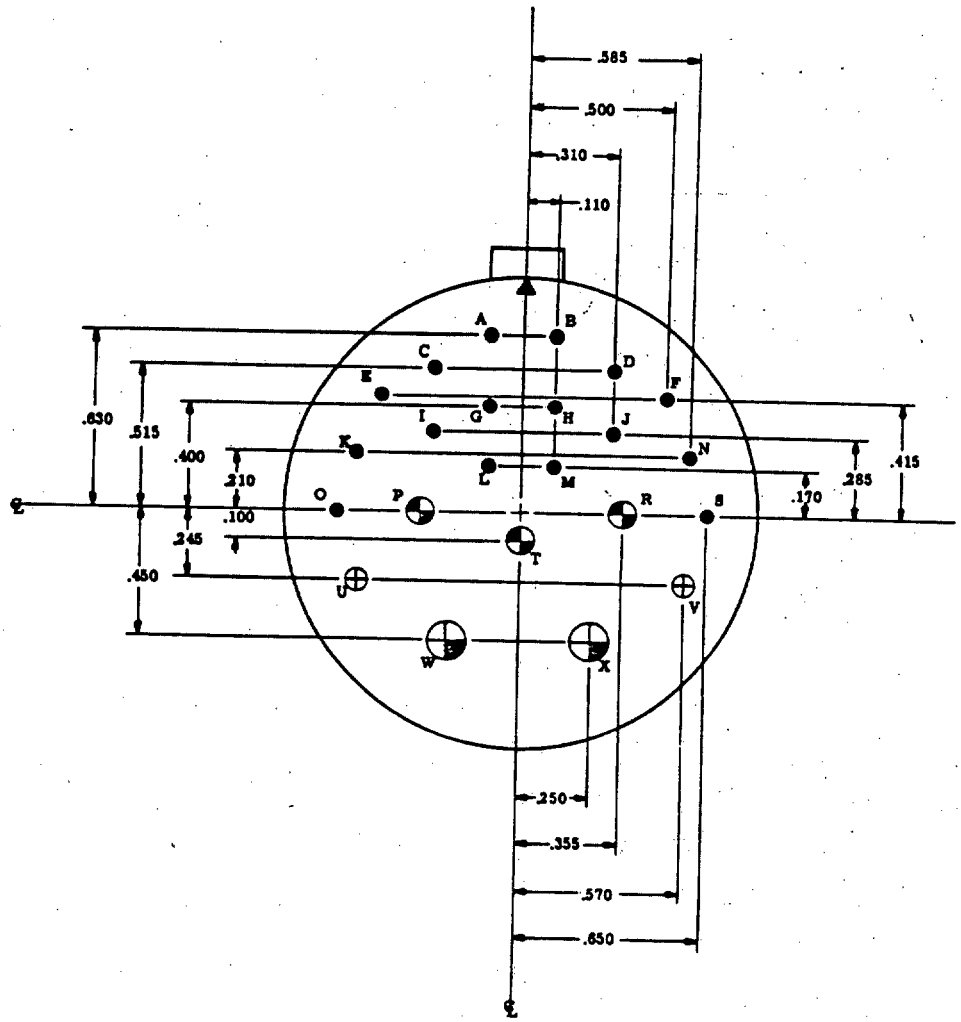


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-5	D	2	0	⊕	A11	MS33689-5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



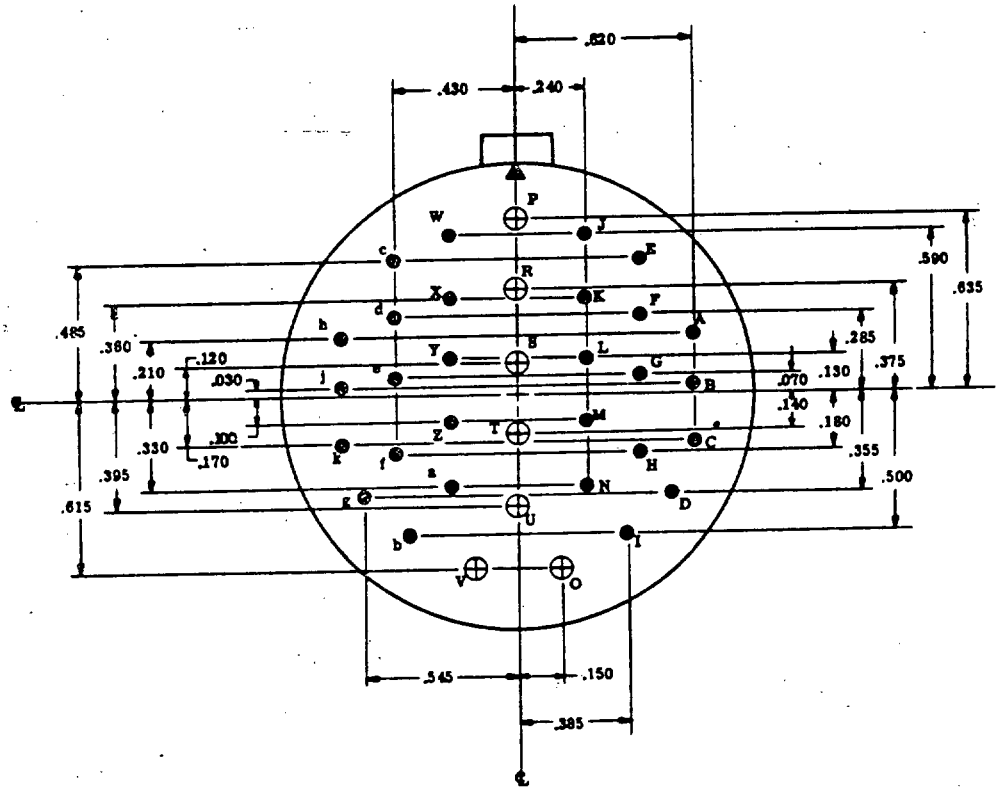
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

* # †

FOR OTHER INSERT POSITIONS SEE 32-16 AND 32-20

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-6	A	16	16		All	MS33689-6
			2	12			
			3	8			
			2	4			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



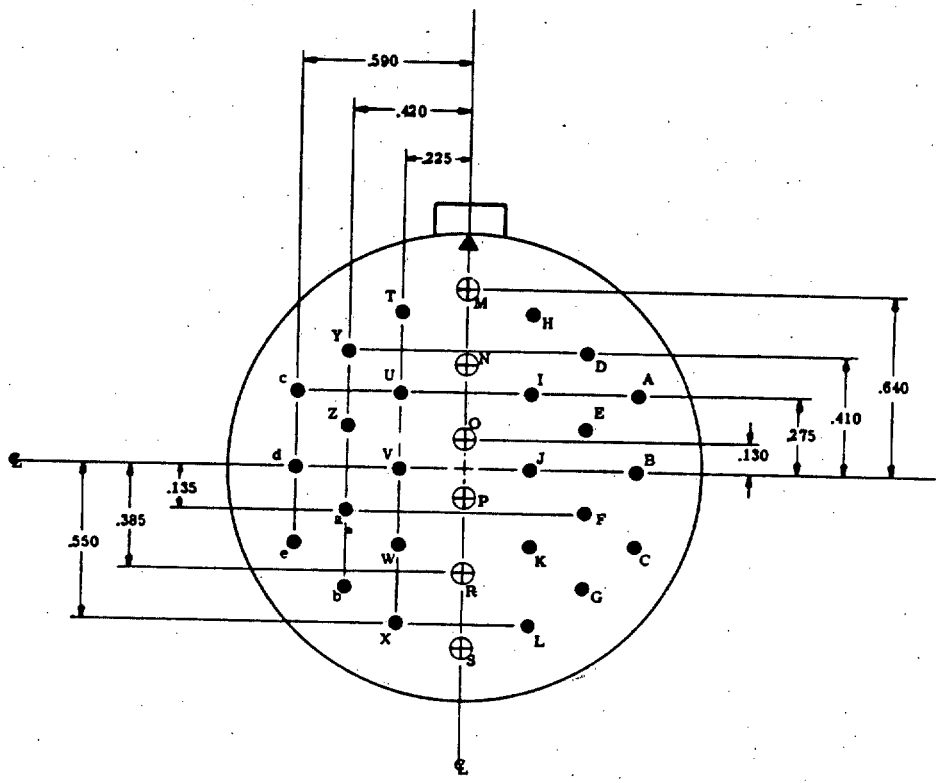
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	125	235	280

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-7	INST	4	16	●	A,B,h,j	MS33689-7
		A	24			All others	
			7	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 13 MAY 1954

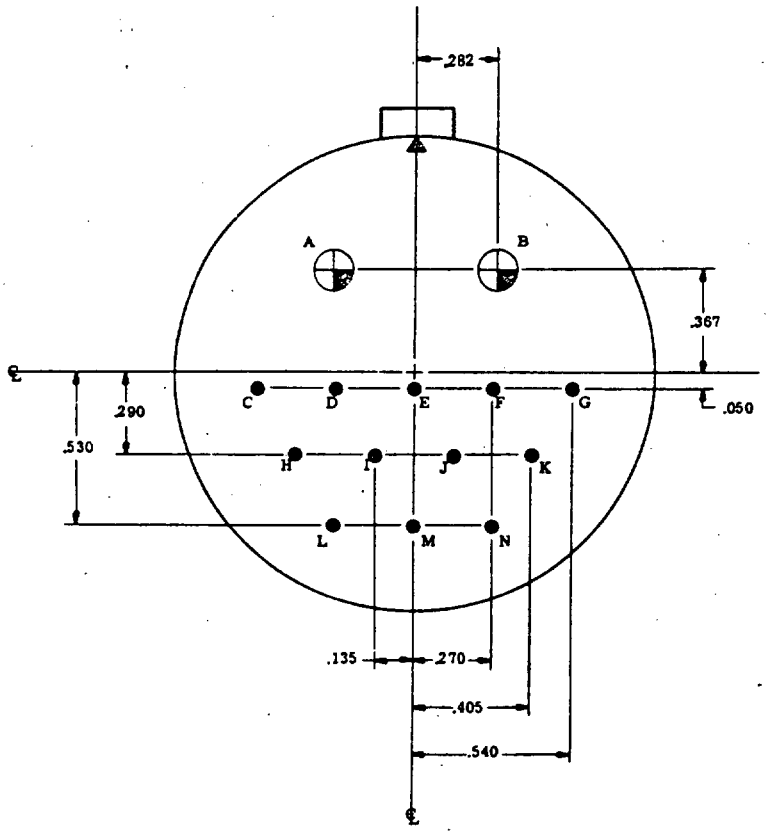


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	125	235	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-8	A	24	16	●	All	MS33689-8
			6	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

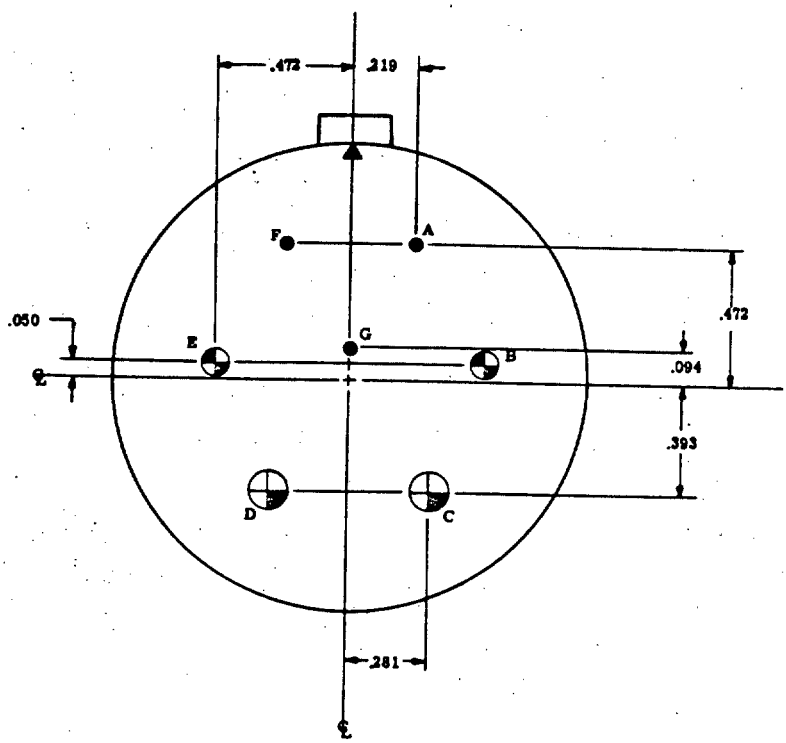


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-9	D	12	16	●	A11	MS33689-9
			2	4	⊕		

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INACTIVE FOR DESIGN AFTER 17 DEC 1958



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-10	E	2	16		A,F	MS33689-10
		B	1			G	
		D	2	8		B,E	
		A	2	4		C,D	

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Check the source to verify that this is the current version before use.

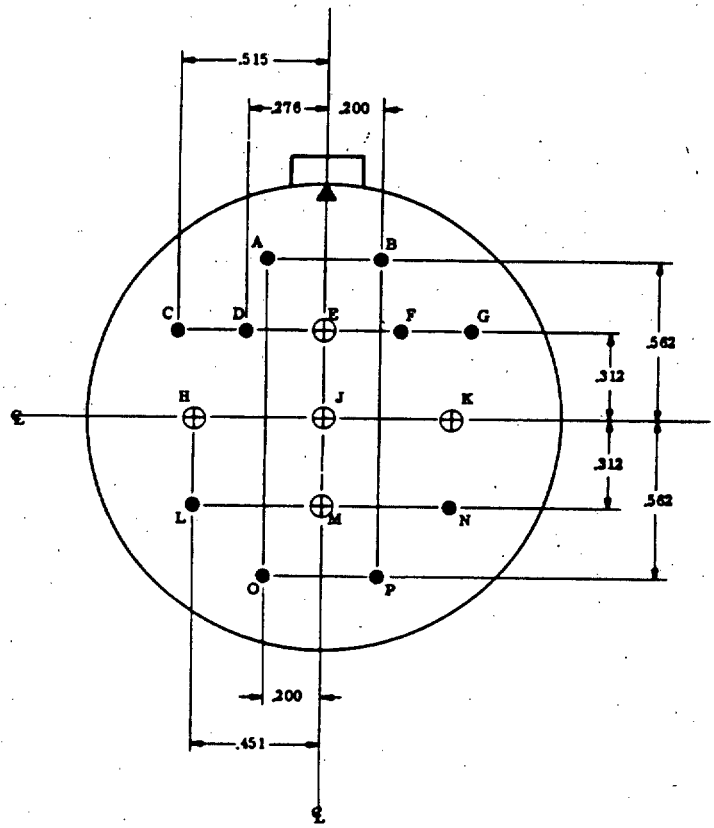
CANCELLED AFTER 26 JUN 1945

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-11	---	---	---	---	---	MS33689-11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 13 MAY 1954

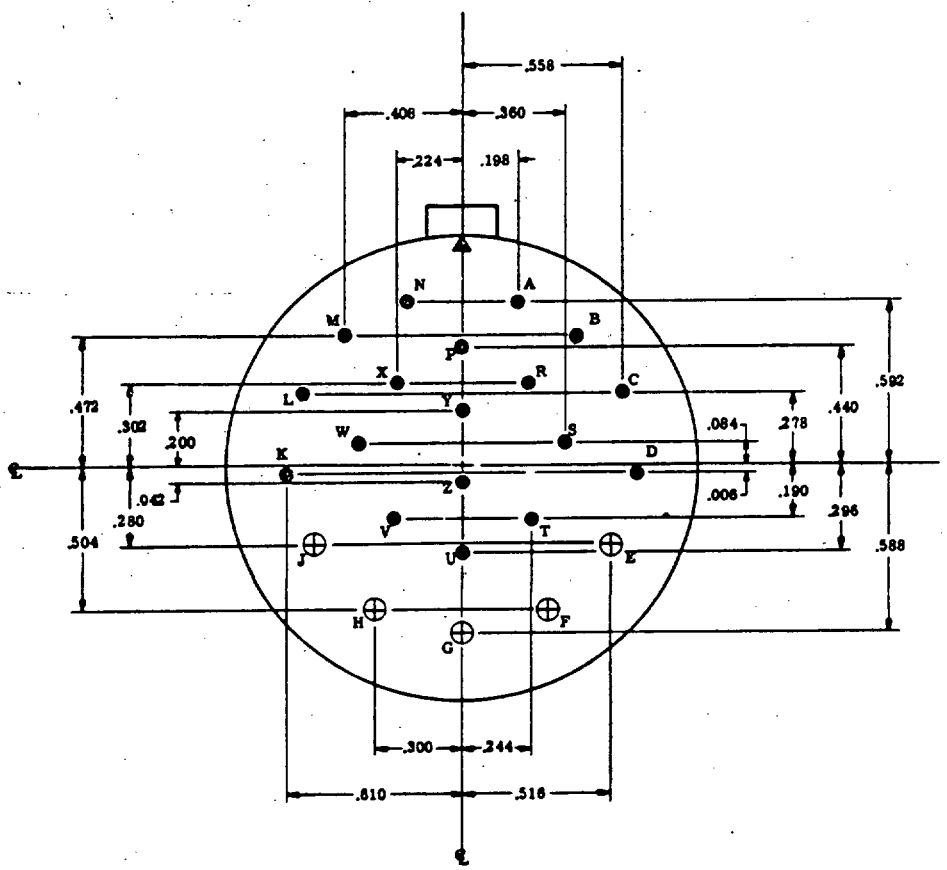


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-12	A	1	12	⊕	E	MS33689-12
		D	4			H,J,K,M	
		A	4	16	●	C,D,F,G	
		D	6			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

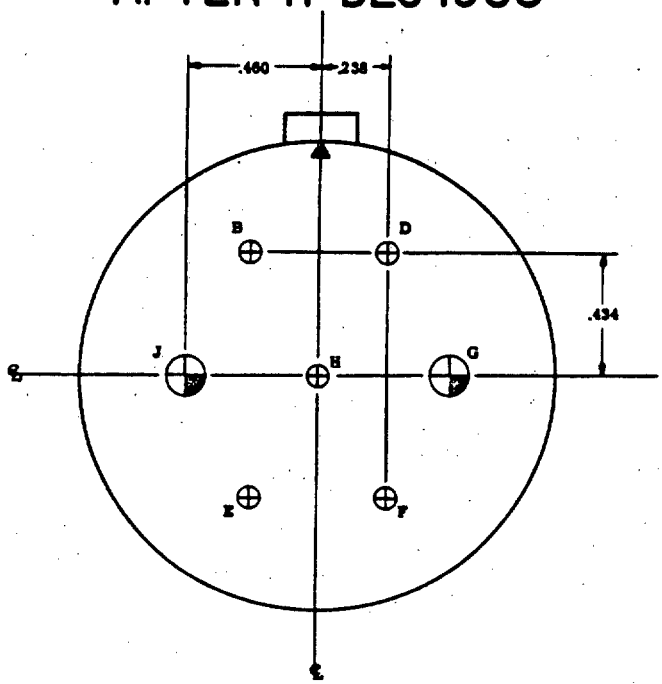


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-13	D	18	16	●	All	MS33689-13
			5	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN
AFTER 17 DEC 1958

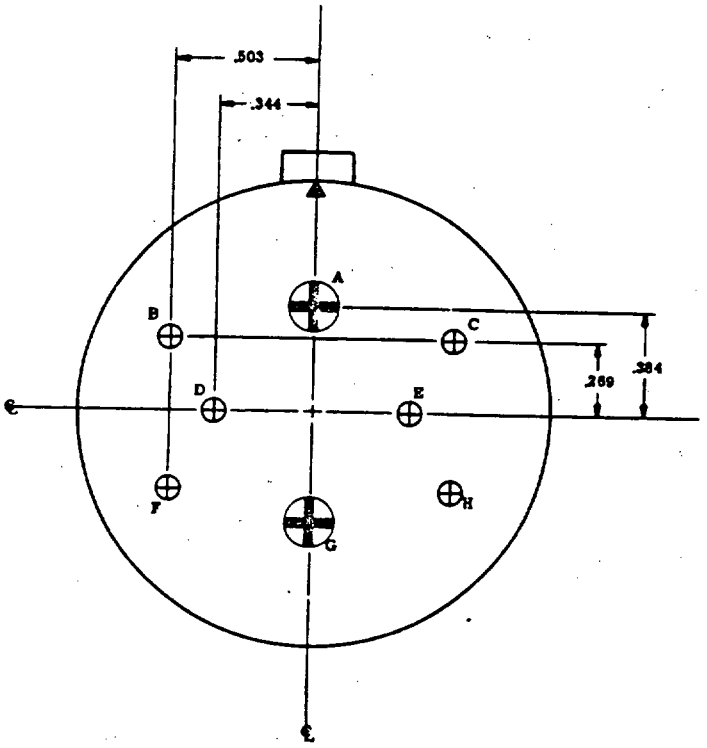


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-14	D	5	12	⊕	All	MS33689-14
			2	4	◔		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

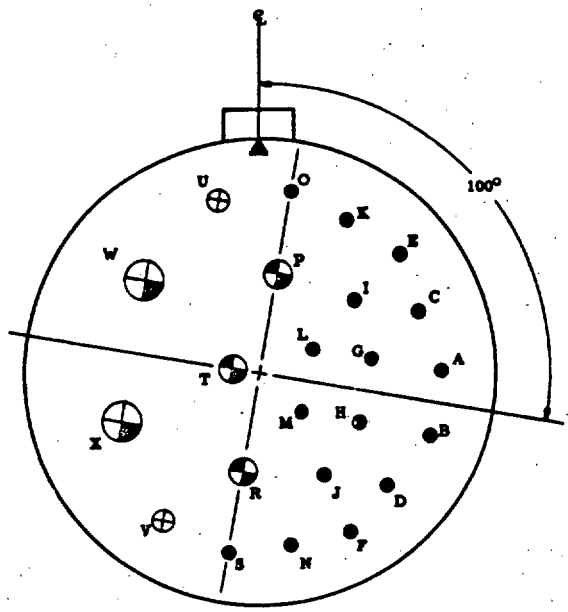


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-15	D	6	12	⊕	All	MS33689-15
			2	0	⊕		

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INACTIVE FOR DESIGN
AFTER 26 JUN 1945



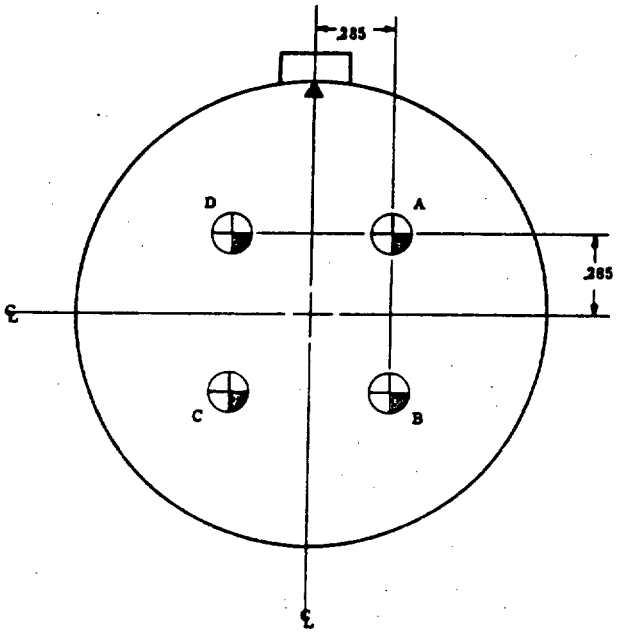
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 32-6 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-16	A	16	16		A11	MS33689-16
			2	12			
			3	8			
			2	4			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

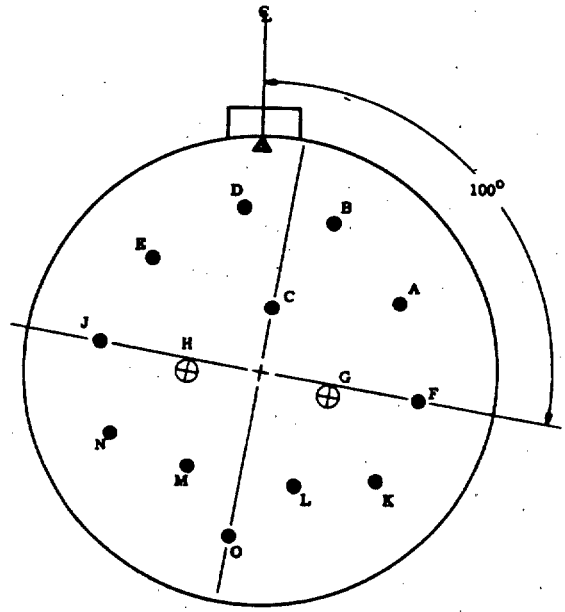


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
45	110	250	——

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-17	D	4	4		A11	MS33689-17

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR DESIGN AFTER 26 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 32-4 EXCEPT FOR POSITION OF INSERT IN SHELL

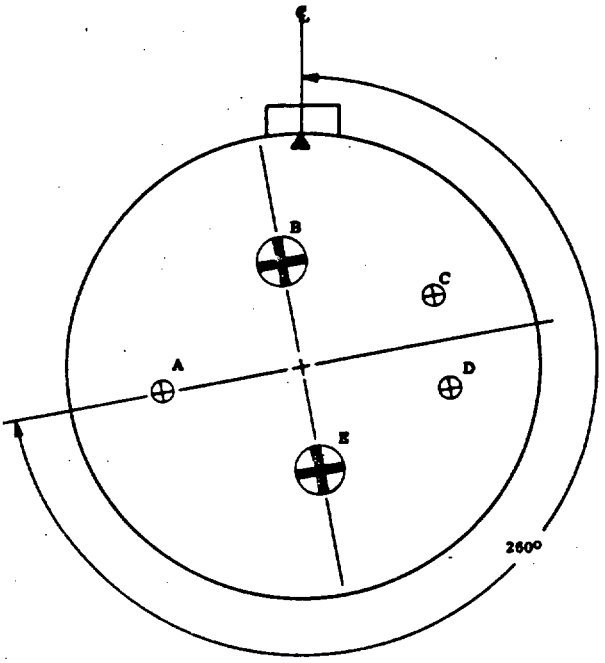
ALTERNATE INSERT POSITIONS. A DEGREES			
W	X	Y	Z

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-18	D	2	12	⊕	G,H	MS33689-18
		A	4	16	●	F,J,K,N	
		D	8			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN
AFTER 26 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 32-1 EXCEPT FOR POSITION OF INSERT IN SHELL

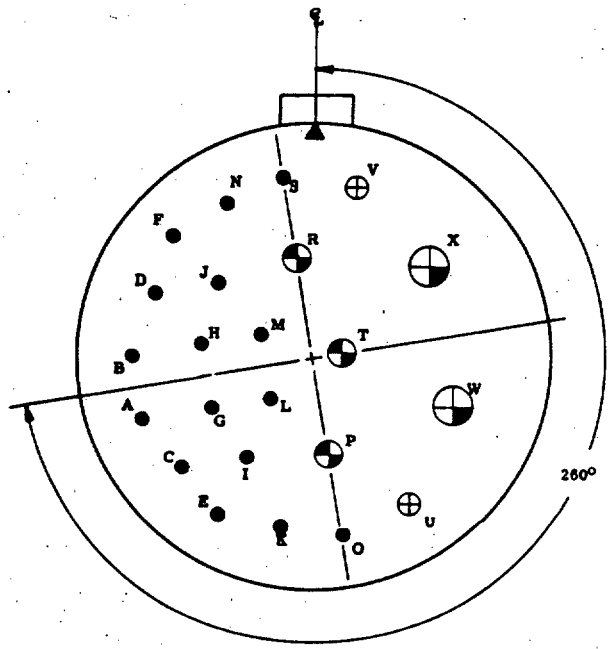
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
———	———	———	———

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-19	E	1	12	⊕	A	MS33689-19
		D	2			D,C	
			2	0	⊕	B,E	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 26 JUN 1945



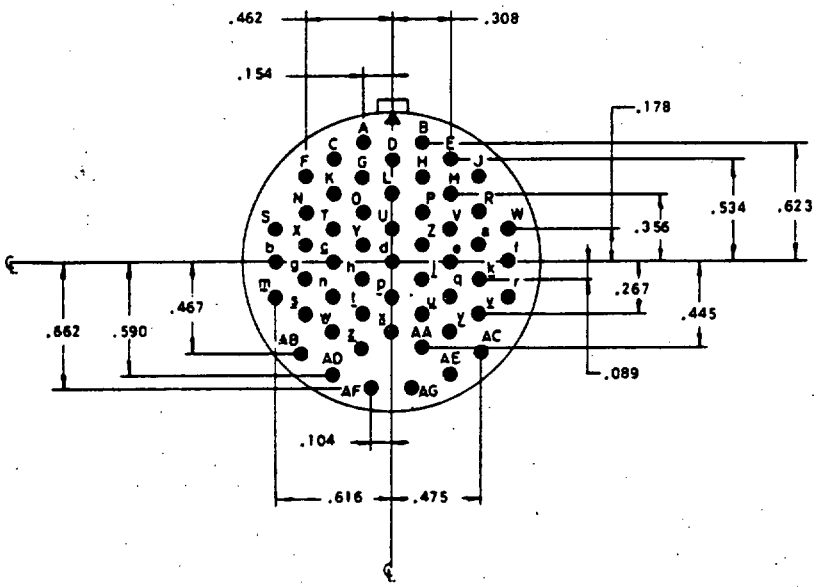
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 32-6 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-20	A	16	16	●	All	MS33689-20
			2	12	⊕		
			3	8	◕		
			2	4	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



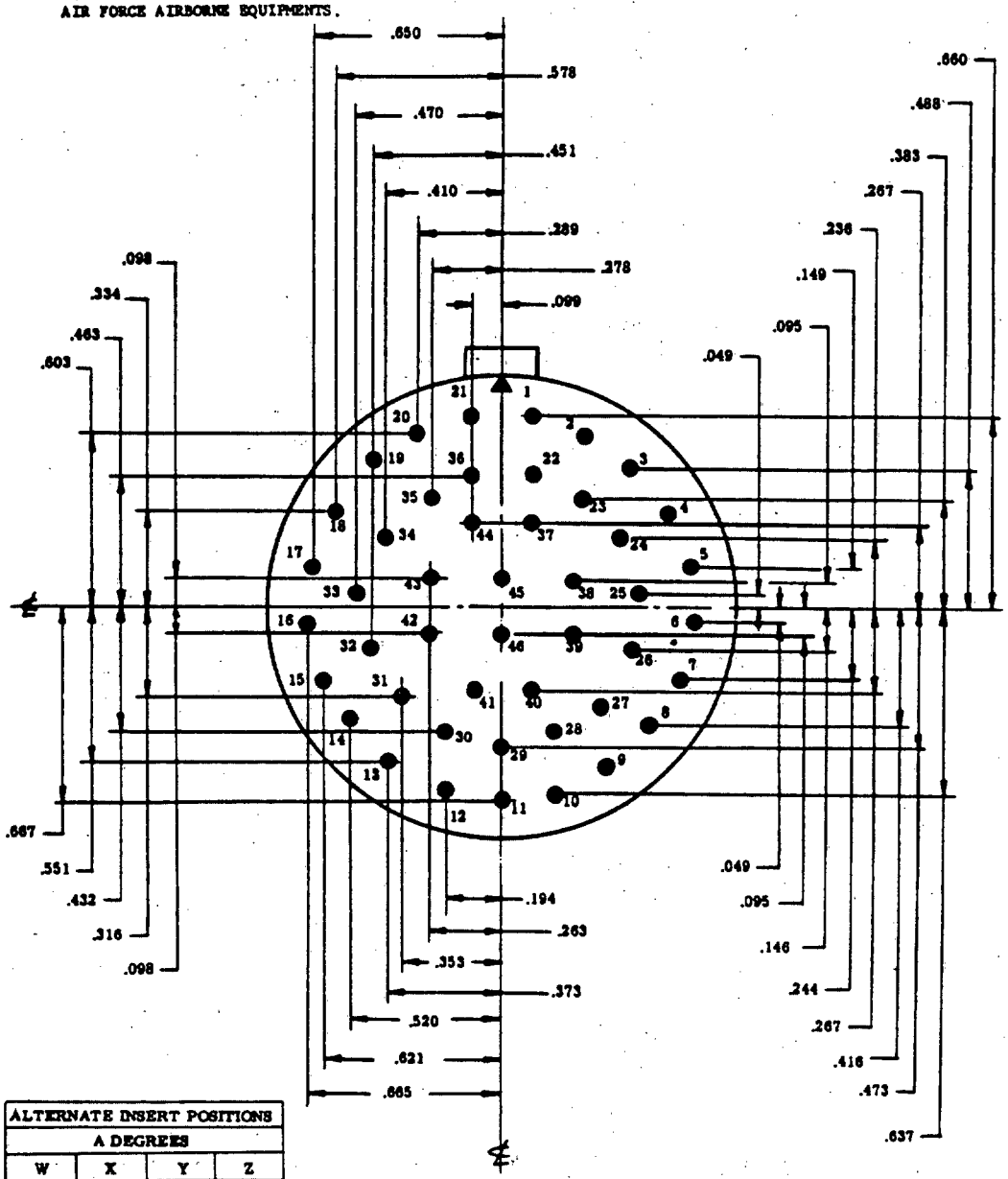
ALTERNATE INSERT POSITION					
A DEGREES					
U	V	W	X	Y	Z
100	260	80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-22	A	54	16	●	A11	MS33689-22

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

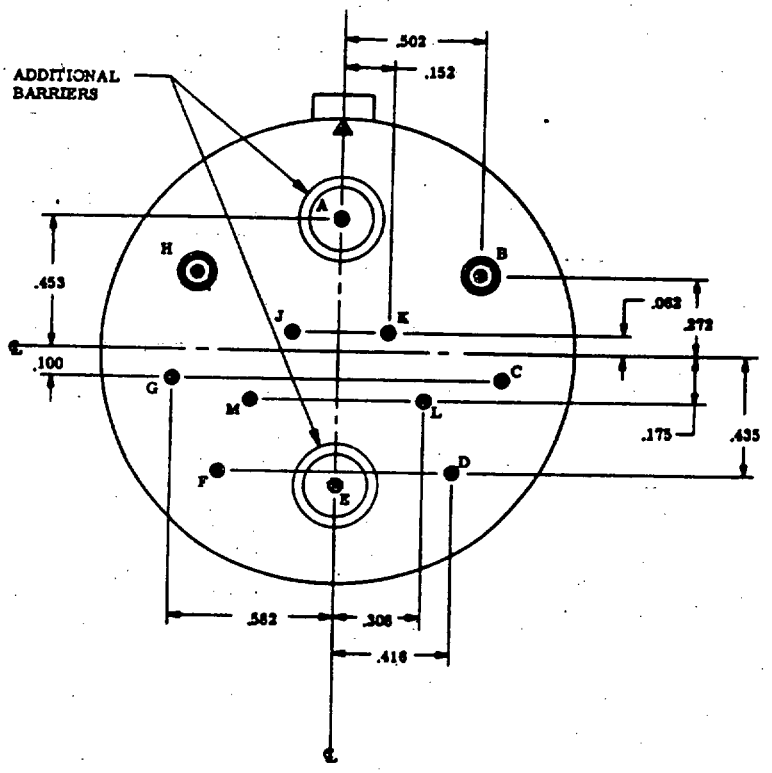
NOT APPLICABLE FOR DESIGN OF
AIR FORCE AIRBORNE EQUIPMENTS.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
36	-	-	-

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-73	A	46	16	●	All	MS33689-73

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

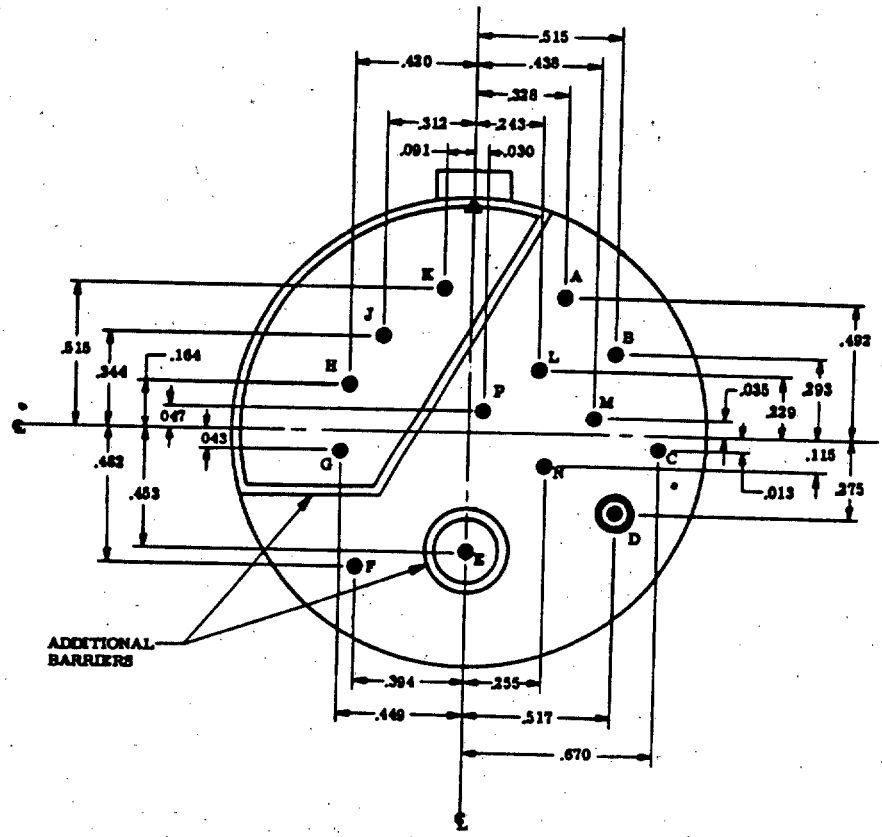


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992 or MIL-C-83723)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-101		2	Coax (MS33695)	◎	B,H	MS33689-101
		C	2	16	●	A,E	
		D	8			All others	

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992 or MIL-C-83723)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-102		1	Coax (MS33695)	◉	D	MS33689-102
		A	6	16	●	C,G,H, J,M,N	
		B	1			E	
		D	6			All others	

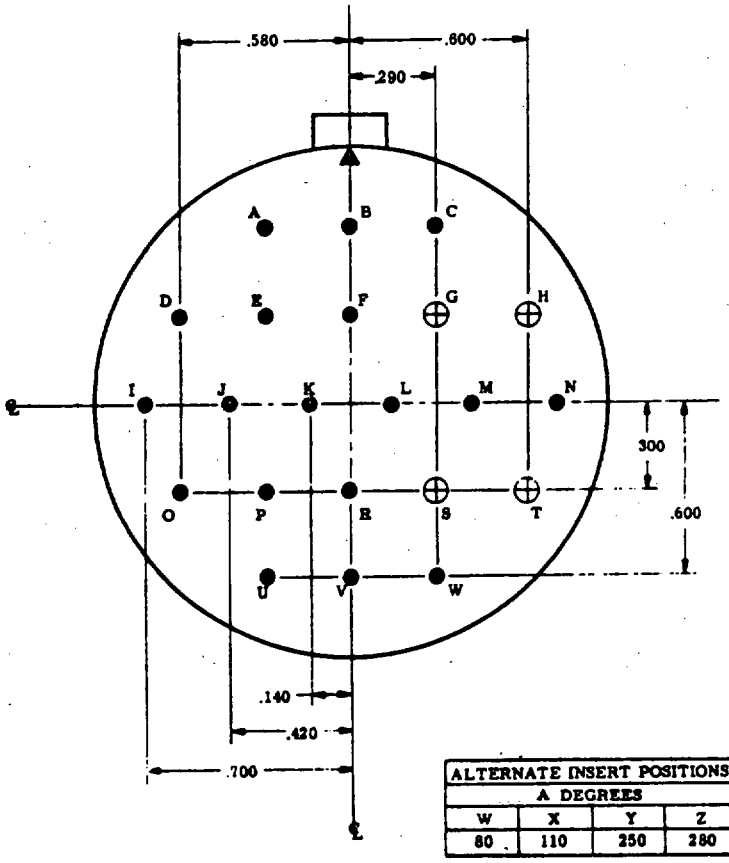
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SECTION 120

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 36

INACTIVE FOR DESIGN
AFTER 13 MAY 1954



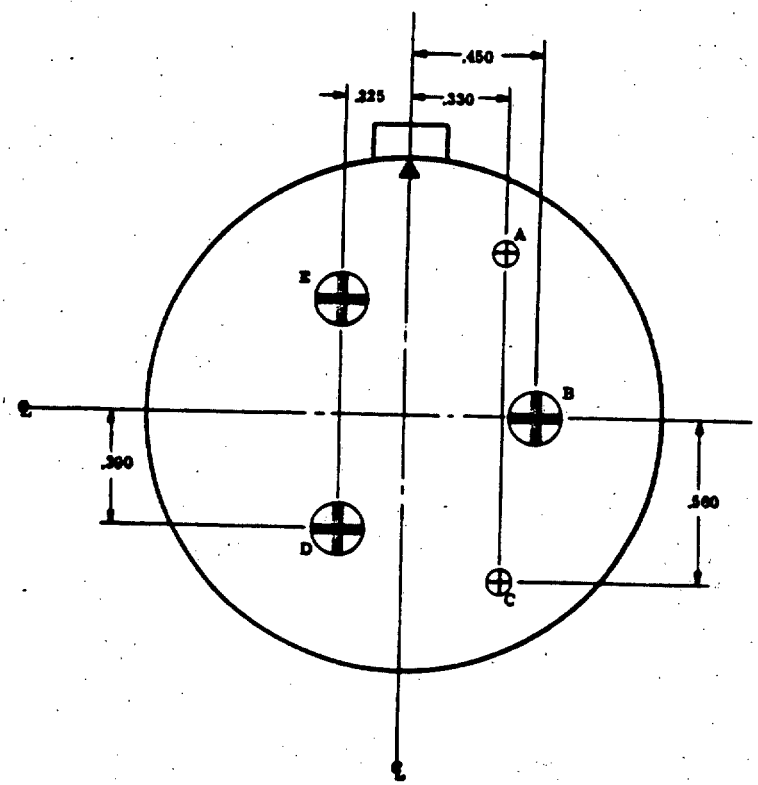
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-1	D	18	16	●	A11	MS33690-1
			4	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 28 DEC 1949

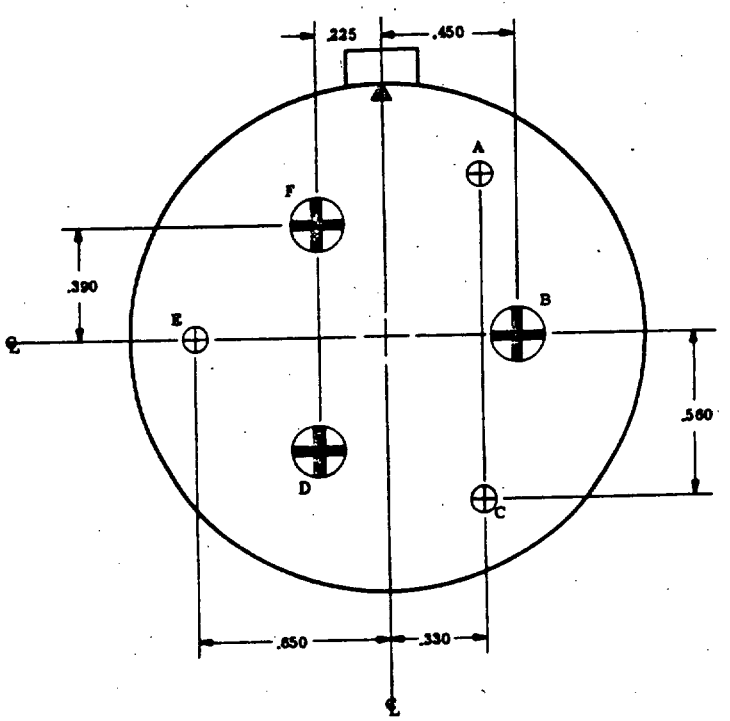


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-2	D	2	12	⊕	All	MS33690-2
			3	0	⊕		

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Check the source to verify that this is the current version before use.

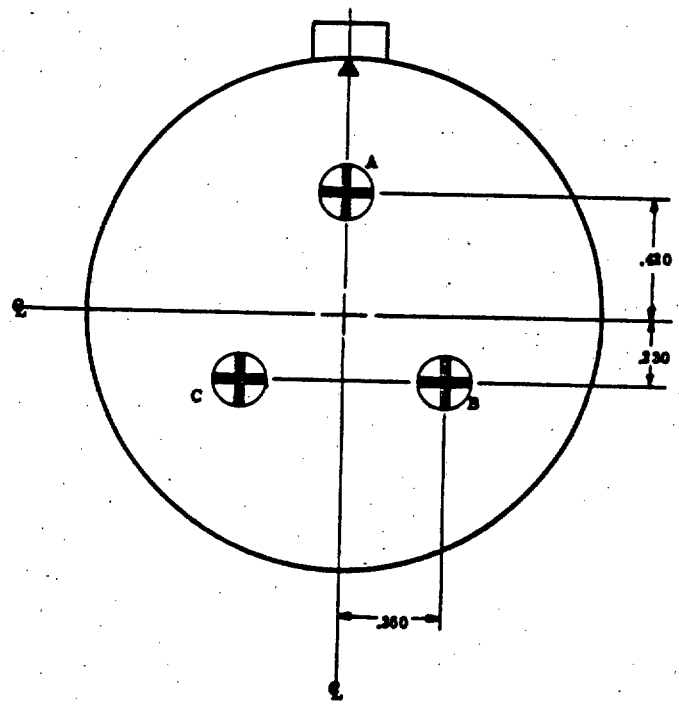


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-3	D	3	12	⊕	A11	MS33690-3
			3	0	⊕		

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INACTIVE FOR DESIGN AFTER 13 MAY 1954

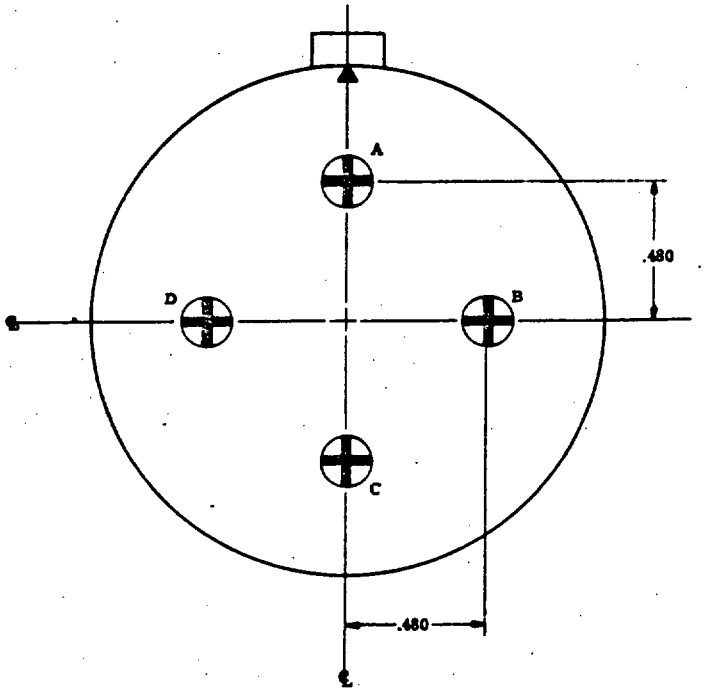


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-4	D	1	0	⊕	A	MS33690-4
		A	2			B,C	

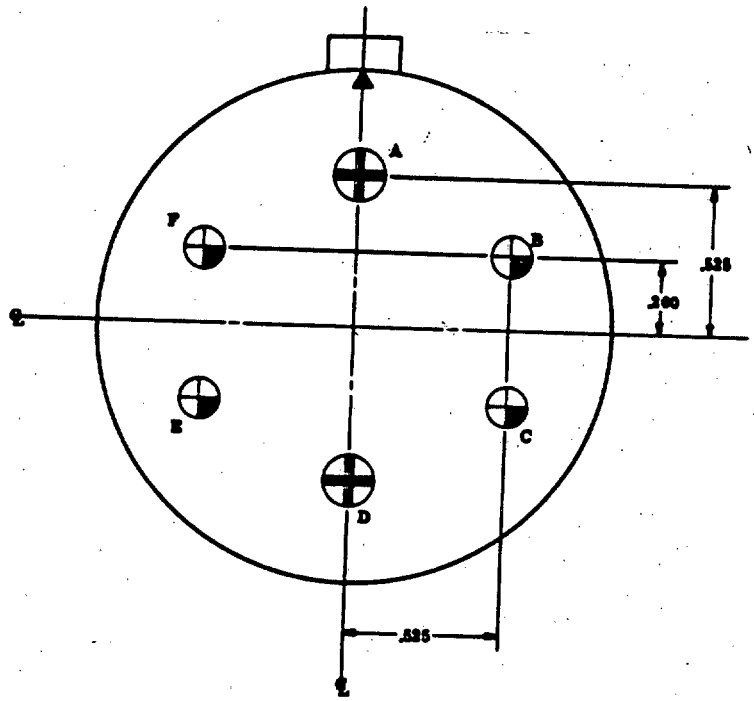
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	120	240	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-5	A	4	0	⊕	A11	MS33690-5

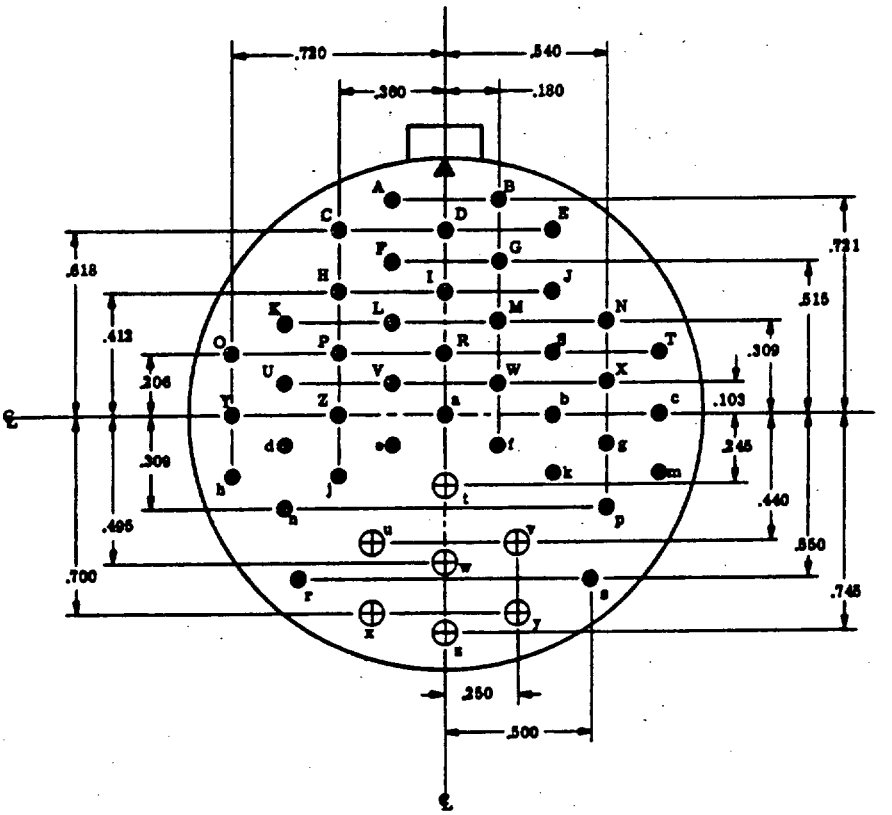
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-6	A	4	4	⊕	A11	MS33690-6
			2	0	⊕		

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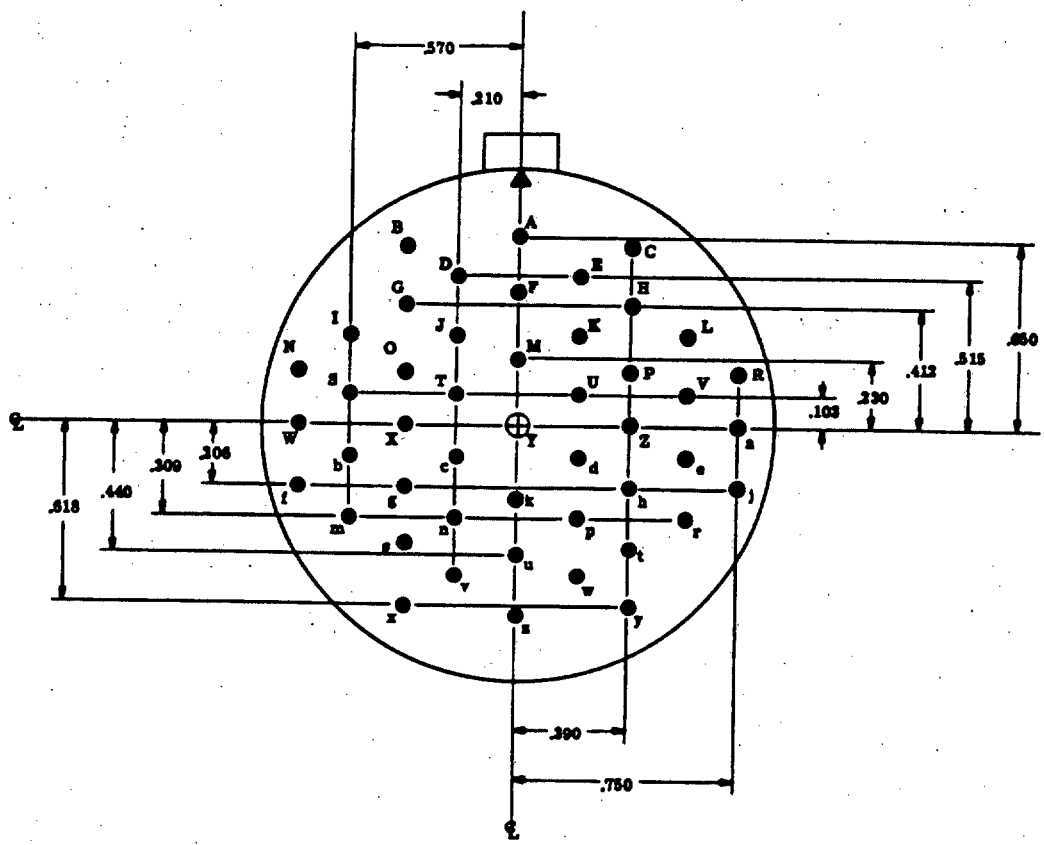
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITIONS SEE 36-16 AND 36-17

* # †

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-7	A	40	16	●	A11	MS3369 -7
			7	12	⊕		

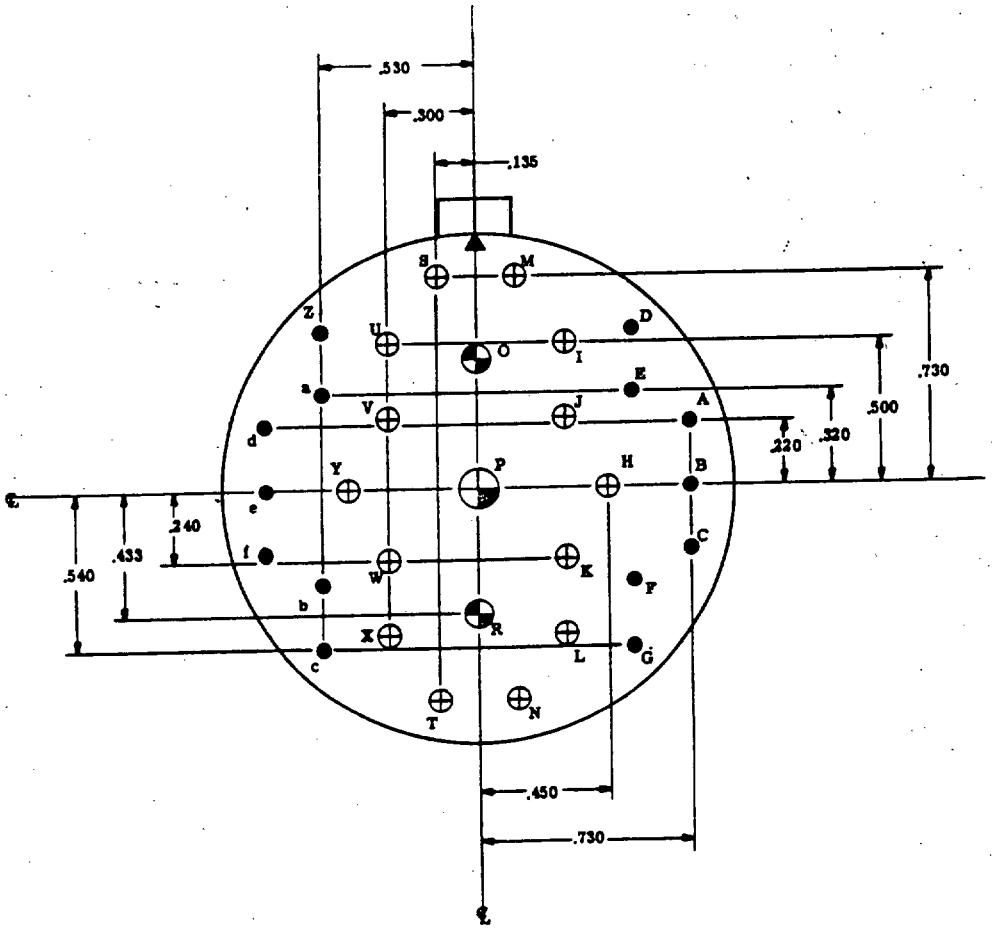
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-8	A	46	16	●	All	MS33690-8
			1	12	⊕		

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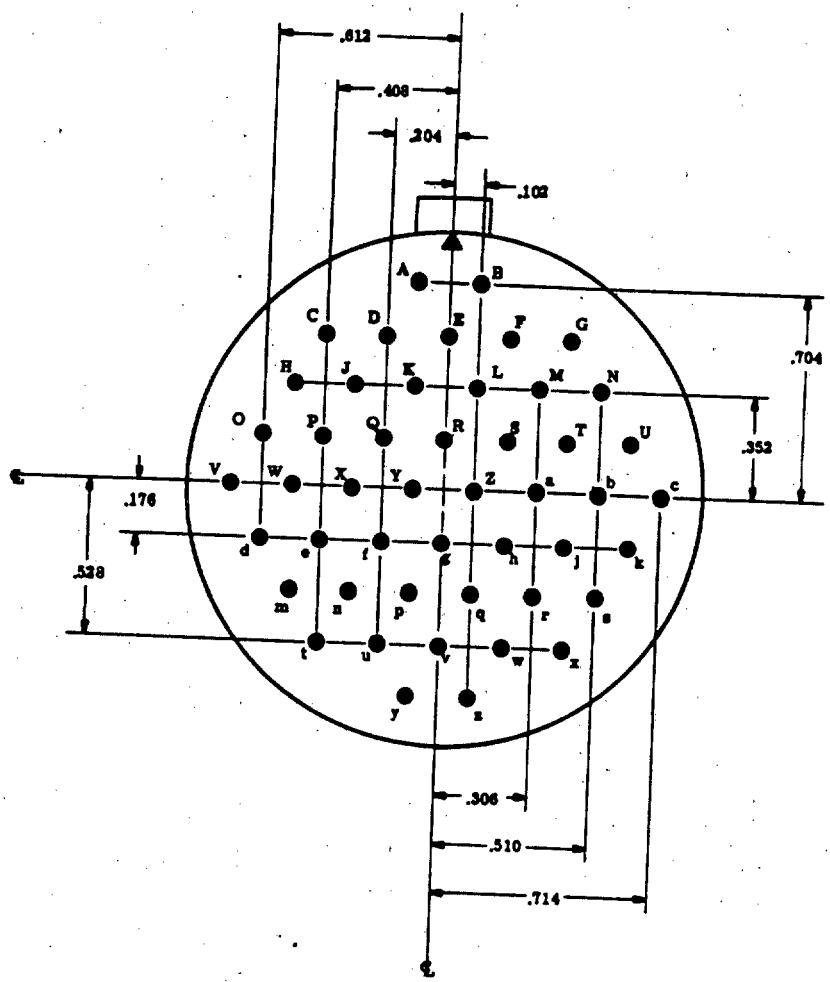


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	125	235	280

FOR OTHER INSERT POSITIONS SEE 36-18 AND 36-21

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-9	A	14	16	●	A11	MS33690-9
			14	12	⊕		
			2	8	◕		
			1	4	⊕		

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Check the source to verify that this is the current version before use.



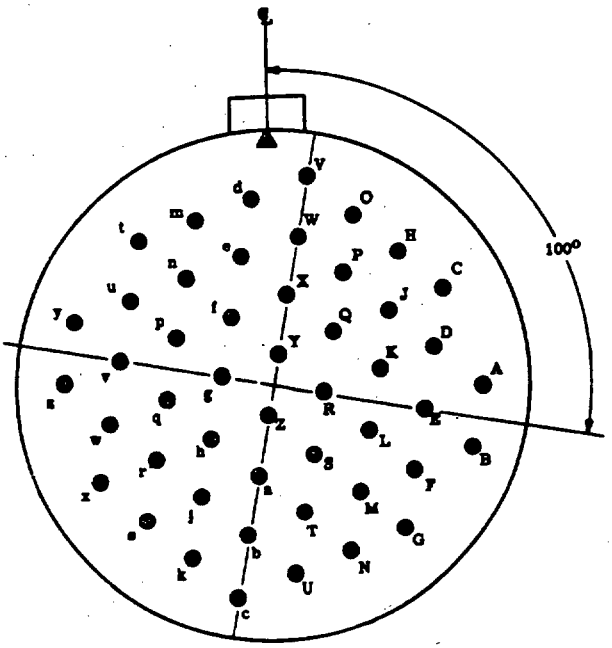
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	125	235	280

FOR OTHER INSERT POSITIONS SEE 36-11 AND 36-12

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-10	A	48	16	●	All	MS33690-10

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INACTIVE FOR DESIGN AFTER 15 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-10 EXCEPT FOR POSITION OF INSERT IN SHELL

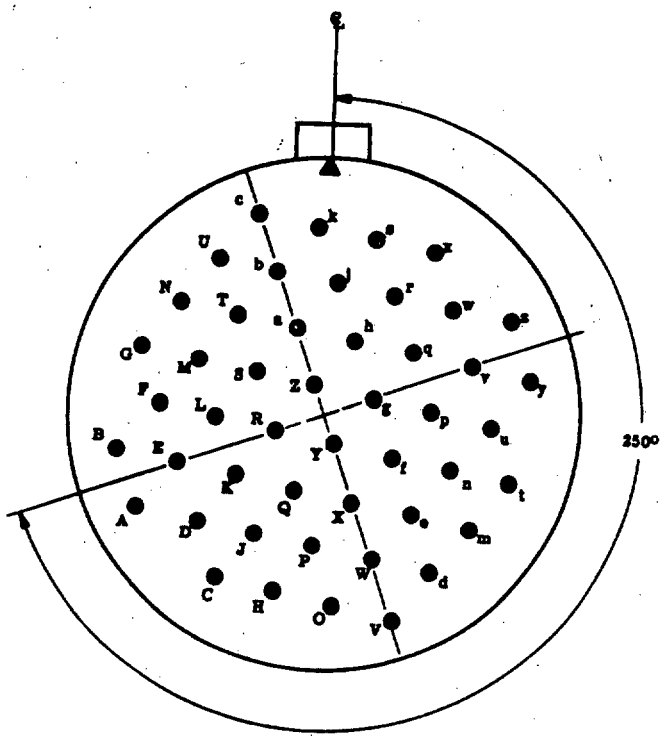
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-11	A	48	16	●	All	MS33690-11

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INACTIVE FOR DESIGN AFTER 15 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-10 EXCEPT FOR POSITION OF INSERT IN SHELL

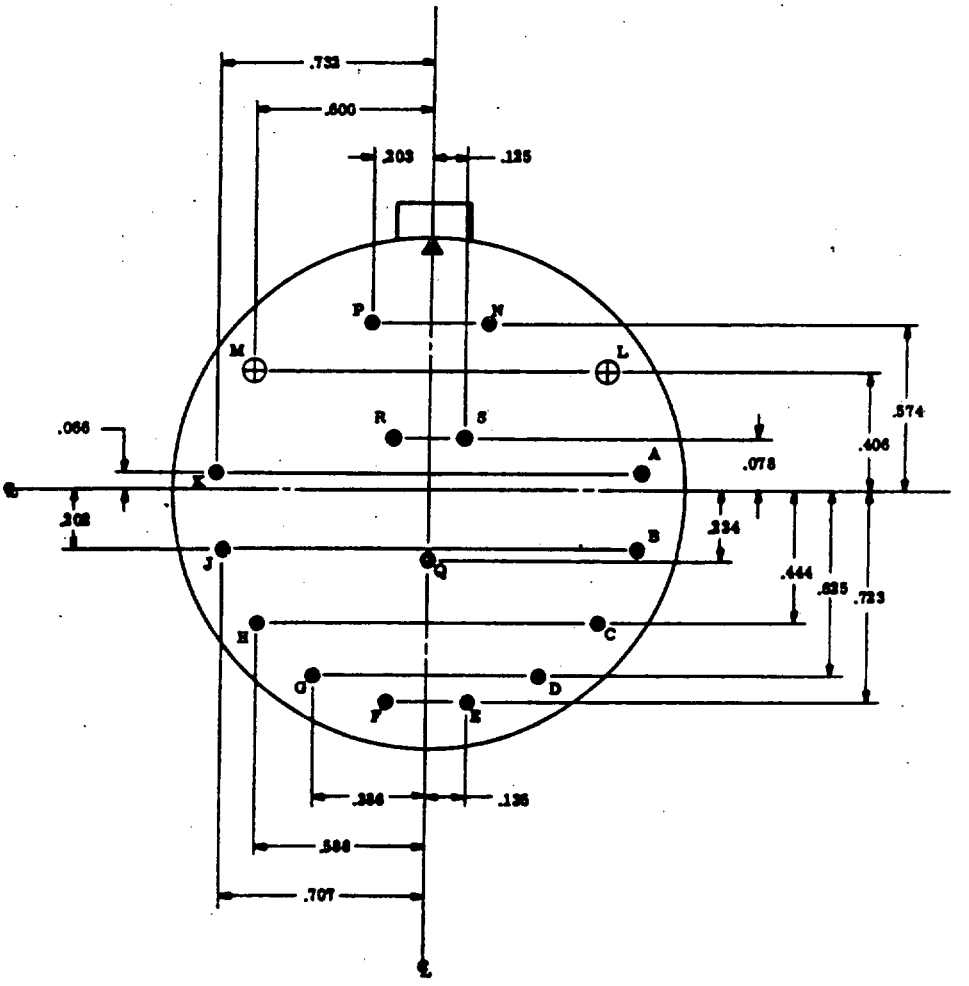
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-12	A	48	16	●	All	MS33690-12

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INACTIVE FOR DESIGN
AFTER 17 DEC 1958



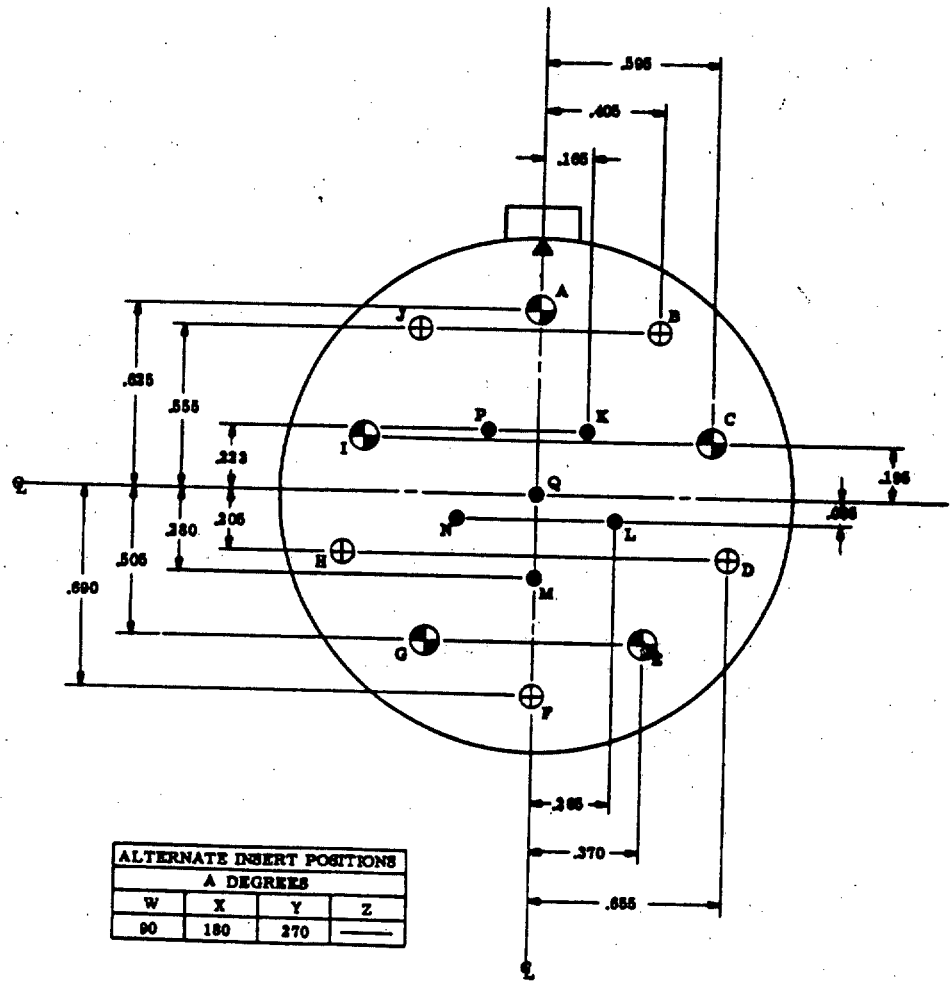
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-13	E	3	16	●	N,P,Q	MS33690-13
		A	12			All others	
			2	12	⊕		

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INACTIVE FOR DESIGN
AFTER 17 DEC 1958

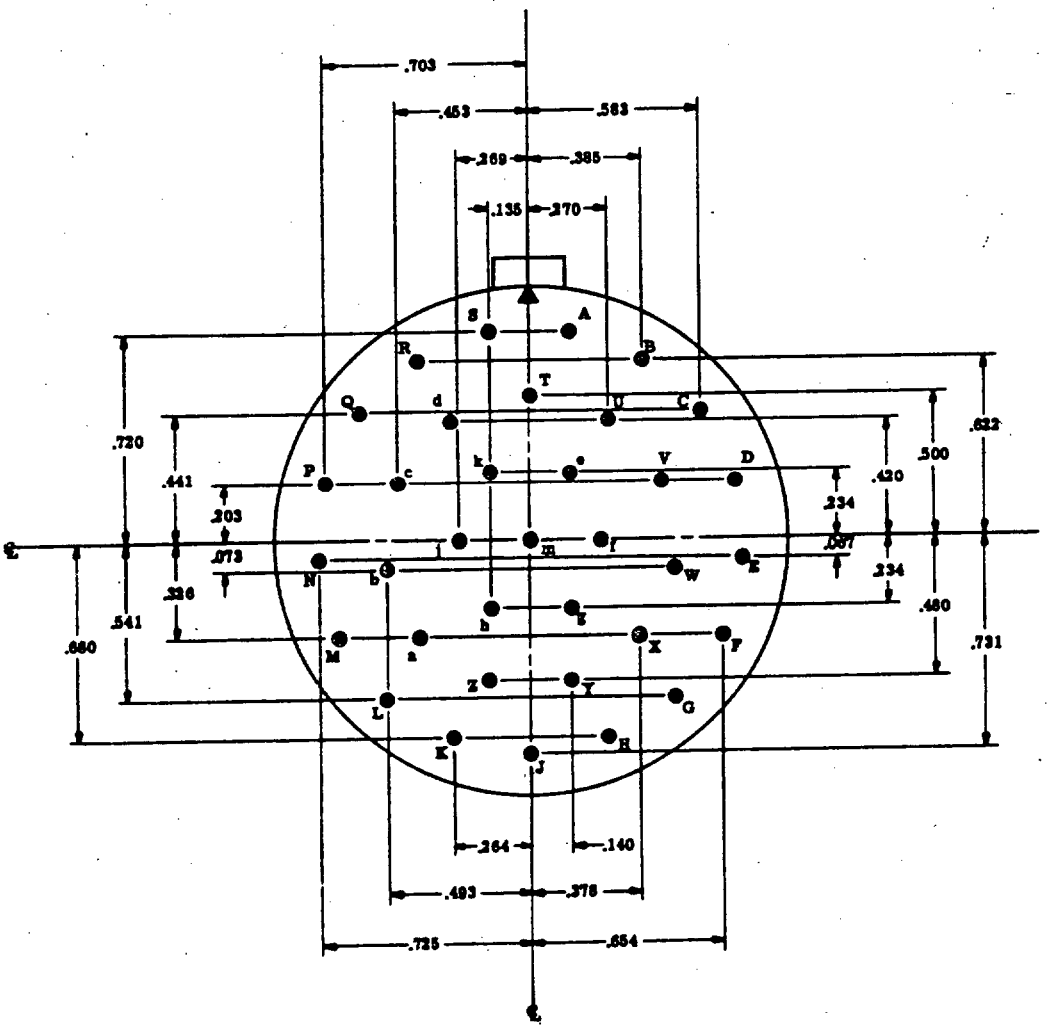


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	——

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-14	D	6	16	●	All	MS33690-14
			5	12	⊕		
			5	8	◕		

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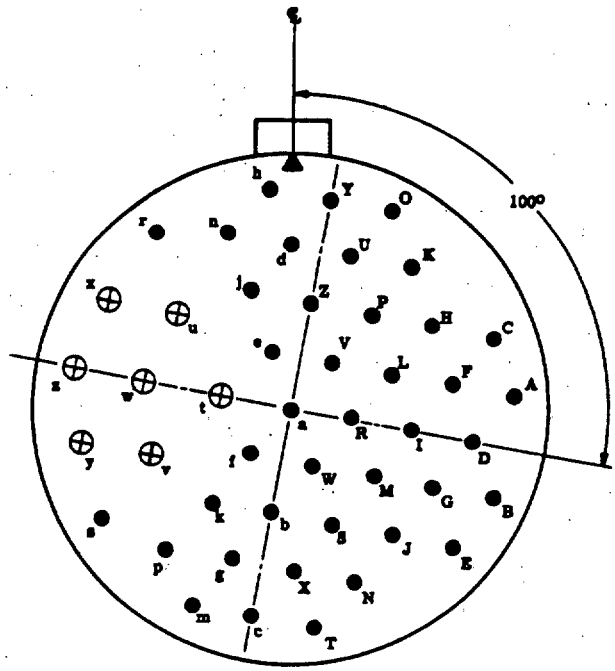


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
60	135	245	305

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-1	D	35	16		M	MS33690-15
		A				All others	

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INACTIVE FOR DESIGN AFTER 15 JUN 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-7 EXCEPT FOR POSITION OF INSERT IN SHELL

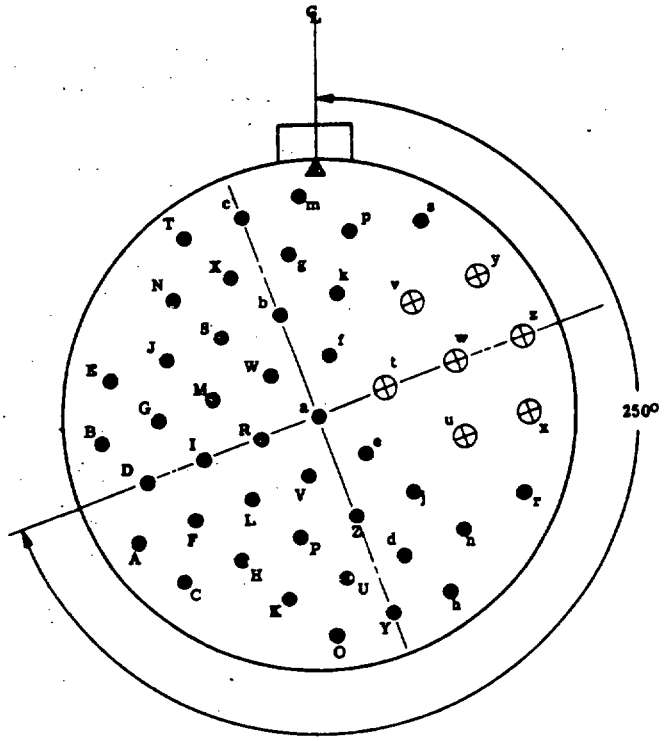
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
———	———	———	———

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-16	A	40	16	●	All	MS33690-16
			7	12	⊕		

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CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-7 EXCEPT FOR POSITION OF INSERT IN SHELL

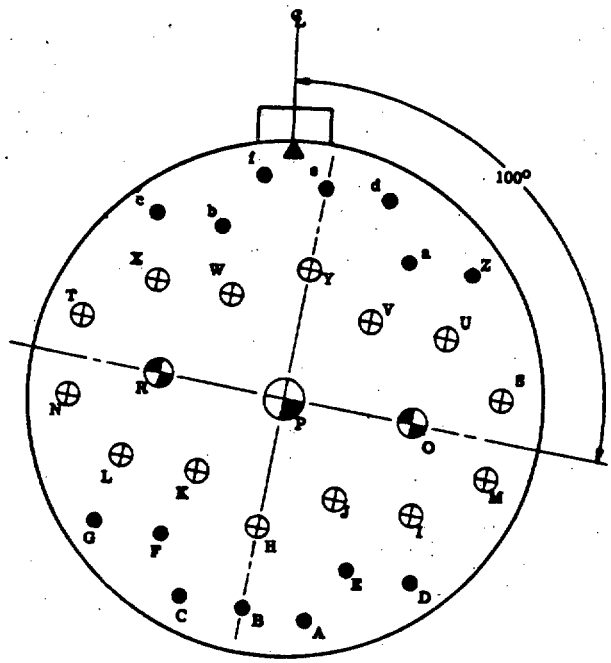
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-17	A	40	16	●	A11	MS33690-17
			7	12	⊕		

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CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-9 EXCEPT FOR POSITION OF INSERT IN SHELL

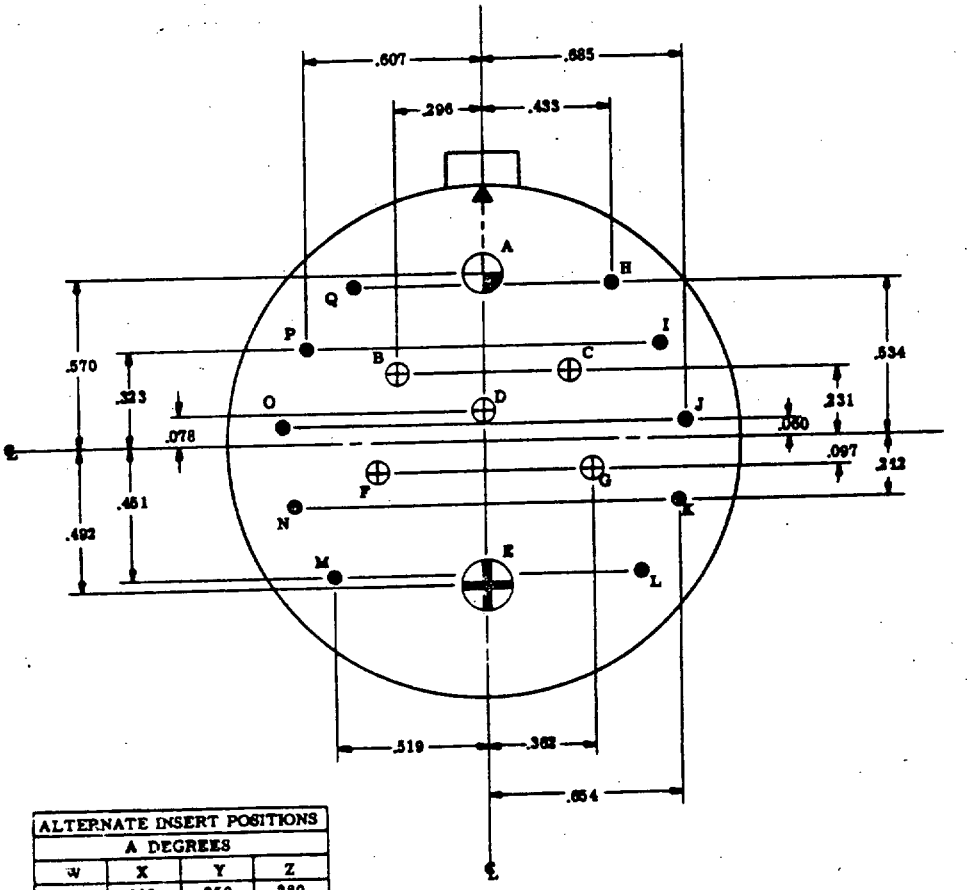
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
———	———	———	———

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-18	A	14	16		All	MS33690-18
			14	12			
			2	8			
			1	4			

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INACTIVE FOR DESIGN
AFTER 17 DEC 1958



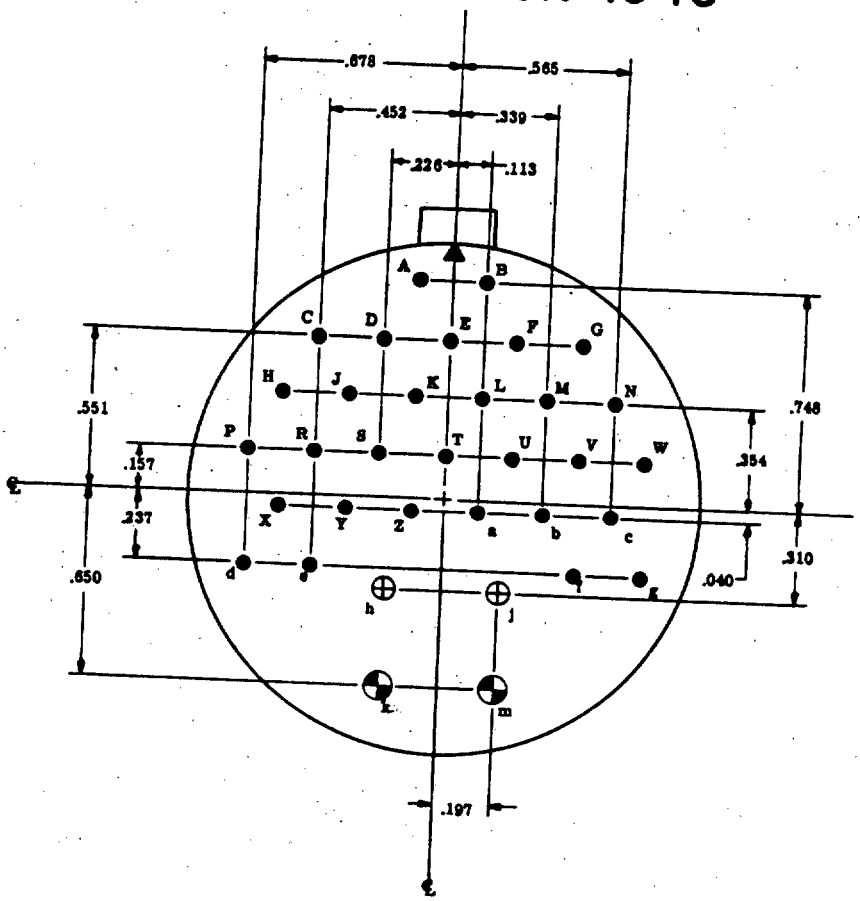
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-19	D	10	16		A11	MS33690-19
			5	12			
			1	4			
			1	0			

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INACTIVE FOR DESIGN
AFTER 15 JUN 1945



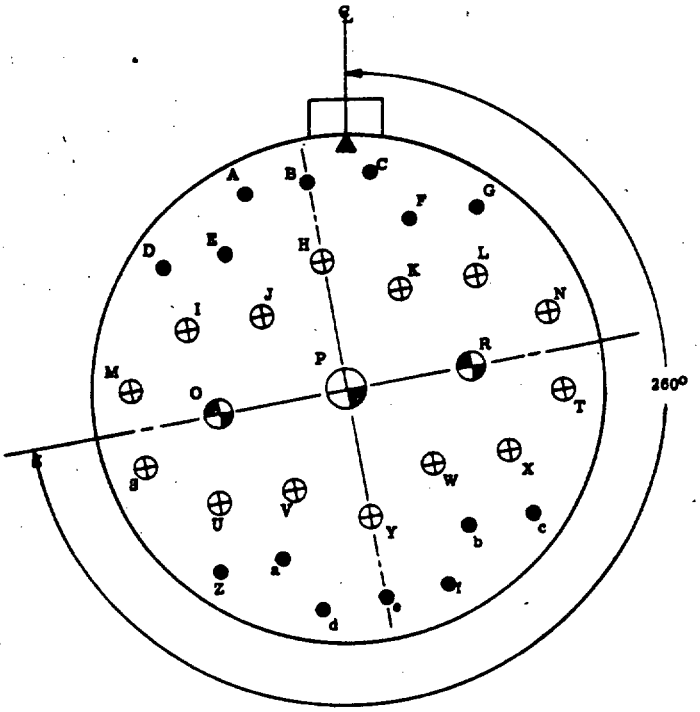
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
——	——	——	——

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-20	A	30	16	●	All	MS33690-20
			2	12	⊕		
			2	8	◒		

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INACTIVE FOR DESIGN AFTER 15 JUN 1945



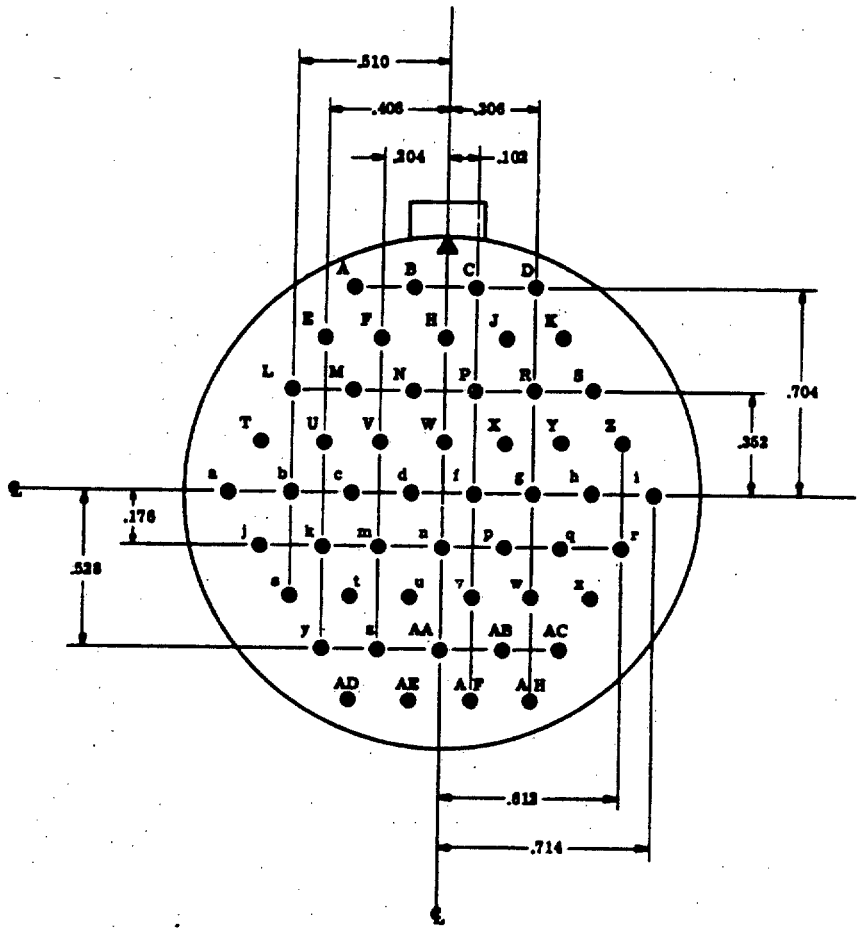
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-9 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-21	A	14	16	●	A11	MS33690-21
			14	12	⊕		
			2	8	◕		
			1	4	⊕		

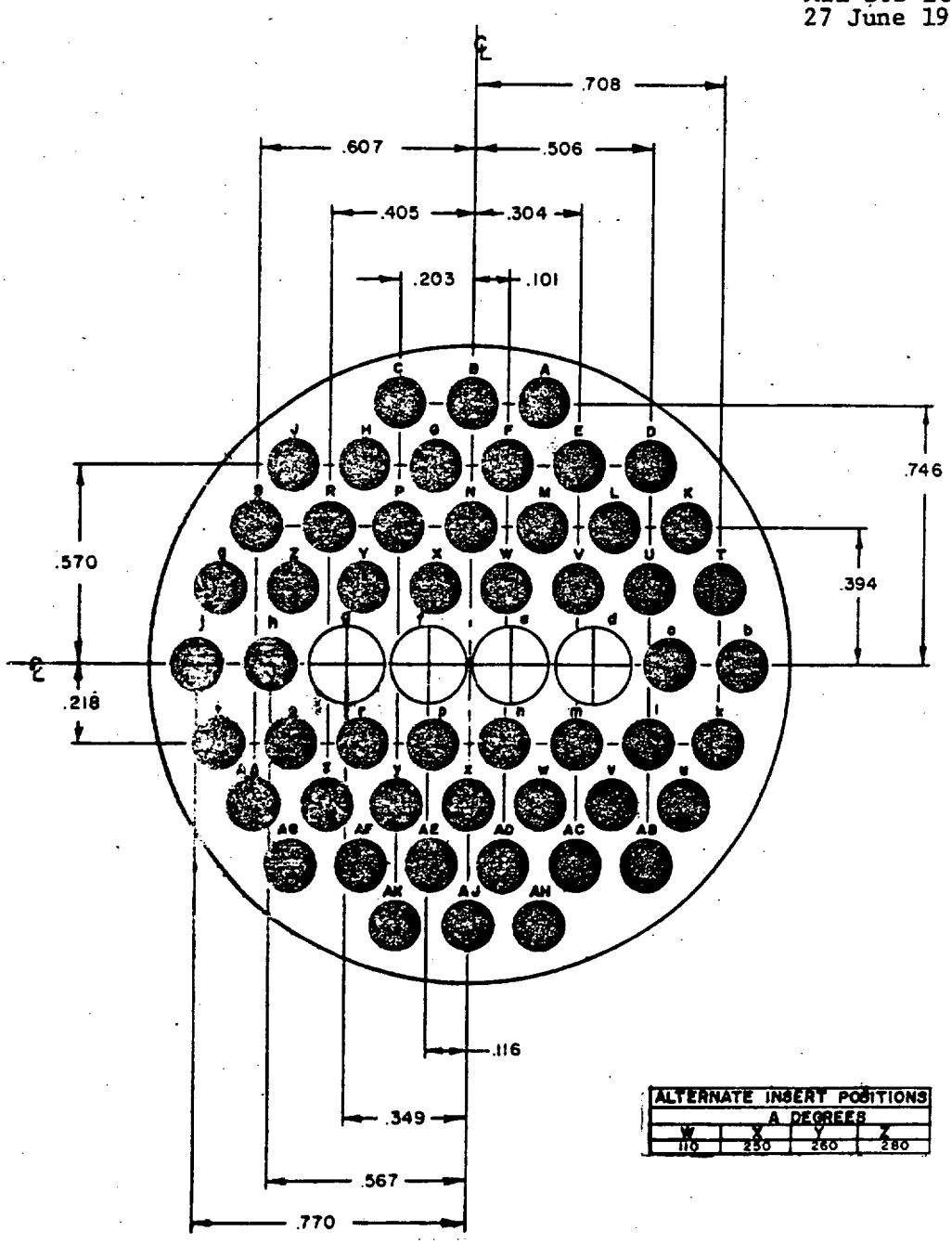
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
72	144	216	288

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-52	A	52	16	●	All	MS33690-52

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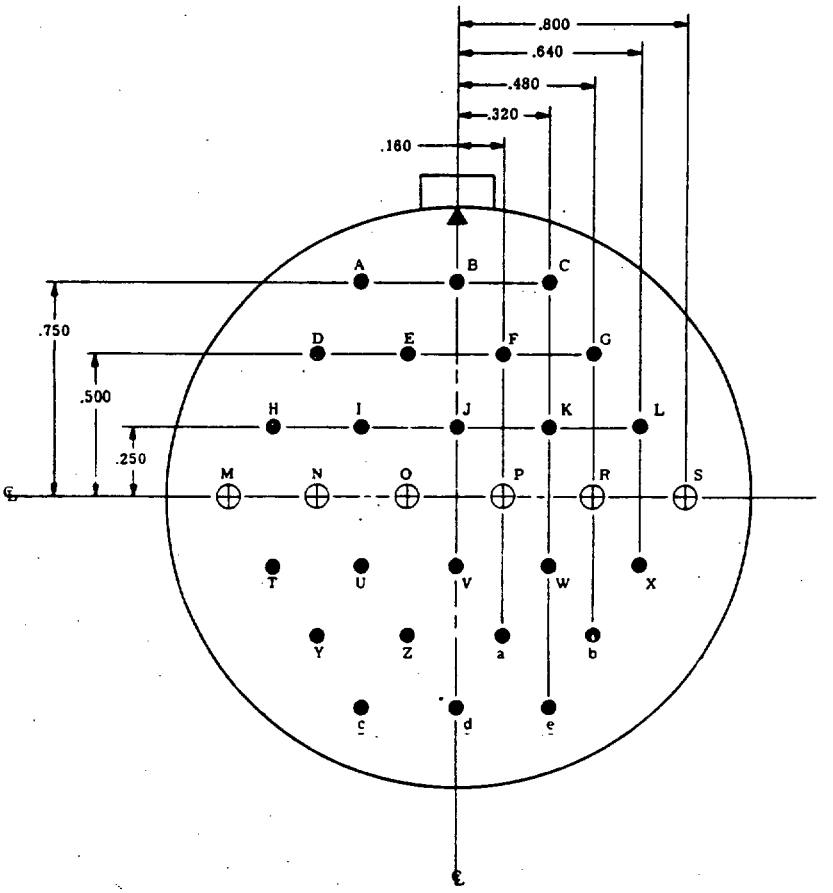
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-66	A	52	16	●	A11	MS33690-66
			4	12	⊕		

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SECTION 130

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

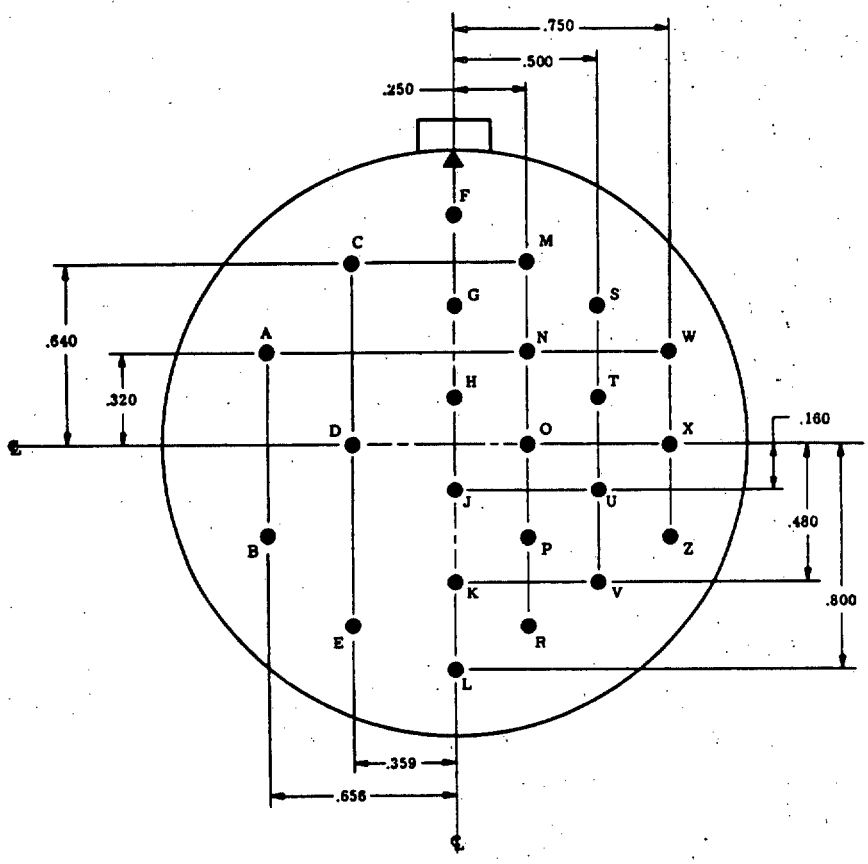
SHELL SIZE 40



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	130	235	300

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-1	D	24	16	●	All	MS33691-1
			6	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



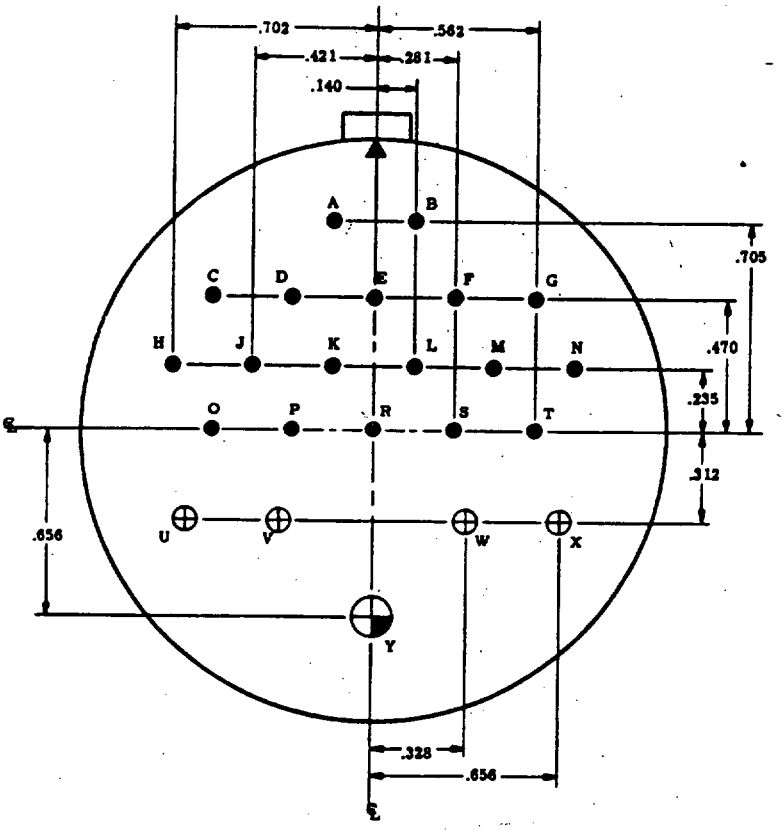
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITION SEE 40-13

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-2	B	5	16		A,B,C,D,E	MS33691-2
		D	18			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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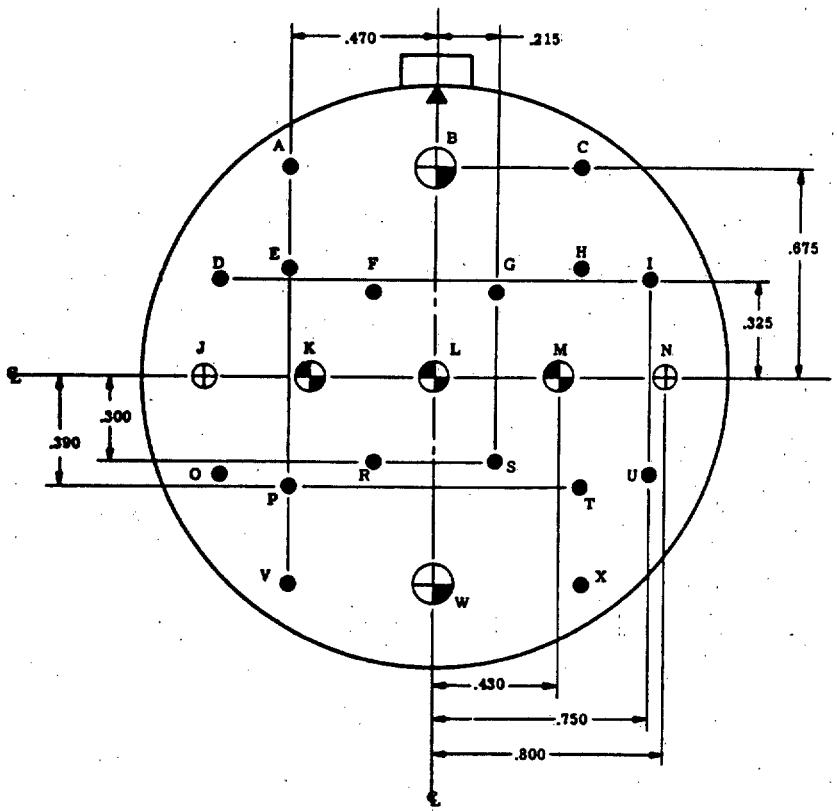


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-3	D	18	16	●	A11	MS33691-3
			4	12	⊕		
			1	4	◒		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

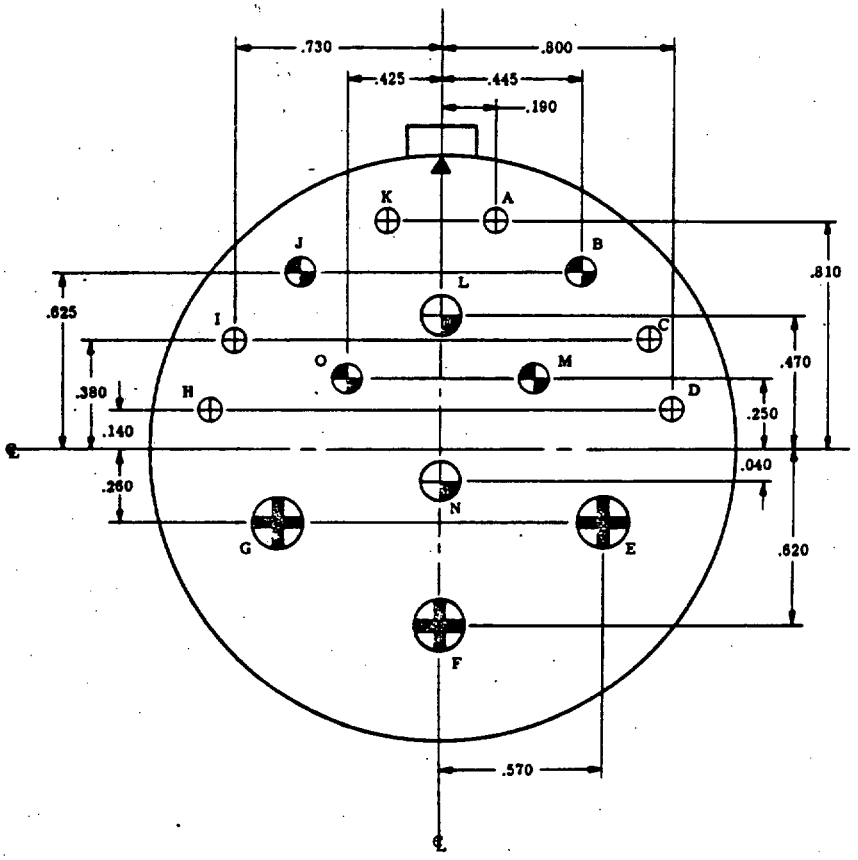


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-4	D	16	16		All	MS33691-4
			2	12			
			3	8			
			2	4			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

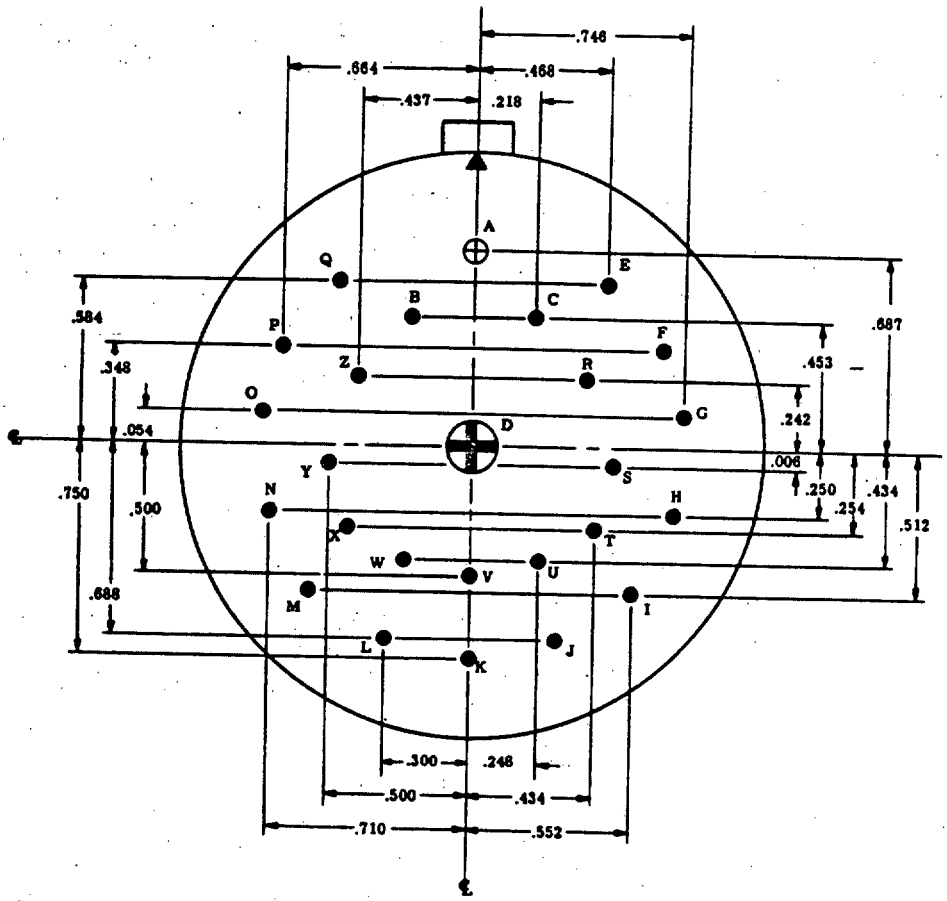


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-5	A	6	12	⊕	A11	MS33691-5
			4	8	◔		
			2	4	◑		
			3	0	◕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

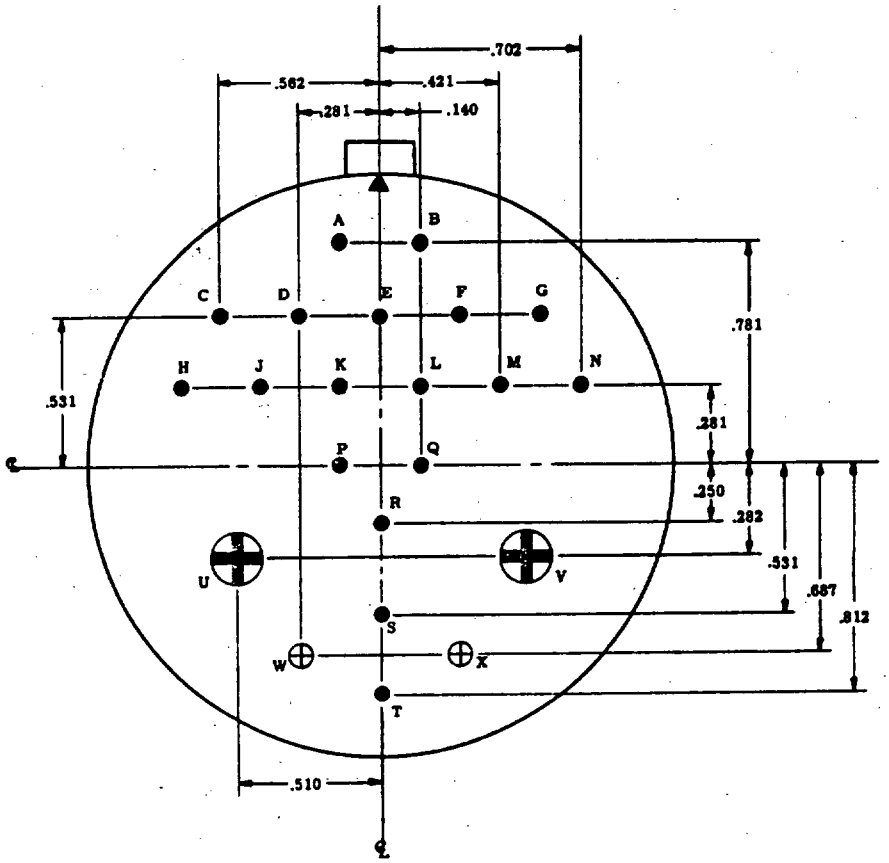


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-6	D	24	16	●	All	MS33691-6
			1	12	⊕		
			1	0	⊕		

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Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-7	A	2	12	⊕	W,X	MS33691-7
			2	0	⊕	U,V	
			2	16	●	P,Q	
		D	16			All others	

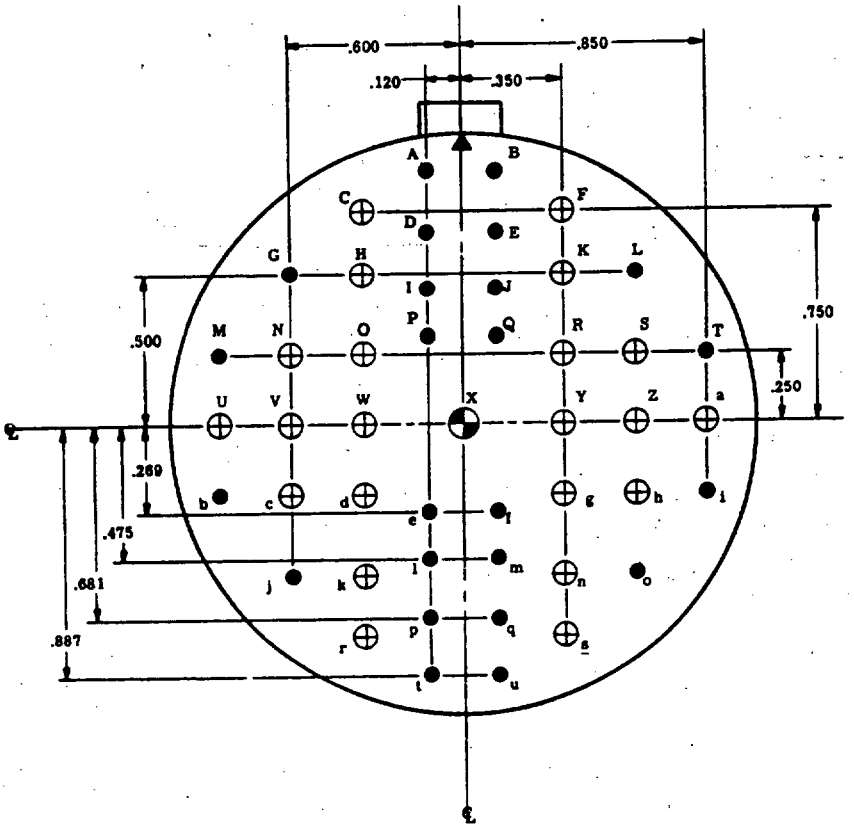
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

THIS ARRANGEMENT NUMBER NOT TO BE USED

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-8	---	---	---	---	---	MS33691-8

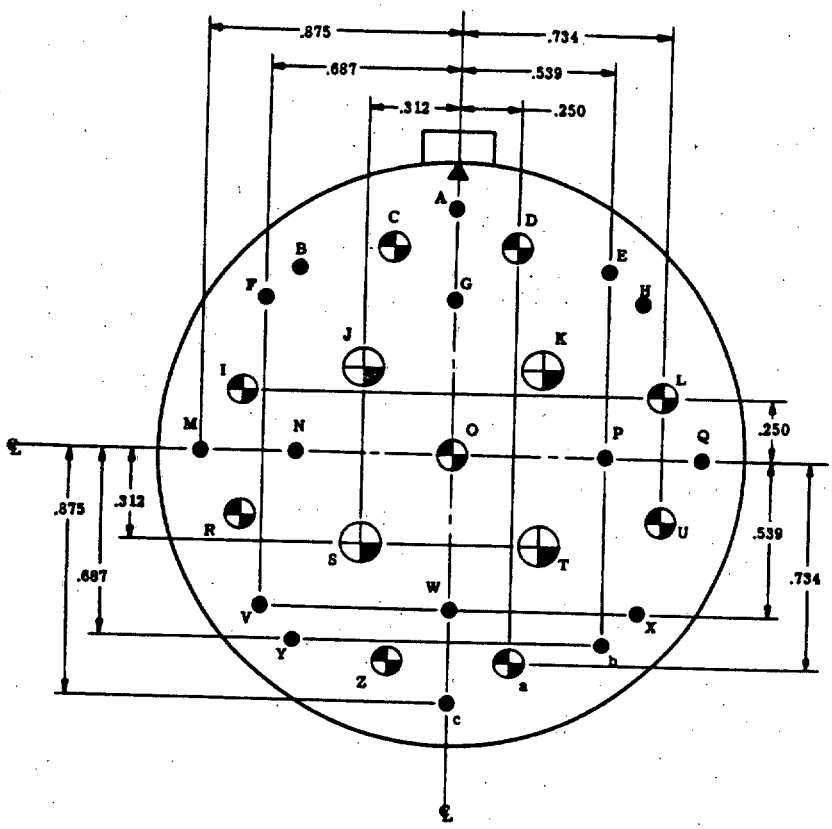
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-9	A	24	16	●	A11	MS33691-9
			22	12	⊕		
			1	8	◔		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

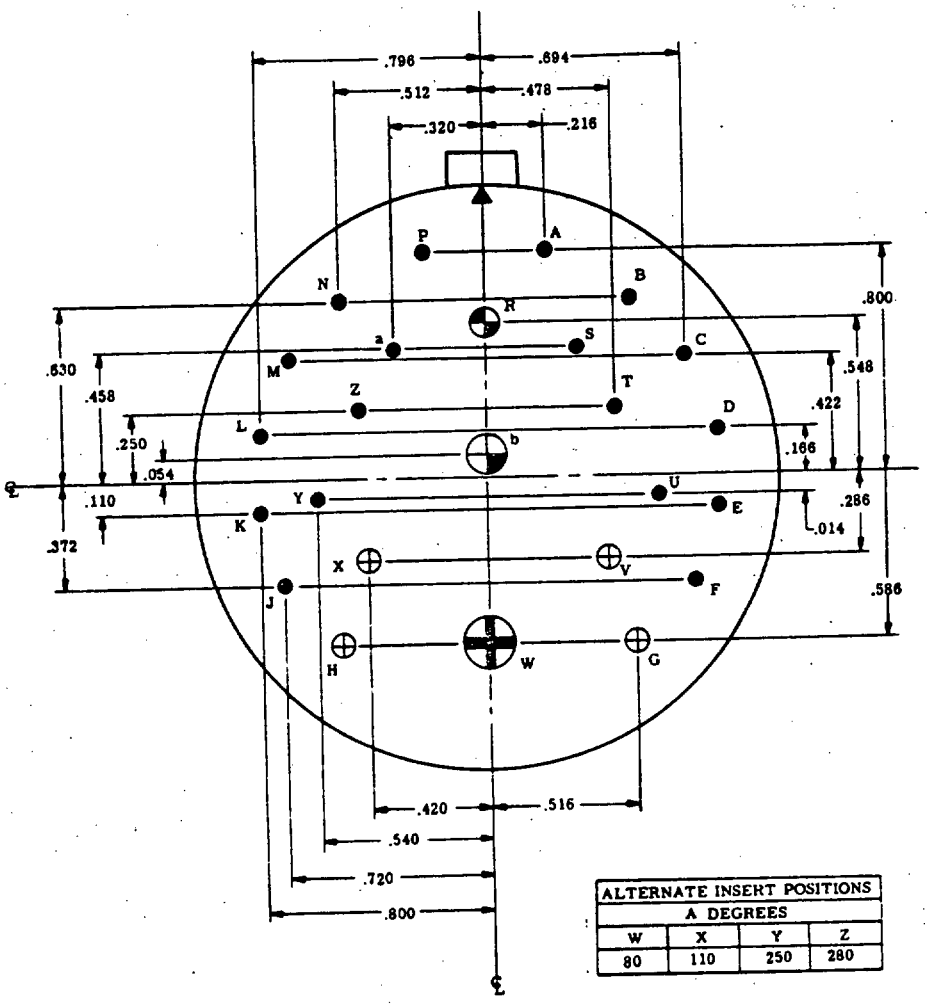


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-10	A	16	16		All	MS33691-10
			9	8			
			4	4			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



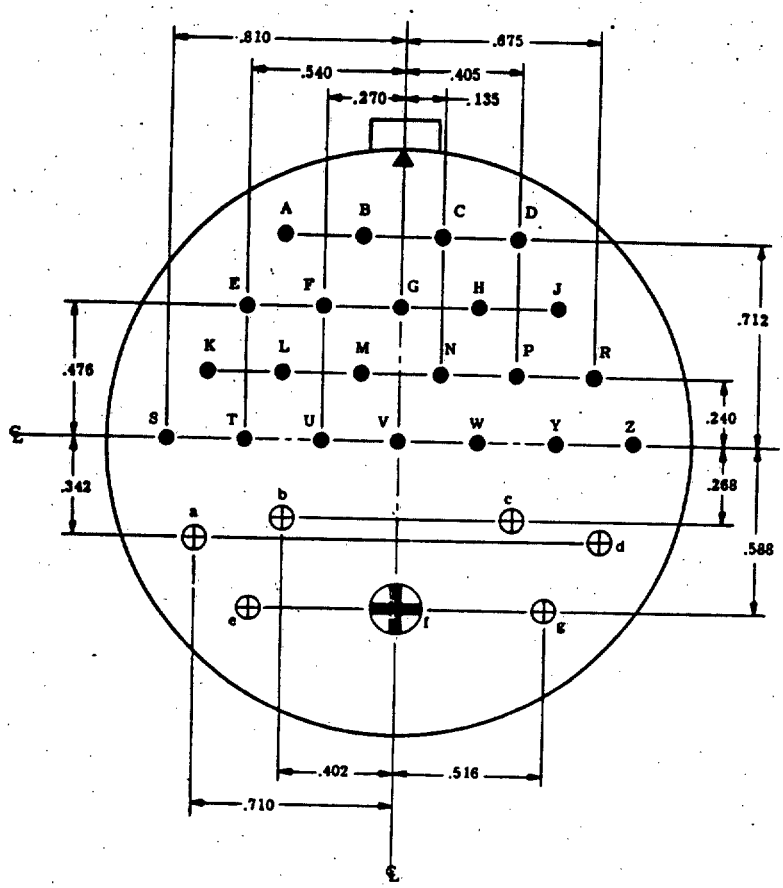
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-11	D	18	16		A11	MS33691-11
			4	12			
			1	8			
			1	4			
			1	0			

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INACTIVE FOR DESIGN AFTER 30 MAY 1945



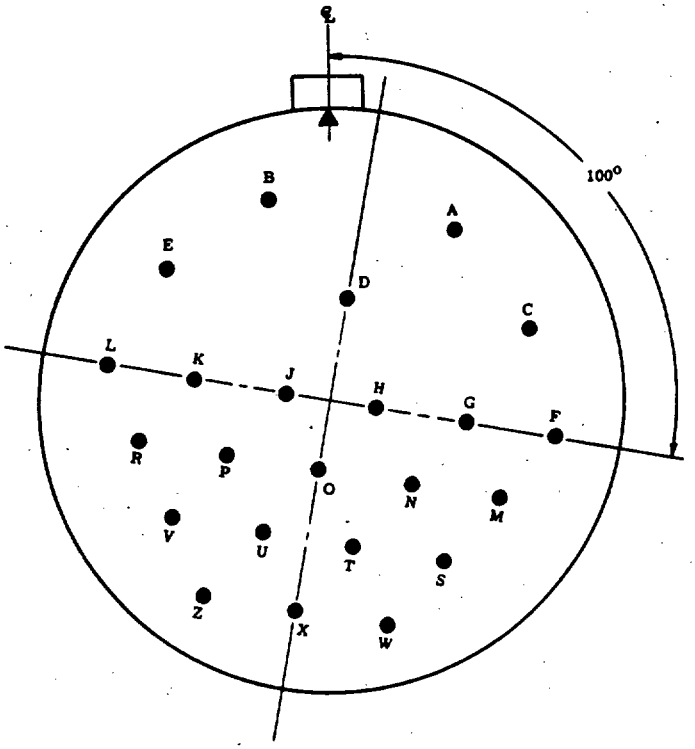
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-12	D	22	16	●	All	MS33691-12
			6	12	⊕		
			1	0	⊕		

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INACTIVE FOR DESIGN AFTER 30 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 40-2 EXCEPT FOR POSITION OF INSERT IN SHELL

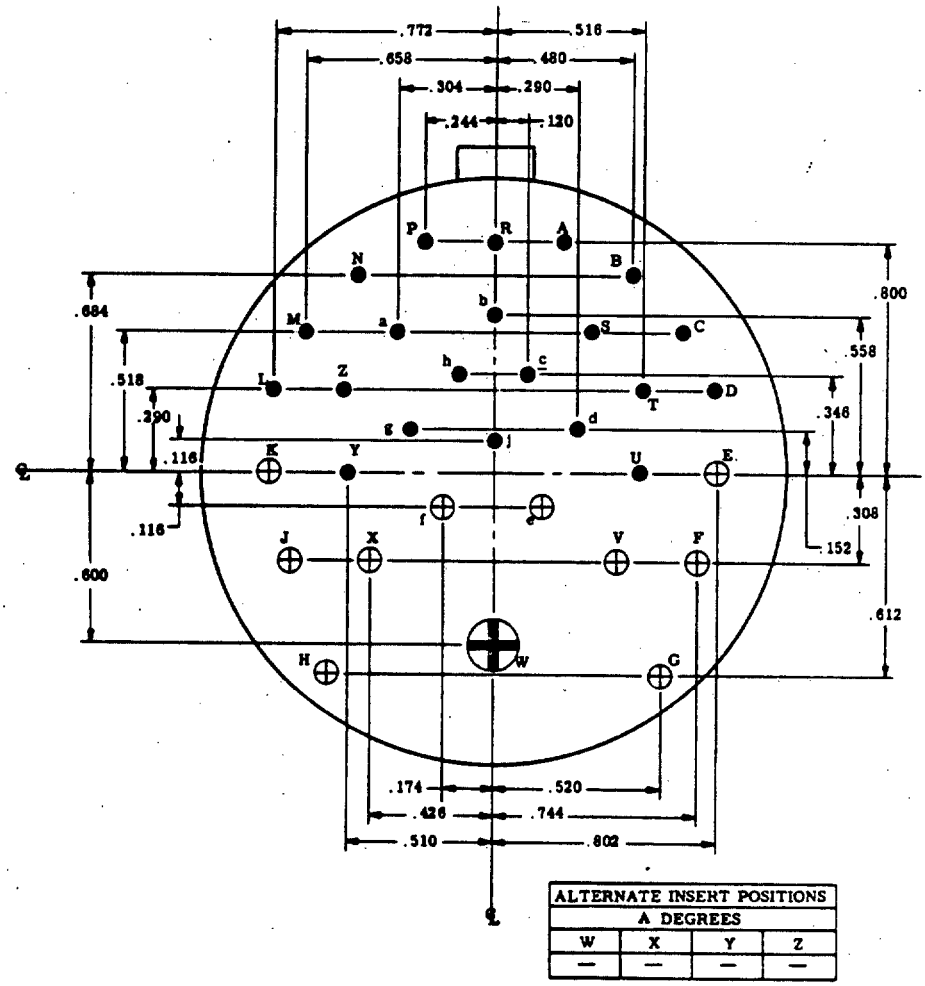
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-13	B	5	16	●	A,B,C,D,E	MS33691-13
		D	18			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 30 MAY 1945

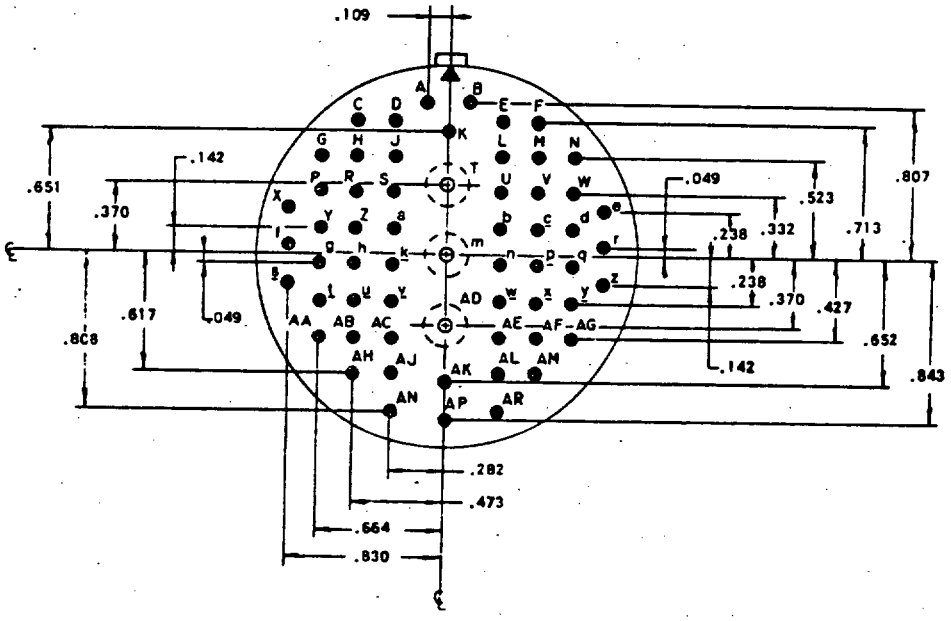


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-14	D	21	16		All	MS33691-14
			10	12			
			1	0			

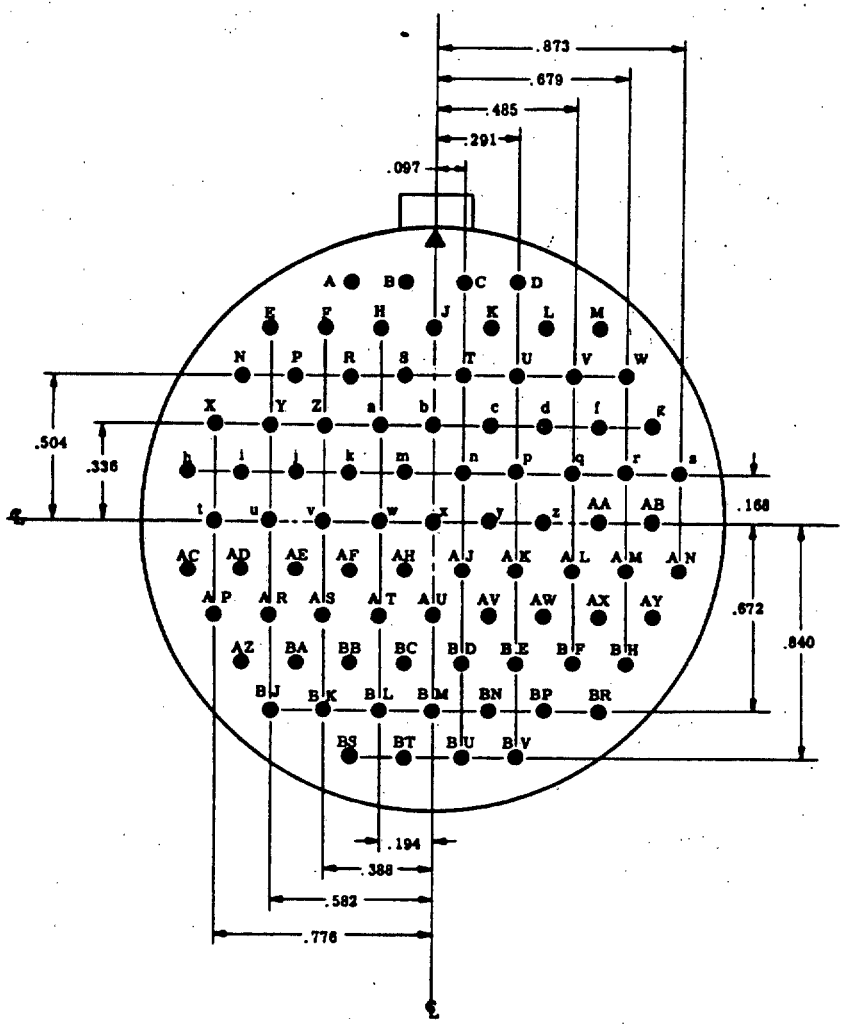
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-15		3	8 (shielded) MS14056-2		T,m,AD	MS33691-15
		INST	57	16		All others	

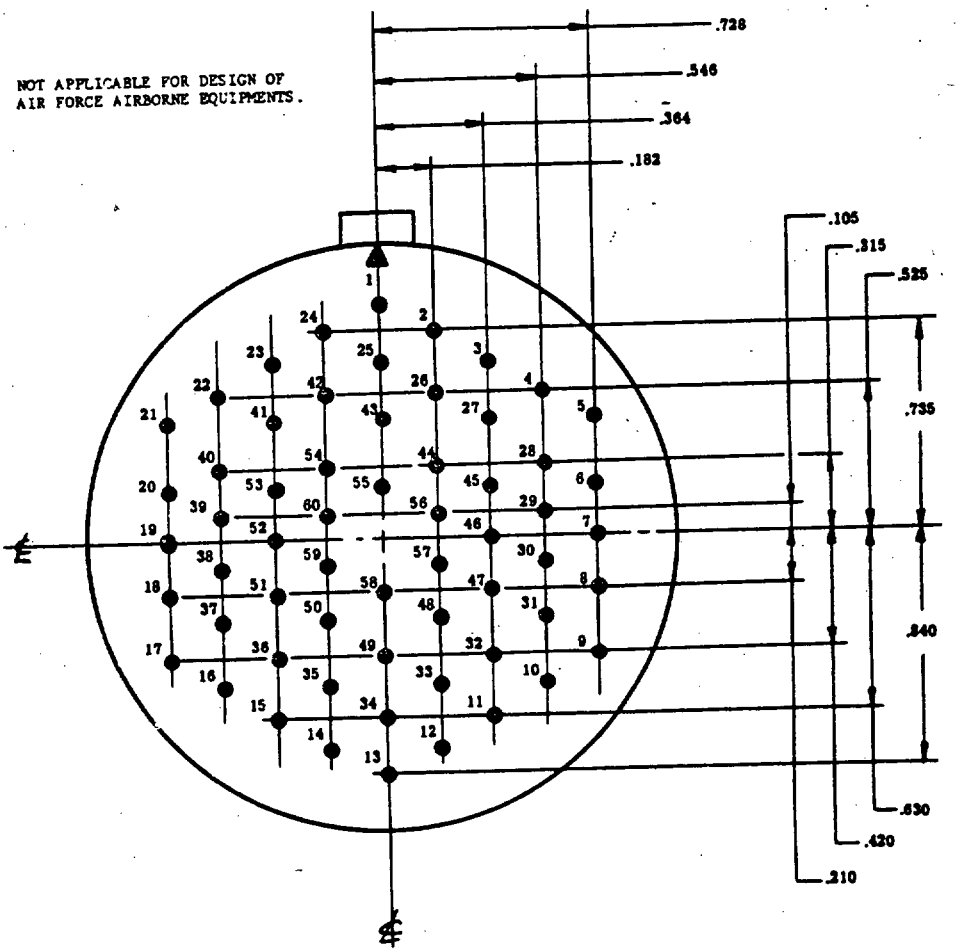
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
72	144	216	288

Shell size	Arrangement number	Service rating.	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-56	A	85	16		A11	MS33691-56

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
30	130	220	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-62	A	60	16		A11	MS33691-62

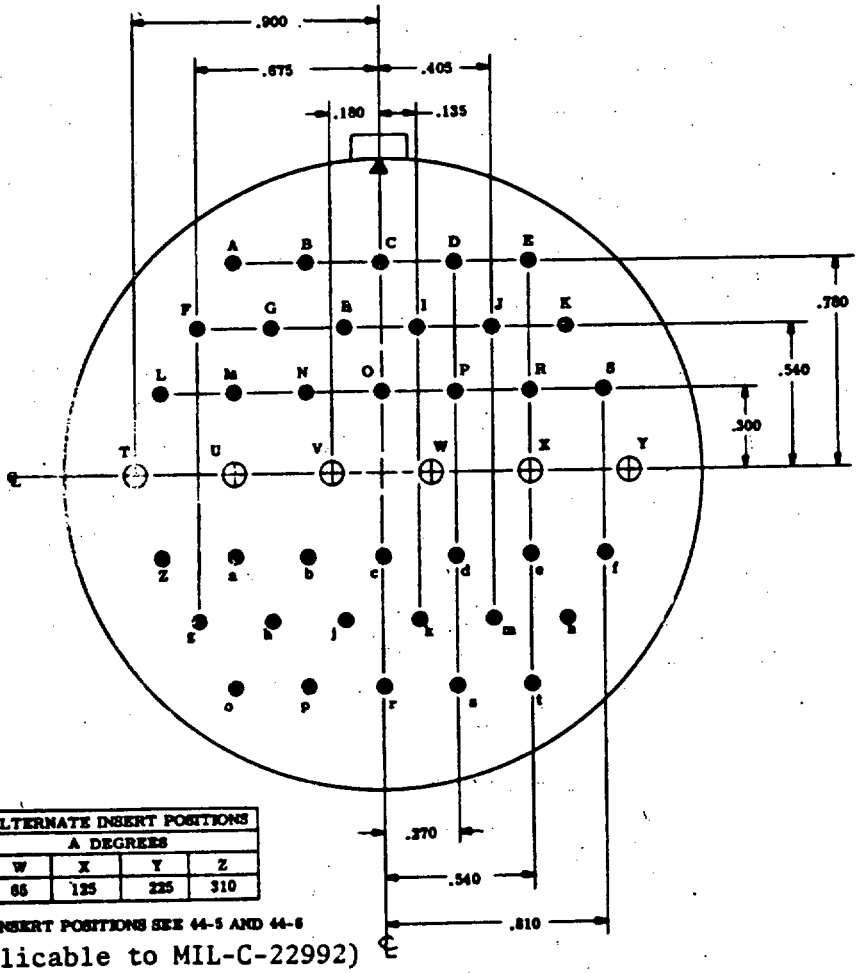
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SECTION 140

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 44

NOTE:
1. Shell size 44 applies to MIL-C-5015, except arrangement number 52 applies also to MIL-C-22992.



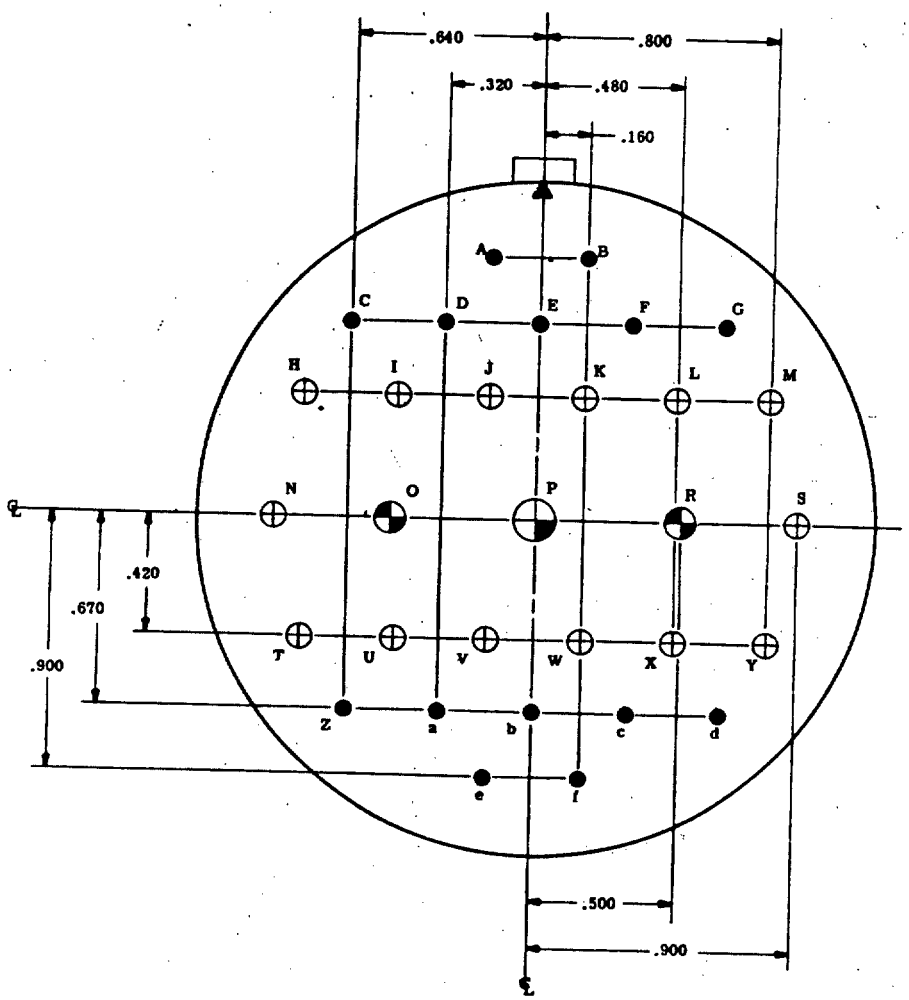
ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
65	125	225	310

FOR OTHER INSERT POSITIONS SEE 44-5 AND 44-6

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-1	D	36	16	●	All	MS33692-1
			6	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

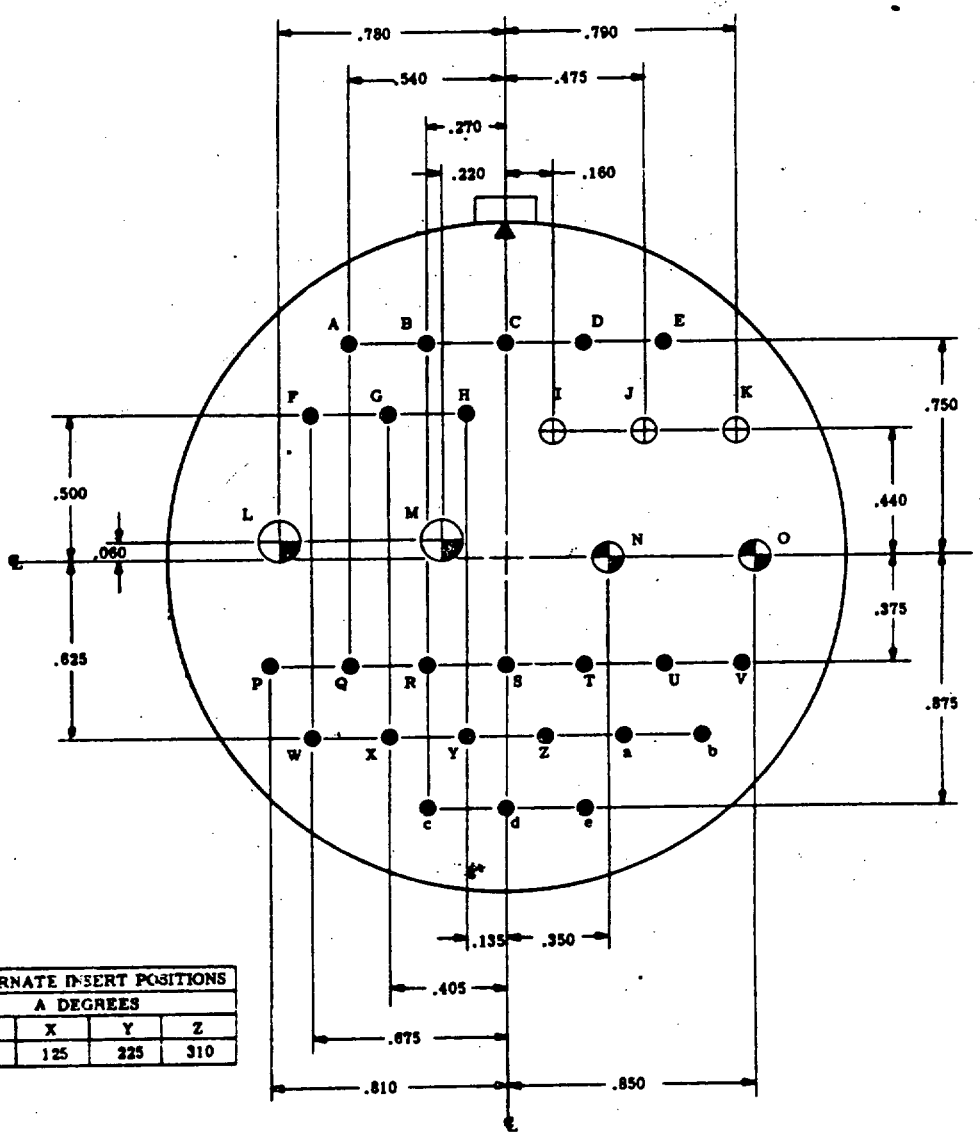


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-2	D	14	16		All	MS33692-2
			14	12			
			2	8			
			1	4			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

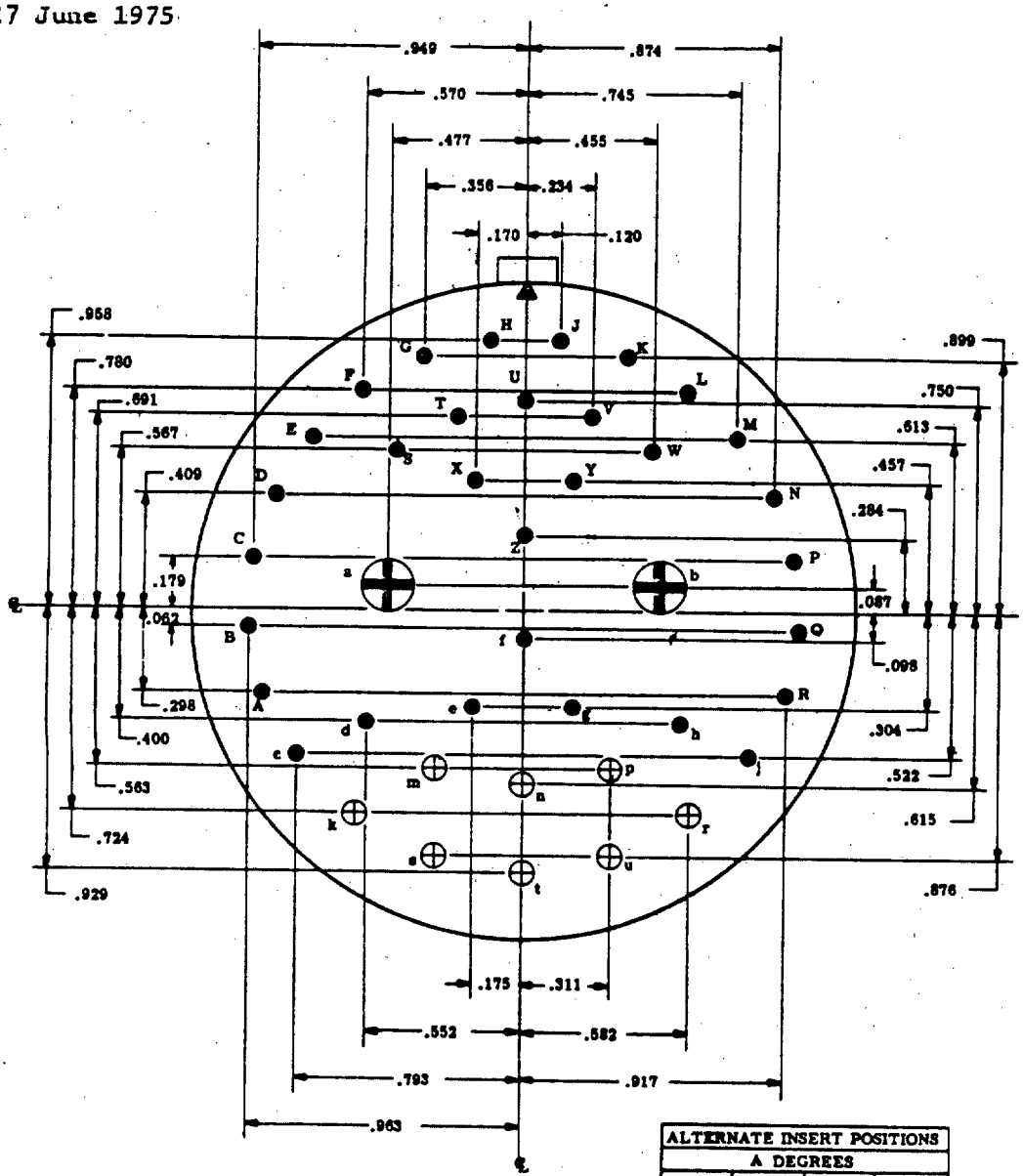


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-3	D	24	16		A11	MS33692-3
			3	12			
			2	8			
			2	4			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



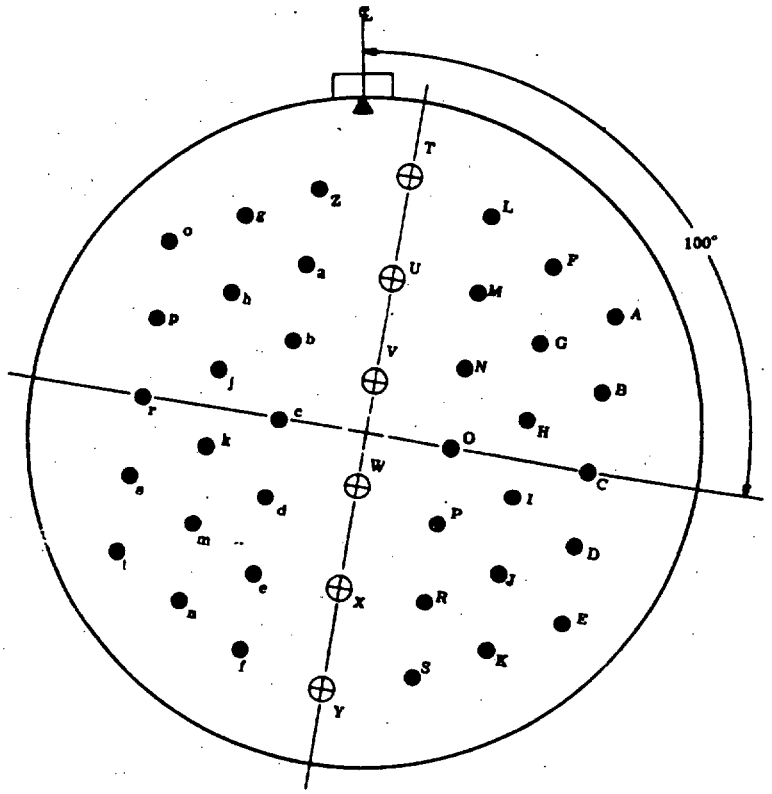
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-4	D	31	16	●	All	MS33692-4
			8	12	⊕		
			2	0	✚		

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INACTIVE FOR DESIGN AFTER 26 MAY 1945



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 44-1 EXCEPT FOR POSITION OF INSERT IN SHELL

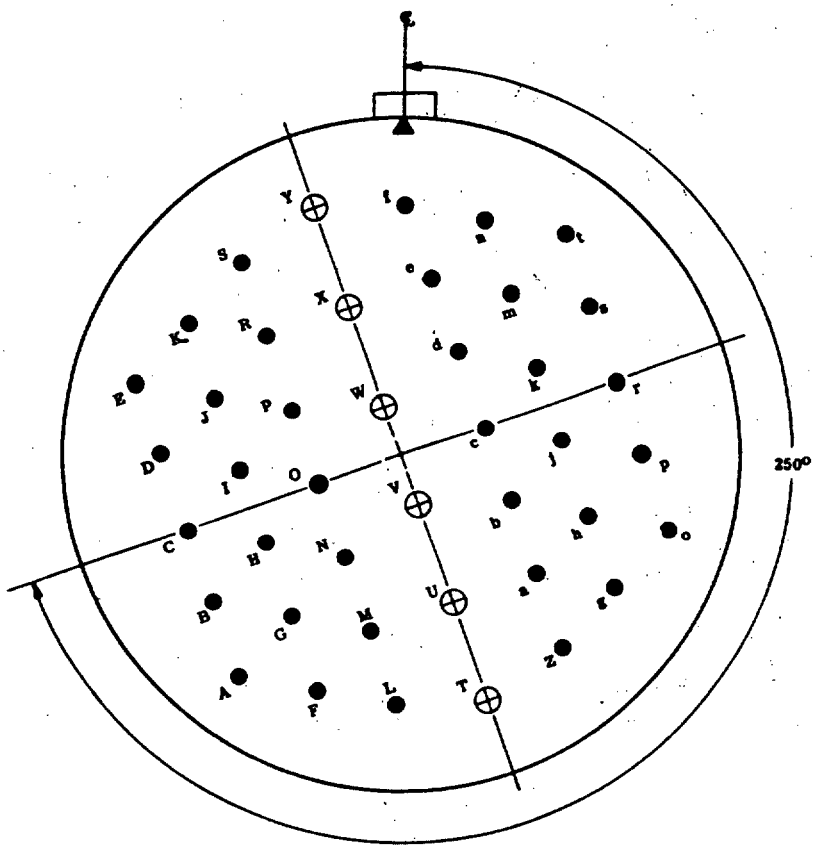
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-5	D	36	16	●	All	MS33692-5
			6	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR DESIGN AFTER 26 MAY 1945



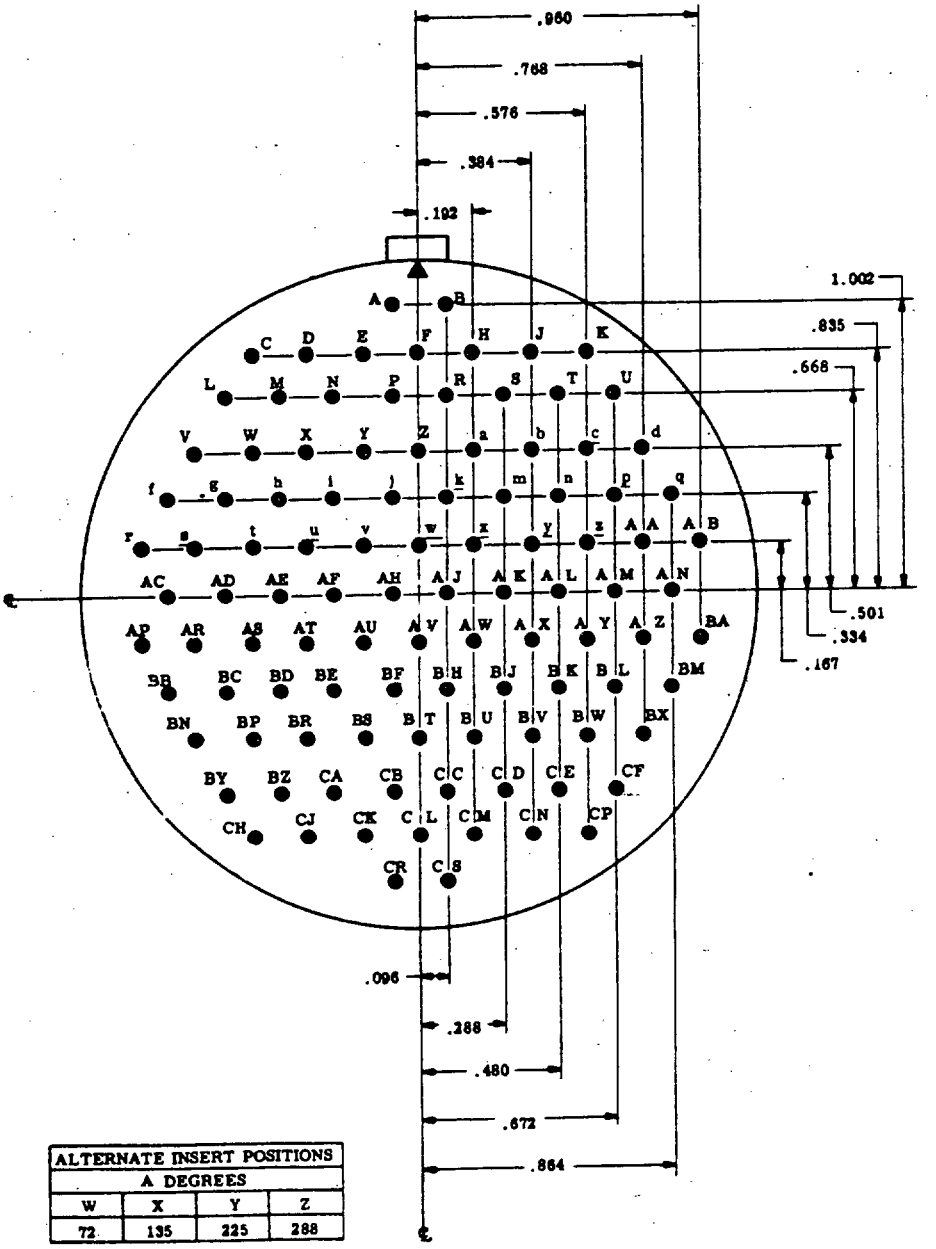
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 44-1 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
—	—	—	—

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-6	D	36	16	●	All	MS33692-6
			6	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS A DEGREES			
W	X	Y	Z
72	135	225	288

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-52	A	104	16	●	A11	MS33692-52

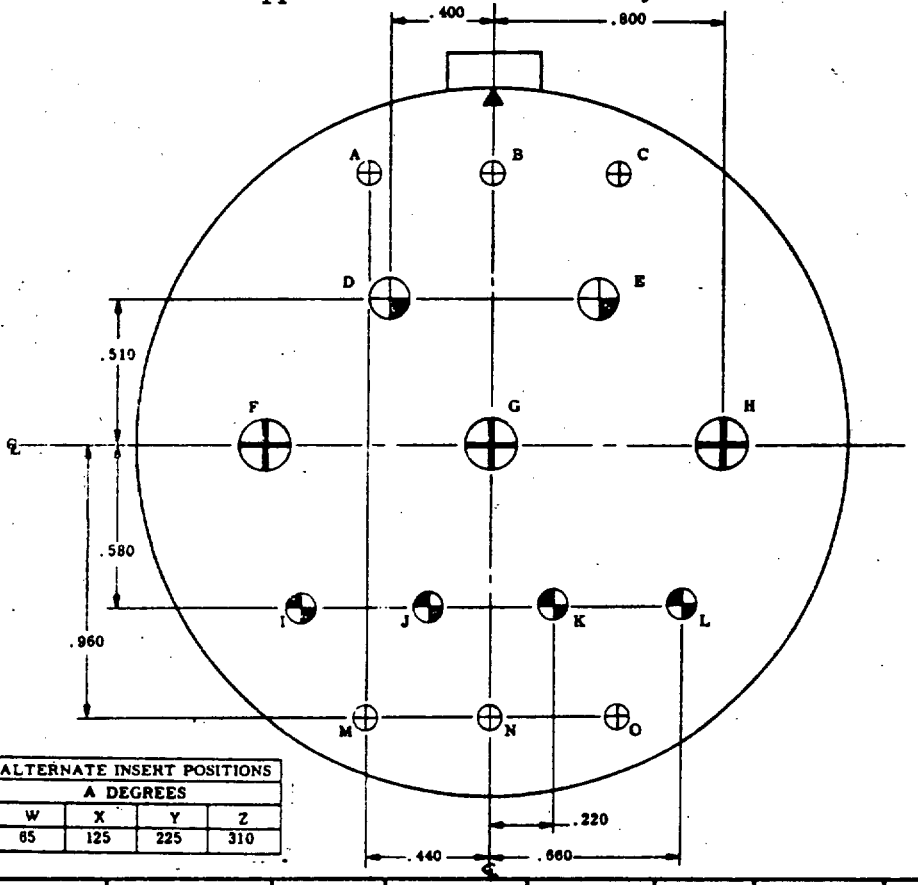
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

SECTION 150

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 48

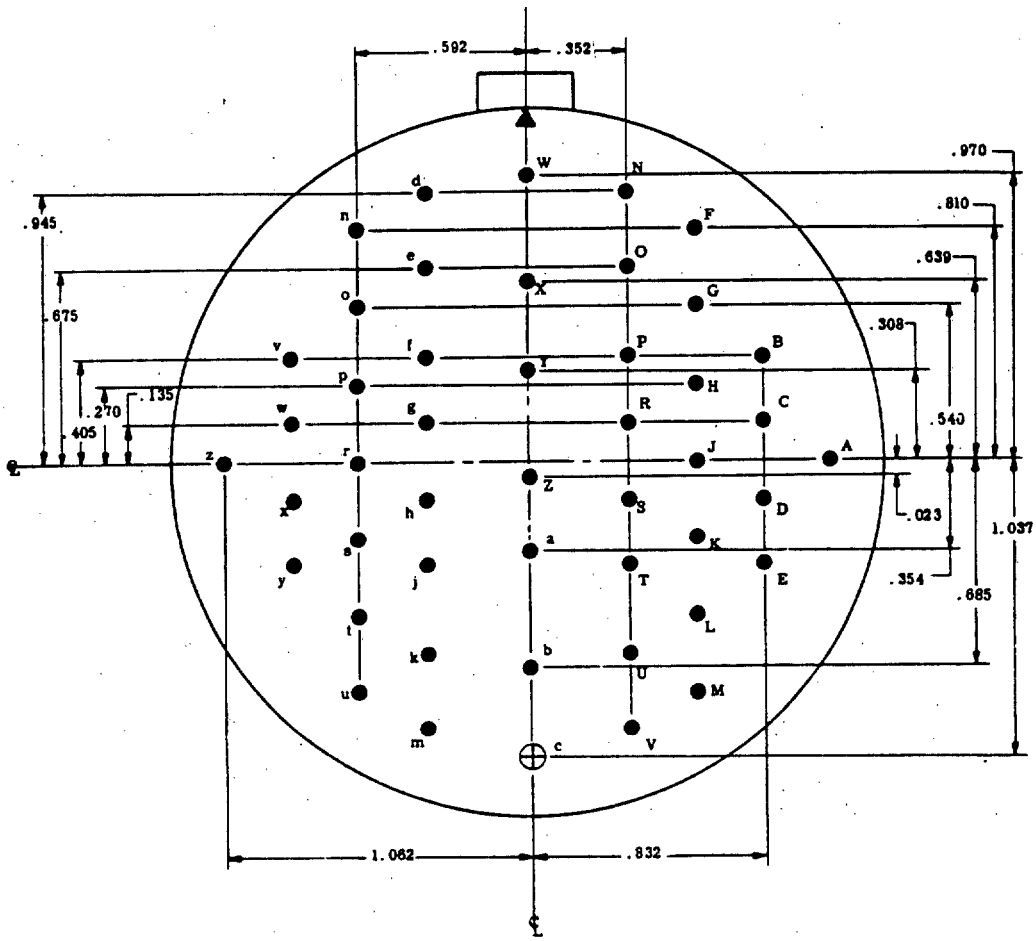
NOTE:
1. Shell size 48 applies to MIL-C-5015 only.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-1	E	2	12	⊕	B,N	MS33693-1
		D	4			A,C,M,O	
			4	8	◔	I,J,K,L	
			2	4	◑	D,E	
			3	0	◕	F,G,H	

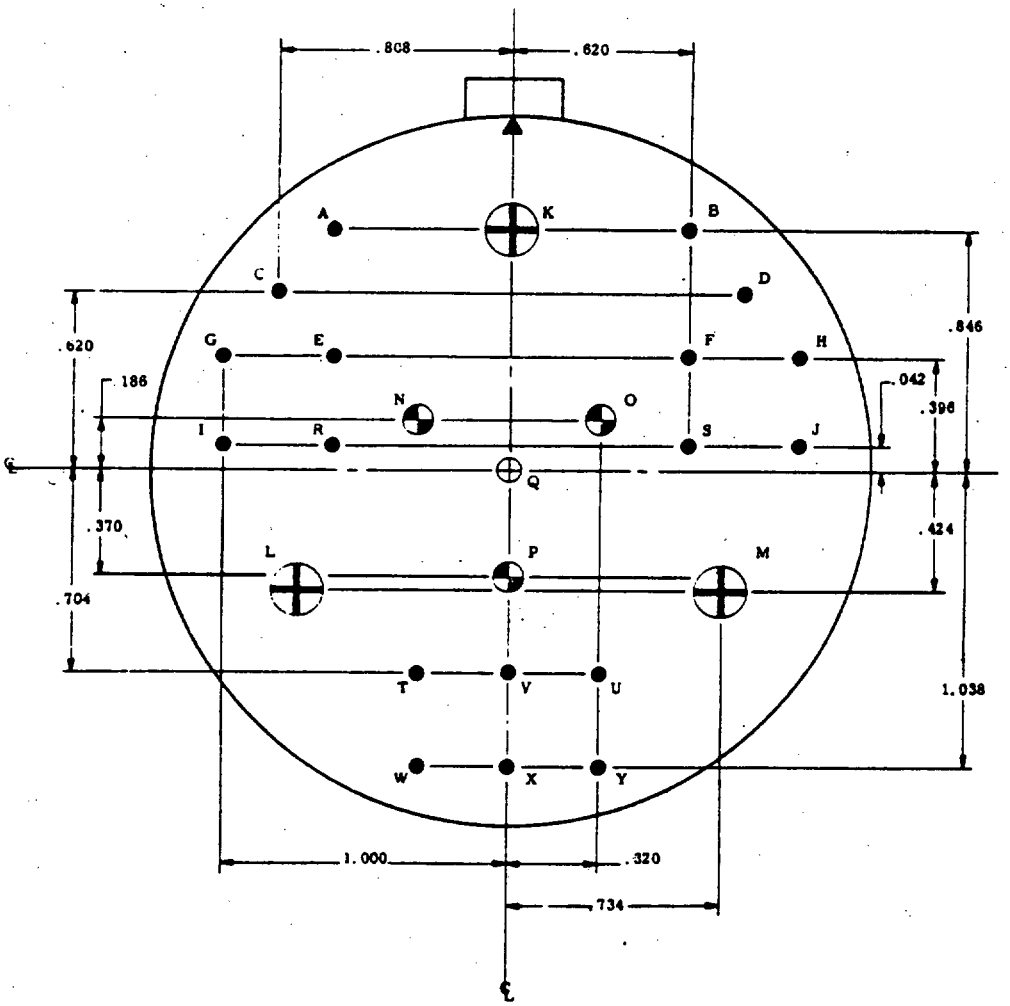
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-2	D	1	12	⊕	C	MS33693-2
		E	6	16	●	W,X,Y,Z, a,b	
		D	40			All others	

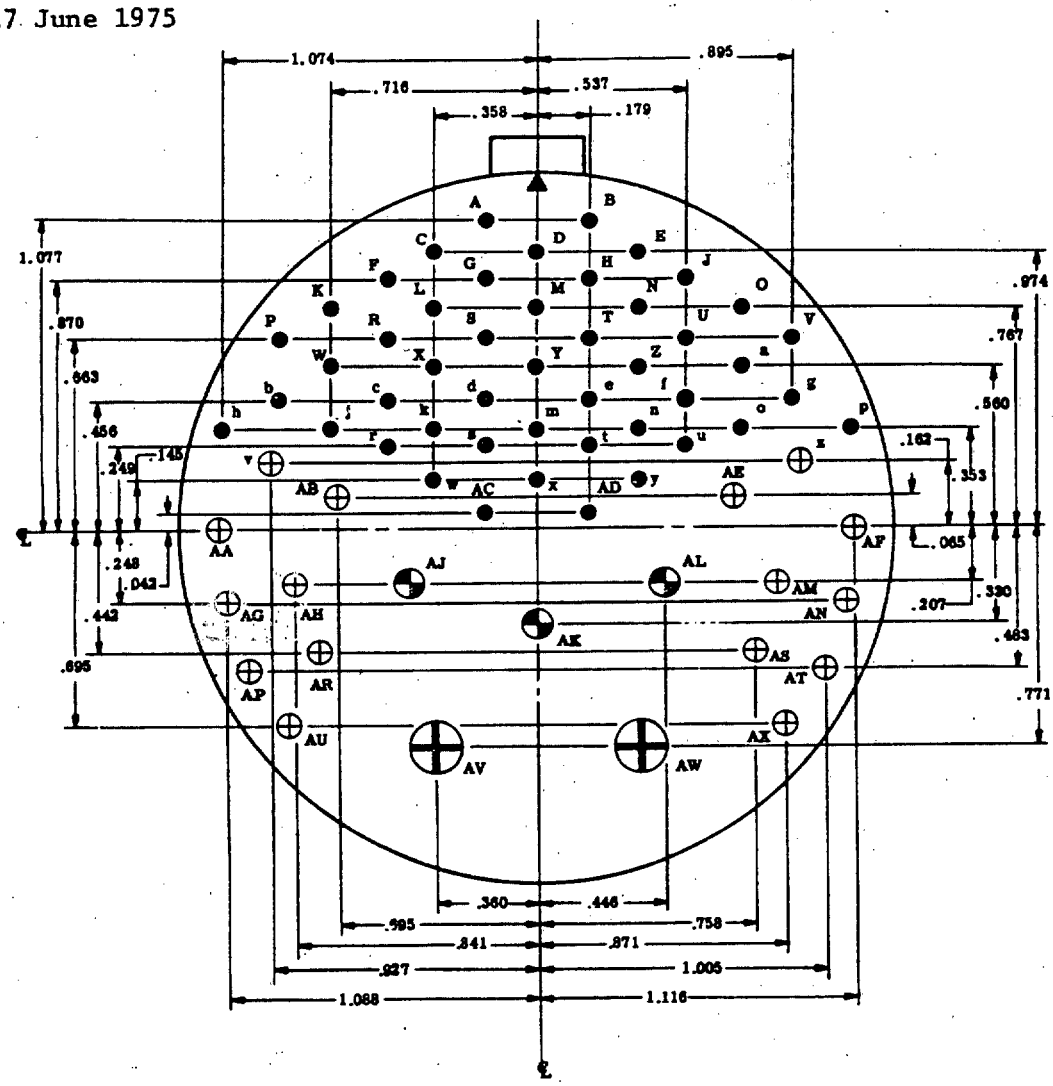
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-3	D	18	16	●	All	MS33693-3
			1	12	⊕		
			3	8	◕		
			3	0	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

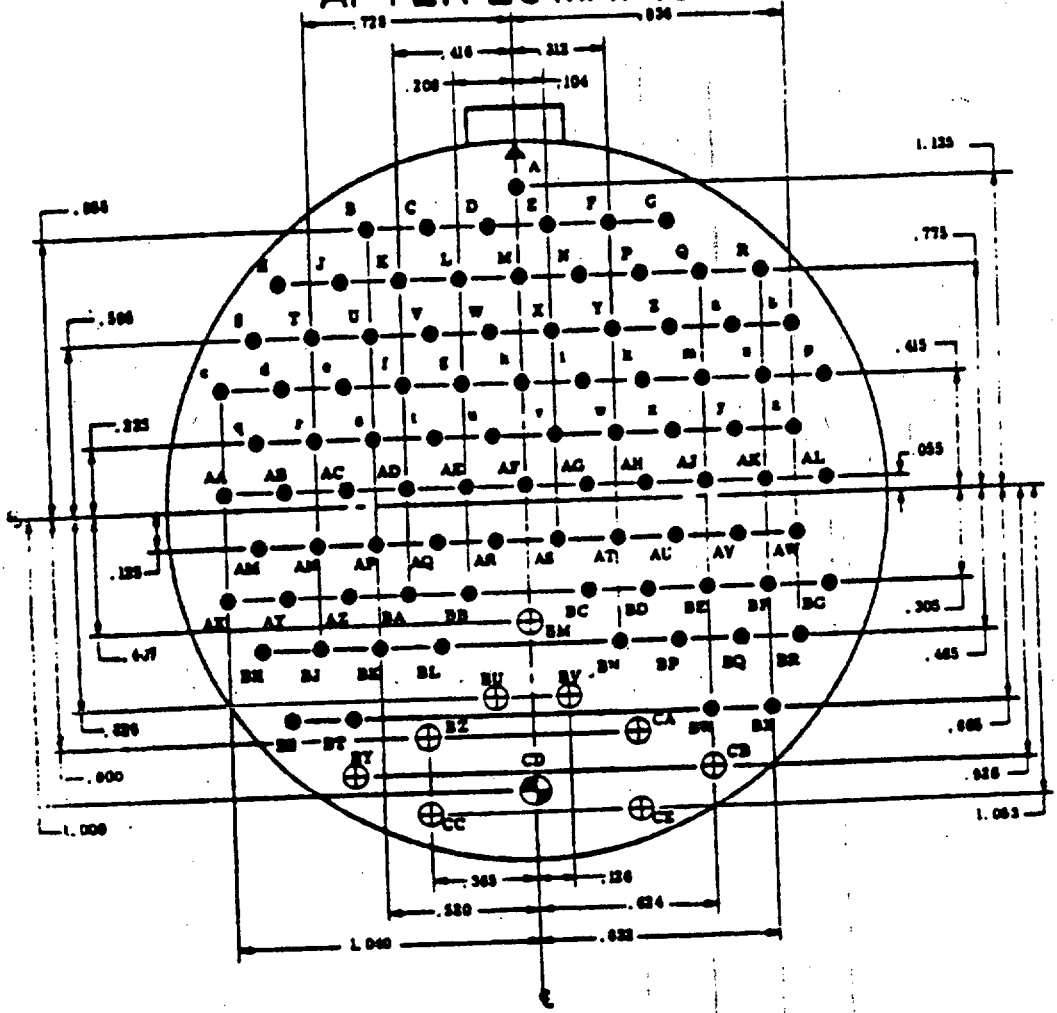


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-4	D	3	8		AJ,AK,AL	MS33693-4
			2	0		AV,AW	
		A	47	16		All others	
			16	12			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

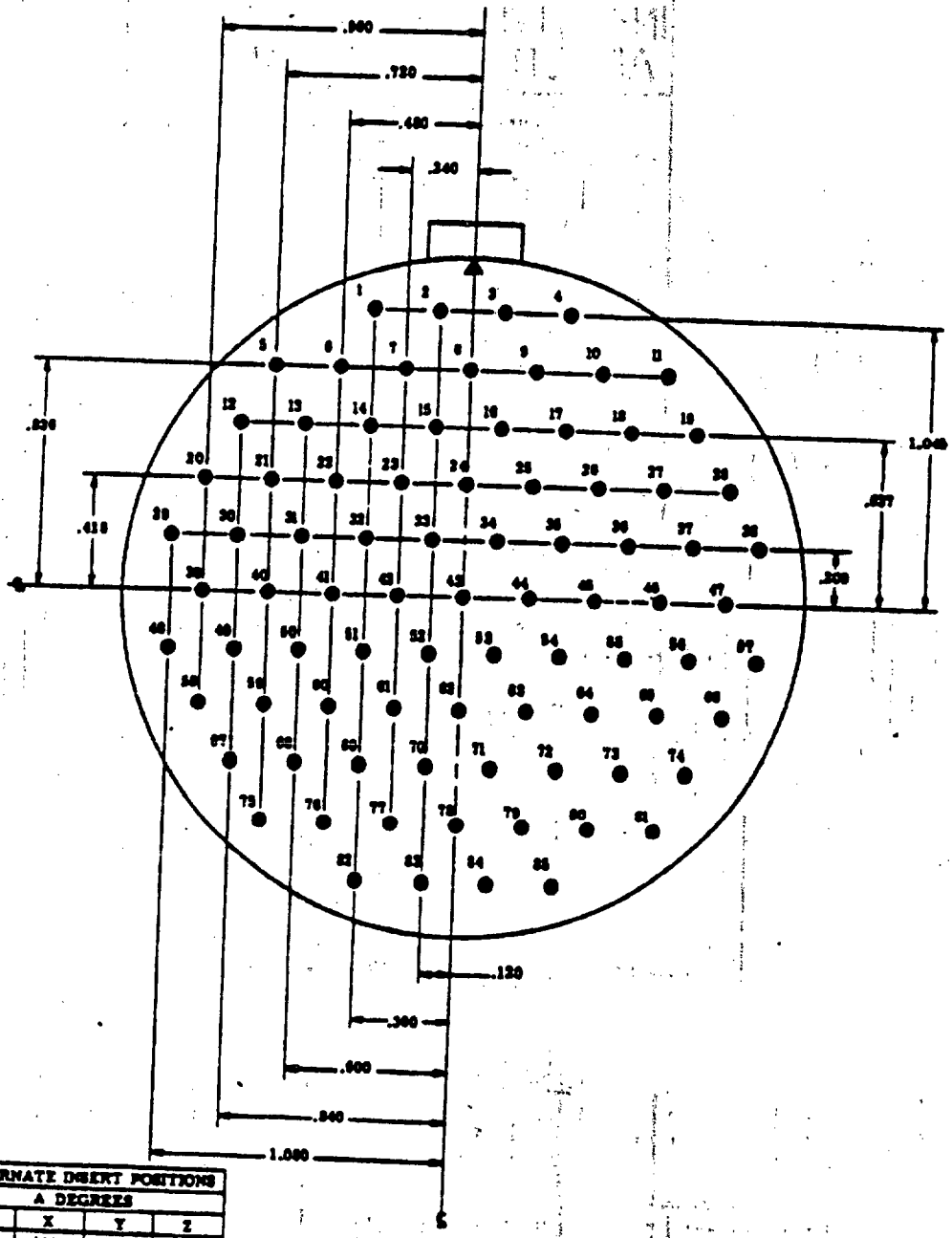
INACTIVE FOR DESIGN
AFTER 26 MAY 1945



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
--	--	--	--

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-5	A	90	16	●	All	MS33693-5
			9	12	⊕		
			1	8	◕		

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
72	144	216	288

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-62	D	35	16	●	All	MS33693-62

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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